LGS EXTRA CLASSES — NACCA CCP MASTERY SERIES

Basic 8 Science

First Term Complete Student Study Pamphlet

Indicator-by-Indicator Lessons, Diagrams, Activities, Misconception Checks & Practice Tests

LEARN • OBSERVE • PRACTISE • APPLY

Built specifically for Basic 8 (JHS 2) learners and Science facilitators across Ghana. Every lesson targets ONE official NaCCA Curriculum Indicator arranged strictly by Strand → Sub-Strand → Content Standard → Indicator for First Term.

5	9	23	184
Strands Covered	Sub-Strands	Indicator Lessons	Practice Questions

What This Complete Study Pamphlet Contains

- **Curriculum-Exact Coverage:** All 23 First Term Indicators scheduled in the Basic 8 Annual Scheme of Learning (Page 62, Weeks 1–13) grounded in NaCCA Science CCP (2023) and Basic 7 prior foundations.
- **Complete Lesson Architecture:** Every lesson includes Learning Outcomes, Key Terms, Clear Study Notes, Comparison Tables, Guided Practical/Analytical Activities, Misconception Alerts, Summary & 8 Practice Questions.
- **12 Labelled Scientific Diagrams:** Custom high-resolution illustrations covering mixtures, separation methods, carbon cycle, air balance, mosquito life cycle, seed beds, crop trials, tooth anatomy, tooth types, hydro dam, biogas digester, and waste hierarchy.
- **Self-Study Back Matter:** Full Answer Key with explanations for all 184 questions, 90-term Alphabetical Glossary, Study Progress Tracker, and Personal Error Log.

Audience: Basic 8 (JHS 2) Learners & Science Facilitators | **Edition:** First Term Edition (2026/2027 Academic Year Alignment)

Curriculum Alignment: Strictly aligned with the NaCCA Science Common Core Programme (CCP) Curriculum (Basic 7–9, 2023) and the Basic 8 Annual Scheme of Learning — First Term (Page 62, Weeks 1–13).

Important Notice, Scope & How to Use This Pamphlet

Orientation for Basic 8 Learners, Teachers & Parents

Strict First Term Scope & Curriculum Integrity Notice

Strictly aligned with the NaCCA Science Common Core Programme (CCP) Curriculum (Basic 7–9, 2023) and the Basic 8 Annual Scheme of Learning — First Term (Page 62, Weeks 1–13).

Scope Notice: This pamphlet covers strictly First Term Basic 8 (JHS 2) Indicators scheduled in the Scheme of Learning (Page 62): 5 Strands, 9 Sub-Strands, 13 Content Standards, and 23 Lesson Indicators across Weeks 1–13. Topics scheduled for Second Term or Third Term (such as Periodic Table elements/compounds, rocks, electric circuits/ capacitors, climate change, or agricultural tools) and curriculum indicators omitted from the Scheme of Learning are intentionally excluded to keep your First Term revision focused and exact.

Note on Sample Classroom Data: Where lessons demonstrate how to read growth charts or community disease tallies (e.g., Figure 2.5 in Lesson 9 and the clinic survey tally in Lesson 22), figures are clearly labelled Illustrative Classroom Sample Data for practice; learners should collect real local measurements during class investigations.

Decoding NaCCA Curriculum Indicator Codes

Decoding NaCCA Indicator Codes (Example: B8.2.3.1.2): B8 = Basic 8 (JHS 2) | .2 = Strand 2 (Cycles) | .3 = Sub-Strand 3 (Crop Production) | .1 = Content Standard 1 | .2 = Indicator 2 (Lesson 8).

Recommended Study Roles

User	Recommended Approach
Basic 8 Learner	Read Learning Outcomes and Key Terms first. Study notes and diagrams, trace Guided Activity steps, check Common Misconceptions, then solve all 8 practice questions BEFORE checking the Answer Key.
Science Facilitator	Use each indicator lesson to structure a lesson matching Weeks 1–13 of the Scheme of Learning. Assign Guided Activities in groups and discuss Common Misconceptions explicitly.
Independent Revision	Work one indicator per 25–35 minute session. Record scores in the Progress Tracker and log mistakes in the Error Log at the back.
Exam Preparation	Use the Quick Reference Guide, Lesson Summaries, Glossary, and Answer Key explanations to drill definitions, energy chains, diagrams, and structured answers.

Six-Step Mastery Method Inside Every Lesson

1. Follow the Curriculum Hierarchy

Lessons are organised strictly by Strand -> Sub-Strand -> Content Standard -> Indicator. Each lesson targets exactly ONE curriculum Indicator so you know the exact skill being tested.

2. Check Learning Outcomes & B7 Foundations First

Read the Learning Outcomes box before starting. Each lesson links back naturally to what you learned in Basic 7 (mixtures, water cycle, housefly, digestion, energy forms, waste, water-borne diseases).

3. Study Notes, Tables & Labelled Diagrams

Read explanations carefully alongside the labelled diagrams and comparison tables. Always trace arrow chains (carbon cycle, mosquito cycle, energy conversions) in order.

4. Work Through the Guided Activity

Every lesson includes a hands-on or analytical activity adapted from NaCCA curriculum exemplars—either perform it safely in class/school garden or study the step-by-step observations and conclusion.

5. Beat Common Misconceptions Early

Pay close attention to the Common Mistakes & Corrections box in every lesson; these highlight the exact examination traps learners frequently fall into.

6. Test Yourself & Verify in the Answer Key

Complete all 4 Multiple-Choice (MCQ) and 4 Short Structured questions at the end of every lesson before checking the complete Answer Key (with brief explanations) at the back of the pamphlet.

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Tip: In the digital PDF, click any line below to jump directly to that Strand, Content Standard, or Indicator lesson.

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Basic 7 Curriculum Foundations & First Term Alignment Map

Bridging Prior Knowledge (CCP B7) to First Term Basic 8 (Scheme of Learning, Page 62)

Each Basic 8 First Term Strand extends core concepts you built in Basic 7 without jumping outside First Term scope:

Strand	Basic 7 Prior Foundation (CCP)	Basic 8 First Term Extension (This Pamphlet)
Strand 1: Diversity of Matter	B7.1.1.1 (Classification & properties of matter; elements, compounds & mixtures)	B8.1.1.1.1–B8.1.1.1.2: Types of mixtures (uniform vs non-uniform) and practical separation methods (Lessons 1–2).
Strand 2: Cycles	B7.2.1.1 (Water cycle), B7.2.2.1 (Housefly metamorphosis), B7.2.3.1 (Soil & plant nutrients), B7.2.4.1 (Domestic animals)	B8.2.1.1–B8.2.4.2: Carbon cycle & air balance, Anopheles mosquito cycle & malaria control, seed beds & crop growth, animal feed & water (Lessons 3–11).
Strand 3: Systems	B7.3.1.1 (Human digestive system organs & digestion stages)	B8.3.1.1.1–B8.3.1.1.3: Mammalian tooth structure, four human tooth types/functions, tooth & gum decay prevention (Lessons 12–14).
Strand 4: Forces & Energy	B7.4.1.1–2 (Forms of energy; heat transfer by conduction, convection, radiation)	B8.4.1.1–B8.4.1.3: Energy conversion chains, hydro power, renewable vs non-renewable & biogas, sustainable management, heat vs temperature (Lessons 15–19).
Strand 5: Humans & Environment	B7.5.1.1 (Waste types & 3Rs), B7.5.2.1–2 (Food/water hygiene & viral/bacterial illnesses)	B8.5.1.1–B8.5.2.1: Sustainable waste hierarchy, community waste plan & evaluation, communicable diseases (Hepatitis etc.) & risk factors (Lessons 20–23).

First Term Scheme of Learning Alignment Checklist (Weeks 1–13)

Week	Str.	Sub-Strand	Content Std	Indicator	Lesson Number & Focus
Wk 1	**S1**	SS1 Materials	B8.1.1.1	**B8.1.1.1.1**	Lesson 1: Types of Mixtures & Their Properties
Wk 2	**S1**	SS1 Materials	B8.1.1.1	**B8.1.1.1.2**	Lesson 2: Separating Mixtures Practically
Wk 3	**S2**	SS1 Earth Science	B8.2.1.1	**B8.2.1.1.1**	Lesson 3: The Process of the Carbon Cycle
Wk 3	**S2**	SS1 Earth Science	B8.2.1.1	**B8.2.1.1.2**	Lesson 4: Environmental Role of the Carbon Cycle
Wk 4	**S2**	SS2 Life Cycle	B8.2.2.1	**B8.2.2.1.1**	Lesson 5: Life Cycle & Economic Importance of Anopheles
Wk 4	**S2**	SS2 Life Cycle	B8.2.2.1	**B8.2.2.1.2**	Lesson 6: Impact of Anopheles & Malaria Control
Wk 12–13	**S2**	SS3 Crop Production	B8.2.3.1	**B8.2.3.1.1**	Lesson 7: Types of Seed Beds & Matching Crops
Wk 12–13	**S2**	SS3 Crop Production	B8.2.3.1	**B8.2.3.1.2**	Lesson 8: Planting Crop Parts on Seed Beds
Wk 12–13	**S2**	SS3 Crop Production	B8.2.3.2	**B8.2.3.2.1**	Lesson 9: Comparing Growth & Flowering Across Seed Beds
Wk 13	**S2**	SS4 Animal Production	B8.2.4.1	**B8.2.4.1.1**	Lesson 10: Animal Feed Types, Nutrients & Labels
Wk 13	**S2**	SS4 Animal Production	B8.2.4.2	**B8.2.4.2.1**	Lesson 11: Importance of Water & Feed to Animal Growth
Wk 5	**S3**	SS1 Human Body	B8.3.1.1	**B8.3.1.1.1**	Lesson 12: External & Internal Parts of a Tooth
Wk 5	**S3**	SS1 Human Body	B8.3.1.1	**B8.3.1.1.2**	Lesson 13: Human Teeth Types & Functions
Wk 6	**S3**	SS1 Human Body	B8.3.1.1	**B8.3.1.1.3**	Lesson 14: Causes & Prevention of Tooth & Gum Decay
Wk 7	**S4**	SS1 Energy	B8.4.1.1	**B8.4.1.1.1**	Lesson 15: Energy Conversion & Everyday Chains
Wk 7	**S4**	SS1 Energy	B8.4.1.1	**B8.4.1.1.2**	Lesson 16: Importance of Conversion & Hydroelectric Power
Wk 8	**S4**	SS1 Energy	B8.4.1.2	**B8.4.1.2.1**	Lesson 17: Renewable & Non-Renewable Energy + Biogas
Wk 8	**S4**	SS1 Energy	B8.4.1.2	**B8.4.1.2.2**	Lesson 18: Sustainable Management of Renewable Energy
Wk 9	**S4**	SS1 Energy	B8.4.1.3	**B8.4.1.3.1**	Lesson 19: Heat and Temperature: Differences & Link
Wk 10	**S5**	SS1 Waste Management	B8.5.1.1	**B8.5.1.1.1**	Lesson 20: Sustainable Waste Management Practices
Wk 10	**S5**	SS1 Waste Management	B8.5.1.1	**B8.5.1.1.2**	Lesson 21: Community Waste Management Plan & Evaluation
Wk 11–12	**S5**	SS2 Human Health	B8.5.2.1	**B8.5.2.1.1**	Lesson 22: Common Communicable Diseases (Hepatitis etc.)
Wk 11–12	**S5**	SS2 Human Health	B8.5.2.1	**B8.5.2.1.2**	Lesson 23: Risk Factors of Communicable Diseases

First Term Science Quick Reference Guide

Essential Facts, Ratios, Chains & Distinctions at a Glance

Clean Dry Air Composition

~78% Nitrogen, ~21% Oxygen (O_2), ~0.03–0.04% Carbon (IV) Oxide (CO_2), ~0.9% Argon/other gases (maintained by photosynthesis & respiration).

Separation Quick Rule

Insoluble solid + liquid -> Filtration or Decantation | Dissolved solid + liquid -> Evaporation (keeps solid) or Simple Distillation (keeps pure liquid) | Immiscible liquids -> Separating funnel | Iron filings + non-magnetic solid -> Magnetisation | Miscible liquids (different boiling points) -> Fractional distillation.

Anopheles vs. Culex Mosquito

Anopheles: single eggs with floats; larvae have NO siphon and lie PARALLEL to water; adults rest at ~45° angle. Culex: eggs in rafts; larvae hang at angle with siphon; adults rest parallel.

Adult Human Dentition

32 permanent teeth total: 8 Incisors (biting/cutting), 4 Canines (tearing), 8 Premolars (crushing/grinding), 12 Molars (heavy grinding). Milk teeth = 20 (no premolars).

Hydroelectric Conversion Chain

Gravitational Potential Energy (dam reservoir) -> Kinetic Energy (penstock & spinning turbine) -> Electrical Energy (generator).

Heat vs. Temperature SI Units

Heat = energy (SI unit: Joule, J; depends on mass; measured with calorimeter). Temperature = hotness/coldness (SI unit: Kelvin, K, commonly °C; independent of mass; measured with thermometer).

Waste Priority Order

1. Reduce -> 2. Reuse -> 3. Recycle / Upcycle -> 4. Compost / Biogas Energy Recovery -> 5. Safe Disposal (Never open-burn plastics!).

BASIC 8 SCIENCE • FIRST TERM

STRAND 1: DIVERSITY OF MATTER

Matter makes up everything around us—solids, liquids, and gases. Building on your Basic 7 study of the states of matter, First Term Basic 8 examines how different materials combine physically to form mixtures, how to distinguish solutions, colloids, and suspensions, and how to design scientific methods to separate mixtures cleanly.

Sub-Strand 1: Materials

Content Standard B8.1.1.1: Demonstrate knowledge of types of mixtures, and understanding of the processes of scientific ways of separating the components of mixtures.

LESSON 1 Types of Mixtures and Their Characteristics

B8.1.1.1.1 • Week 1

Target Indicator: Identify types of mixtures by name and characteristics.
Scheme Topic: Mixtures | **Resources:** Powder, pebbles, bottle tops, salt, sugar, sand, gari, gravel, oil, water, groundnut paste

Learning Outcomes — By the end of this lesson, you should be able to:

- Group everyday materials into solids and liquids before combining them.
- Explain what a mixture is and distinguish between homogeneous and heterogeneous mixtures.
- Compare and contrast a solute and a solvent using physical characteristics.
- Identify solutions, suspensions, and colloids by name, appearance, and behaviour with light (the Tyndall effect).

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Mixture	A physical combination of two or more substances in which each substance keeps its own chemical identity and properties.
Homogeneous mixture	A mixture that looks uniform throughout because its components are evenly distributed (e.g., salt dissolved in water).
Heterogeneous mixture	A mixture whose parts are unevenly distributed and can often be seen separately (e.g., sand and gravel; oil and water).
Solute	The substance that dissolves in a solvent to form a solution (e.g., table salt or sugar).
Solvent	The substance (usually a liquid) that dissolves a solute (e.g., water — often called the universal solvent).
Solution	A clear, homogeneous mixture formed when a solute dissolves completely in a solvent.
Suspension	A heterogeneous mixture in which solid particles remain visible and settle to the bottom when left undisturbed (e.g., groundnut paste in water).
Colloid	A mixture with medium-sized particles that do not dissolve completely yet stay suspended without settling, and scatter a beam of light (Tyndall effect).

Clear Study Notes

1. Grouping Everyday Materials into Solids and Liquids

From Basic 7 (B7.1.1.1), you learned that materials exist as solids, liquids, or gases. Before forming mixtures in the laboratory or kitchen, we classify common materials by their physical state:

When any two or more of these materials are put together without a chemical reaction taking place, they form a mixture. For instance, mixing gari and sugar leaves both tasting sweet and gritty—neither turns into a brand-new substance.

Physical State	Everyday Examples (from Scheme & Curriculum)	Key Physical Features
Solids	Powder, pebbles, bottle tops, salt, sugar, sand, gari, gravel	Fixed shape and volume; particles tightly packed
Liquids	Water, cooking oil, vinegar, fruit juice	Fixed volume, takes shape of container; flows easily

2. Homogeneous vs. Heterogeneous Mixtures

Depending on how evenly the components spread out, mixtures are grouped into two broad categories:

Feature	Homogeneous Mixture	Heterogeneous Mixture
Appearance	Uniform throughout; looks like a single phase	Non-uniform; separate parts or layers are visible
Particle distribution	Evenly mixed at the microscopic level	Unevenly mixed; patches or layers form

Feature	Homogeneous Mixture	Heterogeneous Mixture
Examples	Salt solution, sugar solution, vinegar solution, clear fruit juice, clean air	Sand and gravel, pebbles and bottle tops, sand and water, oil and water, gari and water

3. Solutes, Solvents, and Solutions

When sugar or salt is stirred into water, it disappears from view to form a clear liquid called a solution. The solid that dissolves (salt or sugar) is the solute, while the liquid that does the dissolving (water) is the solvent.

Notice the contrast: a solute is usually present in smaller amount and changes into the dissolved state, whereas a solvent is present in larger amount and provides the dissolving medium. Common solutions include salt solution (salt in water), sugar solution (sugar in water), vinegar solution (ethanoic acid in water), and filtered fruit juice (sugars, acids, and vitamins dissolved in water).

Solid–solid mixtures such as sand and sugar or sugar and salt are NOT solutions unless dissolved in a liquid; because you can still tell the grains apart with a hand lens, dry sand and sugar is a heterogeneous solid–solid mixture.

4. Comparing Solutions, Colloids, and Suspensions

Not every solid mixed into water dissolves! Depending on particle size, mixtures in liquids fall into three types: true solutions, colloids, and suspensions. When you stir groundnut paste into a glass of water, it looks cloudy and thick—if you leave the glass undisturbed, heavier particles gradually settle at the bottom. That is a suspension.

By contrast, mixing a few drops of milk or cooked starch into water creates a colloid. Its particles are larger than dissolved salt particles in a solution, yet small enough to stay suspended indefinitely without settling. If you shine a torchlight through all three in a darkened room, the colloid visibly scatters the light beam—this scattering of light by suspended particles is called the colloidal effect (Tyndall effect).

Property	True Solution	Colloid	Suspension
Particle size	Smallest (ions/molecules)	Medium (1–1000 nm)	Largest (visible bits)
Appearance	Clear & transparent	Cloudy or milky; looks uniform	Cloudy/muddy; clearly uneven
Settles on standing?	No	No	Yes — forms sediment at bottom
Passes through filter paper?	Yes, completely	Yes, passes through	No — solid trapped as residue
Tyndall (light-scattering) effect	No (beam invisible)	Yes (bright visible beam)	Scatters or blocks light while stirred
Ghanaian examples	Salt water, sugar water, vinegar	Milk in water, starch water, soapy water	Groundnut paste in water, muddy water, chalk in water

Comparison of Solution, Colloid, and Suspension (with Light Beam Test)

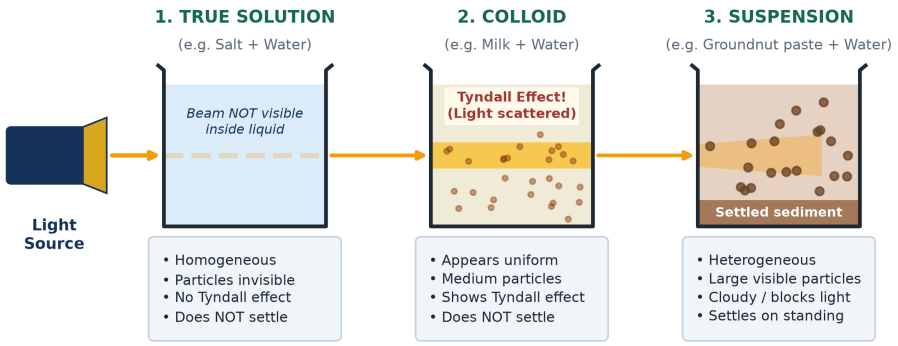


Figure 1.1: Comparing a true solution, a colloid showing the Tyndall effect, and a settling suspension.

Guided Practical Investigation: Testing Mixtures & Observing the Colloidal (Tyndall) Effect

Aim: To classify five mixtures by appearance, settling behaviour, and light scattering.

Materials: 5 transparent glasses/beakers, water, table salt, cooking oil, dry sand, powdered milk, groundnut paste, spoon, and a torchlight.

Procedure / Steps:

1. Label the five glasses A–E and half-fill each with clean water.

2. Stir 1 teaspoon of: salt into A, cooking oil into B, sand into C, 5 drops of milk into D, and 1/2 teaspoon of groundnut paste into E.

3. Observe each glass immediately and again after standing undisturbed for 10 minutes.

4. In a dimly lit corner, shine a narrow torchlight beam sideways through glasses A, D, and E.

Key Observations / Findings:

• Glass A (salt + water): Clear and uniform; nothing settles; light beam is NOT visible inside the water → Homogeneous Solution.

• Glass B (oil + water): Oil floats as a separate top layer → Heterogeneous liquid–liquid mixture (immiscible liquids).

• Glass C (sand + water): Sand sinks quickly to the bottom → Heterogeneous Suspension.

• Glass D (milk + water): Remains evenly milky without settling; torch beam shines clearly across the liquid → Colloid (shows Tyndall effect).

• Glass E (groundnut paste + water): Thick, cloudy mixture that separates and leaves paste particles settling on standing → Suspension.

Scientific Conclusion: Mixtures differ in uniformity, settling behaviour, and light scattering depending on the solubility and particle size of their components.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Calling every cloudy liquid a solution. Correction: A true solution is always clear (though it may be coloured, like tea or diluted Sobolo after fine filtering). Cloudy mixtures like groundnut paste in water are suspensions or colloids.
• Mistake: Mixing up solute and solvent. Correction: Water is the solvent (it dissolves); salt or sugar is the solute (it gets dissolved).
• Mistake: Thinking oil and water form a solution because both are liquids. Correction: Oil and water do not dissolve in each other—they are immiscible liquids forming a heterogeneous mixture.

Lesson Summary — Key Points to Remember

✓ Materials can be grouped into solids and liquids; combining them physically forms a mixture.
✓ Homogeneous mixtures (e.g., salt solution) appear uniform throughout; heterogeneous mixtures (e.g., sand & gravel, oil & water) show distinct parts.
✓ In a solution, a solute dissolves completely in a solvent.
✓ Suspensions (e.g., groundnut paste in water) have large particles that settle on standing; colloids (e.g., milk in water) do not settle and scatter light (Tyndall effect).

Practice Exercise 1 (B8.1.1.1.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which of the following pairs forms a homogeneous mixture?
A. Sand and gravel B. Cooking oil and water C. Sugar and water (well stirred) D. Pebbles and bottle tops

2. In a vinegar solution made by mixing acetic acid with water, water acts as the:
A. Solute B. Solvent C. Suspension D. Residue

3. Groundnut paste stirred thoroughly into a glass of water is best classified as a:
A. True solution B. Pure compound C. Suspension D. Gaseous mixture

4. Which property reliably distinguishes a colloid from a true solution?
A. A colloid settles rapidly to the bottom B. A colloid scatters a beam of light (Tyndall effect) C. A colloid contains only solid solutes D. A colloid has two clearly separated layers

Part B: Short Structured Questions (Write clear scientific answers)

5. Group the following materials into solids and liquids: gari, oil, bottle tops, vinegar, pebbles, water, sugar, powder.

6. State two differences between a solute and a solvent, giving one example of each.

7. Explain why a mixture of sand and sugar is described as heterogeneous, whereas salt solution is homogeneous.

8. Describe a simple test you can perform using a torchlight to distinguish a glass of dilute milk from a glass of clear sugar water.

LESSON 2

Scientific Methods of Separating Mixtures

B8.1.1.1.2 • Week 2

Target Indicator: Design and perform processes for separating kinds of mixtures.
Scheme Topic: Separation of Mixtures | **Resources:** Salt, sugar, sand, water, cooking oil, sieve, filter paper, funnel, beaker, evaporating dish, distillation flask, magnet

Learning Outcomes — By the end of this lesson, you should be able to:

• Select an appropriate separation method based on the physical properties of the mixture's components.
• Describe and perform sieving, filtration, evaporation, decantation, separating funnel use, and simple distillation.
• Design a multi-step separation plan for mixed materials such as sand and salt.
• Present findings from a separation experiment using a labelled diagram and structured scientific report.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Sieving	Separating solid particles of different sizes by passing the mixture through a mesh or sieve.
Filtration	Separating an insoluble solid from a liquid using a porous barrier such as filter paper.
Residue	The insoluble solid trapped on the filter paper during filtration.
Filtrate	The clear liquid that passes through the filter paper during filtration.
Evaporation	Heating a solution to turn the liquid solvent into vapour so the dissolved solid solute remains behind.
Simple distillation	Separating and recovering a liquid solvent from a solution by boiling (evaporation) followed by cooling the vapour (condensation) into a pure distillate.
Decantation	Carefully pouring off a top liquid layer after heavy solid particles have settled at the bottom.

Key Term	Clear Scientific Meaning
Separating funnel	A pear-shaped glass apparatus with a tap used to separate two immiscible liquids of different densities (such as oil and water).

Clear Study Notes

1. Matching Separation Method to Physical Property

Because substances in a mixture are only physically combined, we separate them by taking advantage of differences in their physical properties—such as particle size, solubility, boiling point, density, or magnetism.

Separation Method	Type of Mixture	Physical Property Used	Practical Example
Handpicking / Winnowing	Solid + Solid (large or light/ heavy)	Size, colour, or density in moving air	Picking pebbles from beans; winnowing chaff from rice/maize
Sieving	Solid + Solid (different particle sizes)	Particle size vs. mesh holes	Separating coarse stones from fine gari or building sand
Magnetic separation	Magnetic + non-magnetic solids	Attraction to a magnet	Removing iron filings from sand or sulphur
Decantation	Heavy insoluble solid + liquid	Density / settling under gravity	Pouring water off settled sand or washed rice
Filtration	Fine insoluble solid + liquid	Particle size (trapped by filter pores)	Separating chalk powder or fine sand from water
Evaporation / Crystallisation	Soluble solid + liquid (keep solid)	Boiling point difference (solvent boils off)	Obtaining salt crystals from salt water
Simple distillation	Soluble solid + liquid (keep pure liquid)	Boiling point + condensation of vapour	Recovering pure distilled water from salt solution
Separating funnel	Two immiscible liquids	Difference in density & non-mixing layers	Separating cooking oil (top) from water (bottom)

2. How Filtration and Simple Distillation Work

In filtration, a folded filter paper rests inside a filter funnel above a beaker. When a mixture like muddy or sandy water is poured in, water molecules pass freely through microscopic pores in the paper to collect below as the clear filtrate, while larger sand grains stay trapped on the paper as the residue.

Evaporation recovers the dissolved salt from salt water, but the water escapes into the air as steam. Whenever you want to keep the pure liquid (e.g., obtaining fresh drinking water from salty seawater), you must use simple distillation: the solution is heated in a distillation flask until the water boils at 100 °C; steam travels through a water-cooled Liebig condenser where cold running water cools it back into liquid drops called the distillate, leaving the salt behind in the flask.

3. Multi-Step Separation (Combining Methods)

Sometimes one method is not enough! Consider a mixture of sand and table salt (or sand and sugar):

Step 1 (Dissolving): Add water and stir well. Salt dissolves (soluble), while sand does not (insoluble).

Step 2 (Filtration): Pour through filter paper. Sand remains on the filter paper as residue; salt solution passes through as filtrate.

Step 3 (Evaporation or Distillation): Heat the salt-solution filtrate gently in an evaporating dish to drive off the water and recover dry salt crystals (or distill it to recover both pure water and salt!).

Note on Sugar + Salt: Both sugar and salt dissolve in water, so water alone cannot separate them easily! However, sugar dissolves in warm ethanol (alcohol) whereas common salt does not—showing why choosing the right solvent matters.

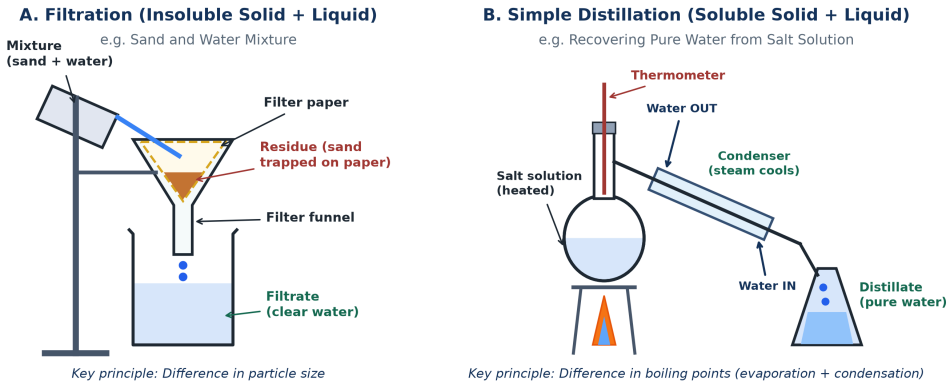


Figure 1.2: Laboratory setups for (A) Filtration and (B) Simple Distillation.

Guided Practical Activity & Report: Separating a Mixture of Sand, Iron Filings, and Salt

Aim: To design and perform a three-step physical separation process and write a complete scientific report.

Materials: Mixture of dry sand, iron filings, and table salt; bar magnet wrapped in paper; beaker; water; stirring rod; filter funnel & filter paper; evaporating dish; Bunsen burner/heat source.

Procedure / Steps:

1. Step 1 (Magnetic separation): Spread the dry mixture on a sheet of paper and pass a paper-wrapped bar magnet over it to lift out the magnetic iron filings.
2. Step 2 (Dissolution & Filtration): Pour the remaining sand + salt into a beaker of water and stir until the salt dissolves. Filter the mixture through folded filter paper into a clean conical flask. Rinse and dry the sand residue on the paper.
3. Step 3 (Evaporation): Pour the clear salt filtrate into an evaporating dish and heat gently until water evaporates and white salt crystals remain.
4. Step 4 (Reporting): Record labelled drawings of the filtration and evaporation setups alongside observations and conclusions.

Key Observations / Findings:
- Iron filings clung cleanly to the magnet.
- Sand remained trapped on the filter paper as brown residue while clear salty water collected as filtrate.
- After heating the filtrate, white crystalline salt remained in the evaporating dish.

Scientific Conclusion: A complex mixture can be separated into its pure components by sequentially applying magnetism, dissolution, filtration, and evaporation.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

- Mistake: Heating salt and sand directly expecting the sand or salt to separate. Correction: Neither sand nor salt evaporates easily; you must first dissolve the salt in water, filter out the sand, and then evaporate the water.
- Mistake: Confusing evaporation with distillation. Correction: Evaporation lets the liquid escape as vapour so you keep only the solid; distillation condenses the vapour back to liquid so you recover the pure liquid solvent.
- Mistake: Thinking filtration can separate dissolved salt from water. Correction: Dissolved salt particles are tiny enough to pass straight through filter-paper pores along with the water.

Lesson Summary — Key Points to Remember

✓ Separation methods rely on physical differences such as particle size, solubility, magnetism, density, and boiling point.
✓ Sieving separates dry solids of different sizes; filtration separates an insoluble solid (residue) from a liquid (filtrate).
✓ Evaporation recovers a soluble solid from a solution, while simple distillation recovers the pure liquid (distillate).
✓ A separating funnel separates immiscible liquids such as oil and water by density.

Practice Exercise 2 (B8.1.1.1.2)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. A trader accidentally spills sand into a bag of table salt. What is the correct order of steps to recover clean, dry salt?
A. Sieving → Magnetic separation B. Add water → Filter → Evaporate filtrate C. Evaporate → Filter → Add water D. Use separating funnel → Filter

2. During filtration of muddy water, the mud left behind on the filter paper is called the:
A. Filtrate B. Distillate C. Solute D. Residue

3. Which apparatus is best suited to separate cooking oil mixed with water?
A. Filter funnel B. Separating funnel C. Sieve D. Bar magnet

4. In simple distillation of salt water, what is the role of the Liebig condenser?
A. To boil the salt solution faster B. To trap sand grains C. To cool steam back into liquid water D. To dissolve more salt

Part B: Short Structured Questions (Write clear scientific answers)

5. State the most suitable method for separating each mixture: (a) stones and gari, (b) chalk dust and water, (c) iron nails and sawdust, (d) pure water from seawater.

6. Why does filtration fail to separate sugar from a sugar solution?

7. In the distillation diagram (Figure 1.2B), why does cold water enter at the lower end of the condenser and leave at the upper end?

8. Outline the five standard headings a student should include when writing a scientific report on a mixture-separation experiment.

BASIC 8 SCIENCE • FIRST TERM

STRAND 2: CYCLES

Cycles in nature are repeated patterns of change that sustain ecosystems, agriculture, and life. Building on your Basic 7 study of the water cycle and the housefly, First Term Basic 8 investigates the carbon cycle, the life cycle and control of the Anopheles mosquito, seed-bed crop production, and farm animal nutrition and water requirements.

Sub-Strand 1: Earth Science

Content Standard B8.2.1.1: Demonstrate understanding of the process of Carbon cycle as an example of repeated pattern of change in nature and how it relates to the environment.

LESSON 3

The Process of the Carbon Cycle

B8.2.1.1.1 • Week 3

Target Indicator: Explain the process of the carbon cycle.
Scheme Topic: The Carbon Cycle | **Resources:** Pictures and charts of the carbon cycle

Learning Outcomes — By the end of this lesson, you should be able to:

- Identify the main reservoirs and stages of the carbon cycle from charts and diagrams.
- Construct a clear flow chart tracing how carbon moves continuously through nature.
- Explain the four core processes in the carbon cycle: photosynthesis, respiration, burning (combustion), and decay (decomposition).
- Give reasons why the carbon cycle is described as a repeated pattern of change.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Carbon cycle	The continuous natural circulation of carbon between the atmosphere, living organisms, soils, oceans, and fossil fuels.
Photosynthesis	The process by which green plants use sunlight, water, and Carbon (IV) Oxide (CO_2) to make glucose (food) and oxygen.
Respiration	The process in living cells of plants, animals, and microbes that breaks down food to release energy and gives off CO_2.
Combustion (Burning)	The rapid reaction of carbon-containing fuels (wood, charcoal, petrol, coal) with oxygen, releasing heat, light, and CO_2.
Decay (Decomposition)	The breakdown of dead plants, dead animals, and organic waste by decomposers (bacteria and fungi), releasing CO_2.
Fossil fuels	Carbon-rich fuels (coal, crude oil, natural gas) formed over millions of years from buried remains of ancient organisms.

Clear Study Notes

1. What Is Carbon and Where Is It Stored?

Carbon is an essential chemical element found in all living things (carbohydrates, proteins, fats, DNA), in the air as Carbon (IV) Oxide gas (CO_2), dissolved in oceans and rivers, and locked underground in rocks and fossil fuels (coal, crude oil, natural gas). Just like water in the water cycle (Basic 7), carbon moves endlessly from place to place without disappearing.

2. The Four Core Processes Driving the Carbon Cycle

Four major biological and physical processes shift carbon into and out of the atmosphere:

Process	Direction of Carbon Movement	How It Happens in Nature
(a) Photosynthesis	Atmosphere (CO_2) → Green Plants	Plants absorb CO_2 through leaf stomata and convert it with water and sunlight into glucose and starch, locking carbon into plant tissues.
(b) Respiration	Plants, Animals & Microbes → Atmosphere (CO_2)	Both plants (day and night) and animals break down glucose in cells for energy, releasing CO_2 back into the air.
(c) Burning (Combustion)	Wood, Charcoal & Fossil Fuels → Atmosphere (CO_2)	Burning firewood, charcoal, bush, petrol, diesel, or coal combines stored carbon with oxygen, rapidly releasing large amounts of CO_2.
(d) Decay (Decomposition)	Dead Organisms & Waste → Soil + Atmosphere (CO_2)	Bacteria and fungi feed on dead leaves, dead animals, and dung; as they respire, they release CO_2 into the air and return nutrients to soil. Buried remains over millions of years form fossil fuels.

3. Why the Carbon Cycle Is a Repeated Pattern of Change

In nature, carbon is neither created nor destroyed—it is recycled continuously. A carbon atom in atmospheric CO_2 today may be absorbed by a maize leaf tomorrow, eaten by a chicken or goat next week, breathed back out into the air through respiration, or returned to the soil when a plant dies and decays. Because the removal of CO_2 (mainly by photosynthesis) and the return of CO_2 (by respiration, decay, and burning) repeat endlessly without a fixed start or finish, the carbon cycle is a repeated pattern of change.

The Carbon Cycle: Continuous Circulation of Carbon in Nature

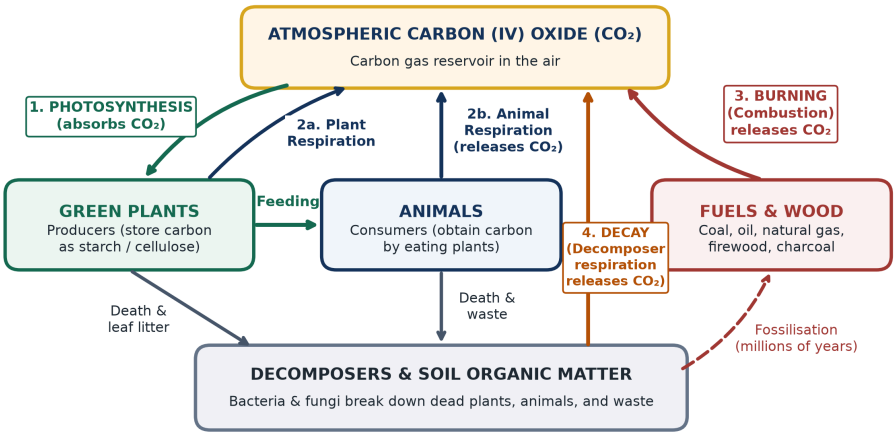


Figure 2.1: Flow chart of the Carbon Cycle showing Photosynthesis, Respiration, Feeding, Decay, and Burning.

Guided Chart Activity: Tracing a Carbon Atom Through a Flowchart

Aim: To construct and narrate a six-step flow chart showing how carbon circulates repeatedly in a Ghanaian farm ecosystem.

Materials: Notebook, ruler, pencils/coloured pens, Figure 2.1.

Procedure / Steps:

1. Draw five labelled boxes: (1) Atmospheric CO_2, (2) Cassava Plant, (3) Goat, (4) Soil Decomposers, (5) Firewood/ Charcoal.

2. Connect the boxes with arrows labelled: Photosynthesis, Feeding, Respiration, Death & Decay, and Burning.

3. Trace two distinct loops: (i) Biological fast loop: Air CO_2 → Cassava (photosynthesis) → Goat (feeding) → Air CO_2 (respiration), and (ii) Soil/fuel loop: Plant/Animal remains → Decomposers (decay → Air CO_2) or Wood → Burning → Air CO_2.

Key Observations / Findings:

• Every arrow that moves carbon OUT of the air (photosynthesis) is eventually balanced by arrows returning CO_2 TO the air (respiration, decay, combustion).

Scientific Conclusion: Carbon flows in closed loops rather than a straight line, confirming that the carbon cycle is a continuous, repeated pattern in nature.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking only animals respire while plants only photosynthesise. Correction: Plants respire 24 hours a day (day and night) to release energy for growth, giving out CO_2—just as animals do! Photosynthesis happens only in daylight.

• Mistake: Thinking burning destroys carbon into nothing. Correction: Burning simply converts solid/liquid carbon in wood or petrol into Carbon (IV) Oxide gas (CO_2) that enters the atmosphere.

Lesson Summary — Key Points to Remember

✓ The carbon cycle is the continuous movement of carbon among the atmosphere, plants, animals, decomposers, and fuels.

✓ Photosynthesis removes CO_2 from the air and stores carbon in plants.

✓ Respiration, decay (decomposition), and burning (combustion) return CO_2 to the atmosphere.

✓ Because carbon circulates endlessly through these repeated stages, the cycle is a repeated pattern of change.

Practice Exercise 3 (B8.2.1.1.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which process is the MAIN way Carbon (IV) Oxide is removed from the atmosphere in the carbon cycle?
A. Respiration B. Combustion C. Photosynthesis D. Decomposition

2. Bacteria and fungi return carbon from dead leaves and dead animals to the air during:
A. Photosynthesis B. Decay (decomposition) C. Transpiration D. Evaporation

3. Which three processes all release Carbon (IV) Oxide directly into the atmosphere?
A. Photosynthesis, respiration, feeding B. Respiration, burning, decay C. Photosynthesis, decay, fossilisation D. Feeding, transpiration, burning

4. Why is the carbon cycle called a 'repeated pattern of change'?
A. Because carbon disappears once used by plants B. Because carbon only moves during rainy seasons C. Because carbon atoms circulate continuously in repeating loops D. Because animals make brand-new carbon atoms

Part B: Short Structured Questions (Write clear scientific answers)

5. Name the four main processes involved in the carbon cycle.

6. Write short notes explaining what happens to carbon during (a) photosynthesis and (b) combustion.

7. Draw or write out a simple 4-step flow chart showing how carbon moves from the atmosphere through a goat and back to the atmosphere.

8. Explain the role of decomposers (bacteria and fungi) in keeping the carbon cycle going.

LESSON 4 Environmental Role of the Carbon Cycle

B8.2.1.1.2 • Week 3

Target Indicator: Describe the role of the carbon cycle to the environment.
Scheme Topic: The Carbon Cycle | **Resources:** Pictures and charts showing air composition, food chains, and greenhouse effect

Learning Outcomes — By the end of this lesson, you should be able to:
- Explain how the carbon cycle maintains the natural balance of gases (CO_2 and O_2) in the air.
- Discuss the roles of both photosynthesis and respiration in plants as part of the carbon cycle.
- Trace how carbon compounds flow through food chains using diagrams.
- Describe the relationship between the carbon cycle, greenhouse gases, and global warming.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Air composition balance	Keeping atmospheric Oxygen (~21%) and Carbon (IV) Oxide (~0.03–0.04%) steady so life on Earth is sustained.
Food chain	A sequence showing how food energy and carbon compounds pass from a producer through consumers to decomposers.
Greenhouse gas	An atmospheric gas (such as Carbon (IV) Oxide, CO_2, or methane, CH_4) that absorbs and traps heat radiated from Earth's surface.
Greenhouse effect	The natural warming of Earth caused by greenhouse gases trapping heat in the atmosphere; without it, Earth would be frozen.

Clear Study Notes

1. Maintaining Balance in the Composition of Air

Clean dry air contains about 78% Nitrogen, 21% Oxygen (O_2), and roughly 0.03–0.04% Carbon (IV) Oxide (CO_2). Although CO_2 makes up a tiny fraction of air, its steady level is vital for life.

The carbon cycle balances O_2 and CO_2 through two linked biological processes:

• During daylight, green plants absorb Carbon (IV) Oxide (CO_2) from the air for photosynthesis and release Oxygen (O_2) as a by-product. Animals and humans breathe in that oxygen.

• Both animals AND plants carry out respiration continuously, breaking down food using oxygen and releasing Carbon (IV) Oxide (CO_2) back into the air for plants to reuse.

If forests are cleared on a massive scale (deforestation), fewer plants remain to absorb CO_2 and release O_2—disturbing the air's natural balance.

2. Effect of the Carbon Cycle on Food Chains

Every living organism needs carbon to build body tissues (carbohydrates, proteins, fats) and release energy. Plants (producers) are the gateway: through photosynthesis, they capture non-living carbon from CO_2 in the air and convert it into living food molecules.

When a herbivore (primary consumer, such as a grasshopper or cow) eats plants, carbon moves into the animal. When a carnivore or omnivore (secondary consumer, such as a toad, bird, or human) eats the primary consumer, the carbon passes onward. Finally, decomposers break down dead remains at every level, returning CO_2 to the air so the food chain can begin again.

3. Greenhouse Gases and the Carbon Cycle

Certain carbon-containing gases in the atmosphere—especially Carbon (IV) Oxide (CO_2) and methane (CH_4)—act like the glass roof of a greenhouse: they let sunlight warm Earth's surface and trap outgoing heat so the planet stays comfortably warm (around 15 °C average rather than −18 °C).

However, human activities such as burning fossil fuels (petrol, diesel, coal), bush burning, and cutting down forests push extra CO_2 into the atmosphere faster than plants and oceans can absorb it. Too much greenhouse gas traps excessive heat, intensifying the greenhouse effect and driving global warming and irregular weather patterns.

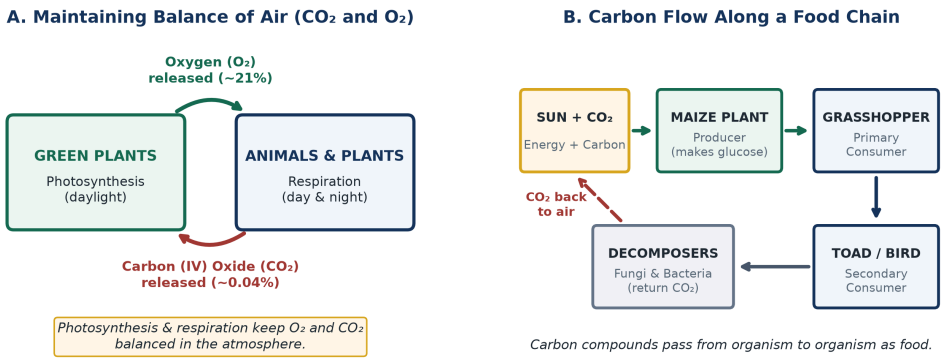


Figure 2.2: (A) Balance of O_2 and CO_2 between photosynthesis and respiration; (B) Carbon flow along a food chain.

Guided Analysis: Predicting Environmental Effects When the Carbon Cycle Is Disturbed

Aim: To evaluate how tree planting and bush burning affect atmospheric CO_2, O_2 balance, food chains, and greenhouse warming.

Materials: Figure 2.2, exercise book, pen.

Procedure / Steps:

1. Scenario A: A community clears a large forest and burns the fallen trees for farmland.

2. Scenario B: Students plant 200 shade and fruit trees around their school compound and avoid burning dry leaves (composting them instead).

3. For each scenario, trace: (1) Effect on CO_2 uptake, (2) Effect on O_2 supply, (3) Effect on heat trapping in the atmosphere, and (4) Effect on local food chains.

Key Observations / Findings:

• Scenario A (burning + clearing): Instantly releases large amounts of CO_2 while removing trees that absorb CO_2 and make O_2 → raises atmospheric CO_2, increases heat trapping (stronger greenhouse effect), and destroys plant producers at the base of food chains.

• Scenario B (tree planting + composting): Growing trees absorb excess CO_2 via photosynthesis, release O_2, support insects/birds in food chains, and composting returns nutrients slowly to soil without smoke.

Scientific Conclusion: Protecting green vegetation and reducing unnecessary combustion keeps the carbon cycle balanced and safeguards Earth's climate and food chains.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking the greenhouse effect is entirely bad. Correction: The natural greenhouse effect is essential to keep Earth warm enough for life; the problem is an enhanced greenhouse effect caused by excess CO_2 and methane from human activities.

• Mistake: Thinking plants do not respire when discussing the air balance. Correction: Plants perform both photosynthesis (daytime, takes in CO_2, gives out O_2) and respiration (day & night, takes in O_2, gives out CO_2). Over a 24-hour day, healthy growing plants absorb more CO_2 than they release.

Lesson Summary — Key Points to Remember

✓ The carbon cycle maintains the atmospheric balance of Oxygen (~21%) and Carbon (IV) Oxide (~0.04%) via photosynthesis and respiration.

✓ Plants convert atmospheric CO_2 into food compounds that pass carbon and energy along food chains to consumers and decomposers.

✓ Carbon-based greenhouse gases (CO_2 and CH_4) regulate Earth's temperature; disrupting the carbon cycle through excessive burning and deforestation increases heat trapping.

Practice Exercise 4 (B8.2.1.1.2)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which two processes directly maintain the natural balance between Oxygen and Carbon (IV) Oxide in the atmosphere?
A. Evaporation and condensation B. Photosynthesis and respiration C. Melting and freezing D. Filtration and distillation

2. In a food chain, which group of organisms first brings carbon from atmospheric CO_2 into living food molecules?
A. Primary consumers B. Secondary consumers C. Producers (green plants) D. Decomposers

3. Which pair consists of carbon-containing greenhouse gases?
A. Nitrogen and Oxygen B. Carbon (IV) Oxide and Methane C. Argon and Helium D. Hydrogen and Neon

4. Why does large-scale bush burning threaten the balance of air?
A. It turns all nitrogen into oxygen B. It releases stored carbon as CO_2 while destroying plants that absorb CO_2 C. It freezes water vapour in clouds D. It stops decomposers from ever working again

Part B: Short Structured Questions (Write clear scientific answers)

5. Explain how plants take part in BOTH sides of the oxygen–carbon (IV) oxide balance in the environment.

6. Use a 4-organism Ghanaian food chain to trace how carbon moves from the air to a bird and back into the environment.

7. Describe the relationship between the carbon cycle and greenhouse gases.
8. Suggest two practical actions your school can take to help keep the carbon cycle balanced.

Sub-Strand 2: Life Cycle of Organisms

Content Standard B8.2.2.1: Demonstrate an activity to show the life cycle of the Anopheles mosquito and show how the effects of the mosquito on humans can be managed.

LESSON 5

Life Cycle and Economic Importance of the Anopheles Mosquito

B8.2.2.1.1 • Week 4

Target Indicator: Describe the life cycle and economic importance of the Anopheles mosquito.
Scheme Topic: Life Cycle Of The Anopheles Mosquito | **Resources:** Pictures, life cycle charts, hand lens, covered/ netted glass jar for safe observation

Learning Outcomes — By the end of this lesson, you should be able to:
• Name and draw the four stages in the complete metamorphosis of the Anopheles mosquito: egg, larva, pupa, and adult.
• State key identifying features of Anopheles eggs, larvae, and adults compared with other mosquitoes.
• Describe safe procedures for observing mosquito development in a netted glass jar.
• Explain the economic importance (both costly negative impacts and minor ecological roles) of the Anopheles mosquito.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Complete metamorphosis	A life cycle with four distinct body forms: Egg → Larva → Pupa → Adult.
Larva (Wriggler)	The active aquatic feeding stage that hatches from the egg; Anopheles larvae lie parallel to the water surface.
Pupa (Tumbler)	The comma-shaped, non-feeding aquatic stage inside which body structures transform into an adult mosquito.
Air floats	Tiny side air-sacs on Anopheles eggs that keep single eggs floating horizontally on water.
Economic importance	The financial, health, agricultural, and ecological effects—harmful or beneficial—that an organism has on society.

Clear Study Notes

1. Complete Metamorphosis of the Anopheles Mosquito

Like the housefly studied in Basic 7 (B7.2.2.1), the Anopheles mosquito undergoes complete metamorphosis with four distinct stages. The first three stages (egg, larva, pupa) live strictly in water, while the adult lives on land and in the air. Under warm Ghanaian temperatures (around 25–30 °C), the entire cycle from egg to flying adult can finish in just 10–14 days!

Stage	Duration	Habitat & Feeding	Key Identifying Features of Anopheles
1. Egg	2–3 days	Laid on clean, still/stagnant water	Boat-shaped with side air floats; laid SINGLY (not stuck together in rafts like Culex); dries out if water disappears.
2. Larva (Wriggler)	5–7 days	Water; feeds on algae & microbes	Has NO long breathing siphon tube, so it lies PARALLEL to the water surface to breathe through spiracles; moults 4 times as it grows.
3. Pupa (Tumbler)	2–3 days	Water; does NOT feed	Comma-shaped; swims with jerky tumbling motion; breathes through short trumpet-shaped tubes on thorax while adult forms inside.
4. Adult (Imago)	2–4 weeks	Air/land; male drinks plant nectar; female takes blood meals	Rests at a distinct ~45° angle with head tilted downward and abdomen pointing up; delicate spotted wings.

2. Economic Importance of the Anopheles Mosquito

When asked about the 'economic importance' of an organism in science, consider both its harmful costs to humans and any minor ecological roles in food webs:

Negative Economic Impacts (Overwhelmingly Harmful): Female Anopheles mosquitoes transmit malaria, causing huge spending by families and the Ministry of Health on clinic visits, diagnostic tests, drugs, and insecticide-treated nets.

Sick learners miss school days, while sick farmers, traders, and workers lose productive work hours, lowering household income and national productivity.

Ecological Roles in Nature: In aquatic ecosystems, mosquito larvae serve as food for small fish (such as tilapia and guppies), tadpoles, and dragonfly nymphs. Adult male mosquitoes (and females when not seeking blood) feed on flower nectar and assist slightly in plant pollination. However, these minor ecological roles can easily be filled by harmless insects—so controlling Anopheles mosquitoes remains vital.

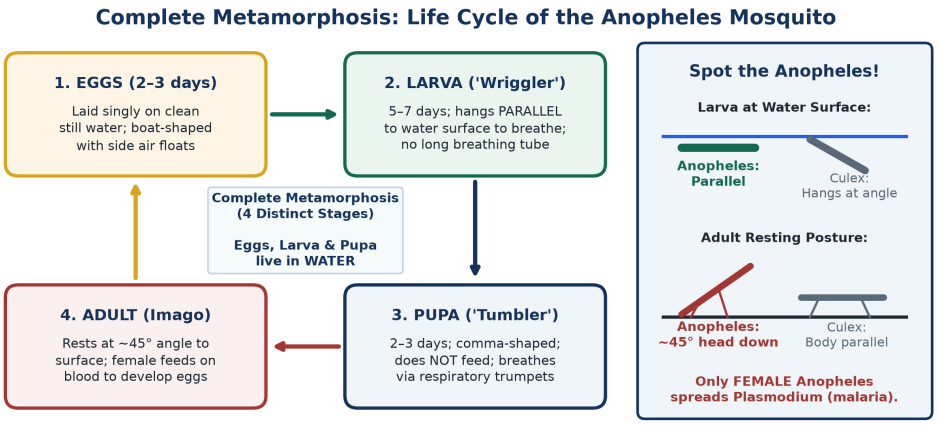


Figure 2.3: Complete metamorphosis of the Anopheles mosquito and how to distinguish it from Culex.

Guided Practical Observation: Safely Studying Mosquito Larvae in a Covered Glass Jar

Aim: To observe and draw the larval and pupal stages of mosquitoes safely without releasing biting adults.

Materials: Clean glass jar, stagnant rainwater containing mosquito eggs/larvae from a puddle or open barrel, fine mosquito-netting cover, rubber band, hand lens, drawing book.

Procedure / Steps:

1. Collect a small sample of stagnant water showing tiny wriggling larvae into the glass jar.

2. Immediately tie fine mosquito netting tightly over the jar mouth with a rubber band so emerging adults CANNOT escape.

3. Use a hand lens through the glass to observe how larvae hang near the water surface and how comma-shaped pupae tumble when tapped gently.

4. Sketch and label the larva, pupa, and any newly emerged adult inside the sealed jar. Afterwards, pour soapy/oiled water or flush safely onto dry sunny soil so no adults escape.

Key Observations / Findings:

• Anopheles larvae lie flat/parallel right under the water film; pupae have a rounded head-thorax and curled tail and do not feed.

Scientific Conclusion: Covering the jar strictly prevents adult escape while letting students directly verify the aquatic stages of complete metamorphosis.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking mosquitoes have only three stages or that larvae turn directly into adults. Correction: Mosquitoes have FOUR stages—the non-feeding pupa stage bridges larva and adult.

• Mistake: Thinking male Anopheles mosquitoes bite humans for blood. Correction: Only FEMALE Anopheles mosquitoes take blood meals (they need blood protein to develop eggs). Males feed exclusively on plant nectar and sugary juices.

Lesson Summary — Key Points to Remember

✓ The Anopheles mosquito develops by complete metamorphosis: Egg → Larva → Pupa → Adult.

✓ Eggs have side floats and are laid singly; larvae lie parallel to the water surface; pupae are comma-shaped non-feeders; adults rest at ~45°.

✓ Economically, Anopheles mosquitoes impose heavy healthcare costs and lost school/work productivity through malaria, though larvae form part of aquatic food webs.

Practice Exercise 5 (B8.2.2.1.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which sequence correctly shows the four stages of the Anopheles mosquito life cycle in order?

A. Egg → Pupa → Larva → Adult B. Larva → Egg → Pupa → Adult C. Egg → Larva → Pupa → Adult D. Pupa → Larva → Egg → Adult

2. How does an Anopheles larva position itself at the water surface to breathe?

A. Head pointing straight down at 90° B. Parallel (flat) along the water surface C. Attached upside down to rocks at the bottom D. Inside an air bubble under mud

3. Which aquatic stage of the Anopheles mosquito does NOT feed?
A. Larva B. Pupa C. Both larva and adult D. Female adult

4. What is the main reason an adult female Anopheles mosquito feeds on human blood?
A. To get water to drink B. To obtain protein needed to develop her eggs C. To cool down her wings D. To digest flower nectar

Part B: Short Structured Questions (Write clear scientific answers)

5. List the four stages in the life cycle of the Anopheles mosquito and state which stages live in water.

6. State two visible differences between an Anopheles mosquito and a Culex mosquito (one at the larval stage and one at the adult stage).

7. State two negative economic impacts and one ecological role of the Anopheles mosquito.

8. Why must a glass jar used to observe mosquito breeding in school always be covered tightly with fine netting?

LESSON 6

Impact of the Anopheles Mosquito and Malaria Control in Ghana

B8.2.2.1.2 • Week 4

Target Indicator: Discuss the impact of the Anopheles mosquito on humans and how it can be controlled.
Scheme Topic: Life Cycle Of The Anopheles Mosquito | **Resources:** Pictures, posters, charts of malaria transmission and control measures

Learning Outcomes — By the end of this lesson, you should be able to:

• Explain the meaning of a biological vector and describe how the female Anopheles mosquito transmits Plasmodium.
• Describe the common symptoms and harmful health effects of malaria on humans.
• Generate practical environmental, biological, chemical, and personal protection measures to control malaria in Ghana by breaking the mosquito life cycle.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Vector	An organism (often an insect) that carries and transmits a disease-causing pathogen from one host to another without necessarily suffering from the disease itself.
Plasmodium	The microscopic single-celled protozoan parasite that causes malaria in humans.
Insecticide-Treated Net (ITN)	A bed net coated with a safe long-lasting insecticide that blocks and kills mosquitoes while a person sleeps.
Larvicide	A substance applied to stagnant water to kill mosquito larvae before they grow into biting adults.

Clear Study Notes

1. The Female Anopheles Mosquito as a Vector of Plasmodium

Malaria is NOT caused directly by mosquito saliva—it is caused by a tiny protozoan parasite called Plasmodium. When a female Anopheles mosquito bites a person already infected with Plasmodium, she sucks up the parasites in their blood. Inside the mosquito, the parasites multiply and travel to her salivary glands. Later, when she bites a healthy person, she injects saliva containing Plasmodium parasites into the bloodstream.

Once inside the human body, Plasmodium travels to the liver and then invades red blood cells, destroying them and releasing toxins. Common symptoms of malaria include:

• High fever (often alternating with chills and severe shivering)
• Throbbing headache and joint/muscle body pains
• Loss of appetite, nausea, vomiting, and extreme tiredness
• Anaemia (pale inner eyelids and palms due to destruction of red blood cells), especially dangerous for young children and pregnant women.

2. Controlling Malaria in Ghana: Breaking the Life Cycle

Because the first three stages of Anopheles live in water and the fourth stage flies at dusk/night, the smartest way to control malaria is to attack multiple stages of the life cycle at once:

Control Approach	Target Stage(s)	Practical Actions in Ghanaian Homes & Communities
Environmental Sanitation	Eggs & Larvae	Drain stagnant puddles, desilt choked gutters, fill potholes, clear empty tins/tyres/coconut shells, cover water tanks tightly, and slash overgrown weeds where adults shelter.
Biological Control	Larvae	Introduce larva-eating fish (such as tilapia or guppies) into ornamental ponds, reservoirs, and water troughs.
Chemical / Surface Control	Larvae, Pupae & Adults	Apply approved larvicides or pour a thin film of kerosene/ oil onto non-drinking stagnant drains (blocks larvae/pupae from breathing air at surface); carry out Indoor Residual Spraying (IRS).

Control Approach	Target Stage(s)	Practical Actions in Ghanaian Homes & Communities
Personal Bite Protection	Adult Female	Sleep every night under an Insecticide-Treated Net (ITN); fit wire mesh screens on windows/doors; wear long-sleeved shirts/trousers in the evening; use approved repellents.
Prompt Testing & Treatment	Plasmodium in Humans	Visit a clinic quickly for a Malaria Rapid Diagnostic Test (RDT) and complete the full prescribed Artemisinin-based Combination Therapy (ACT) dose so mosquitoes cannot pick up parasites from untreated people.

Community Problem-Solving Plan: Eliminating Malaria Breeding Sites Around School

Aim: To inspect a school or home compound, identify mosquito hazards, and match each hazard to a practical control solution.

Materials: Inspection checklist, notebook, pen.

Procedure / Steps:

1. Walk around four zones: (1) Classroom gutters/drains, (2) Water storage polytanks/drums, (3) Bushy corners & discarded containers behind the compound, (4) Dormitory or bedroom sleeping areas.

2. List every standing-water site or adult resting/biting hazard found.

3. Assign the matching control strategy (environmental, biological, physical barrier, or medical).

Key Observations / Findings:

• Sample findings: Choked rainwater drain near canteen → clear silt weekly so water flows freely; Uncovered rainwater drum → fit a tight lid; Torn window mesh → repair wire gauze and sleep under ITNs.

Scientific Conclusion: Combining clean surroundings (removing larval habitats) with ITNs and prompt treatment drastically reduces malaria transmission.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Believing malaria comes from eating oily food, walking in hot sunshine, or drinking dirty water. Correction: Malaria is caused ONLY by the Plasmodium parasite transmitted through the bite of an infected female Anopheles mosquito.

• Mistake: Stopping malaria medicine halfway once fever drops. Correction: Stopping ACT tablets early leaves surviving Plasmodium parasites in the blood, causing the illness to return and encouraging drug resistance.

Lesson Summary — Key Points to Remember

✓ The female Anopheles mosquito is the vector that transmits the Plasmodium parasite, which causes malaria.

✓ Malaria destroys red blood cells and causes fever, chills, headache, vomiting, weakness, and anaemia.

✓ Effective malaria control in Ghana targets eggs/larvae (draining stagnant water, covering containers, fish predators, larvicides) and adults (ITNs, window screens, IRS, clearing bushes) alongside prompt testing and full ACT treatment.

Practice Exercise 6 (B8.2.2.1.2)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. What is the actual disease-causing organism (parasite) transmitted by the female Anopheles mosquito?
A. Virus B. Plasmodium C. Vibrio cholerae D. Housefly larva

2. Why does pouring a thin layer of oil on a non-drinking stagnant pool kill mosquito larvae and pupae?
A. It freezes the water B. It blocks their breathing tubes/spiracles from reaching air at the water surface C. It turns larvae into fish food D. It dissolves their wings

3. Which malaria control measure directly protects humans while they sleep at night?
A. Clearing empty coconut shells B. Sleeping under an Insecticide-Treated Net (ITN) C. Introducing guppies into a pond D. Weeding overgrown lawns

4. In biological control of mosquitoes, small fish such as guppies or tilapia are placed in ponds to:
A. Feed on mosquito larvae and pupae B. Eat adult mosquitoes in mid-air C. Spray insecticides underwater D. Provide blood for female mosquitoes

Part B: Short Structured Questions (Write clear scientific answers)

5. Define the term 'vector' and name the vector and causative parasite of malaria.

6. List four common symptoms of malaria in humans.

7. Explain why draining stagnant water and clearing choked gutters helps control malaria even though it does not kill adult mosquitoes directly.

8. Propose a 3-part community action plan to reduce malaria in your neighbourhood.

Sub-Strand 3: Crop Production

Content Standard B8.2.3.1: Demonstrate knowledge and skills in planting crops on different seed beds.

LESSON 7 **Types of Seed Beds and Matching Crops** **B8.2.3.1.1 • Weeks 12–13**

Target Indicator: Explore the different seed beds for planting crops in your community.
Scheme Topic: Crop Production | **Resources:** Pictures and charts of seed beds, school garden beds, soil samples

Learning Outcomes — By the end of this lesson, you should be able to:
• Identify and describe common seed beds used in Ghanaian farming and school gardens: flat beds, raised beds, sunken beds, ridges, and mounds.
• Compare and contrast the similarities and differences among seed bed types (drainage, moisture retention, aeration).
• Match appropriate seed bed types to specific crops, growth stages (nursery vs field), and local weather/soil conditions.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Seed bed	A specially prepared plot of tilled, levelled, or shaped soil where seeds, seedlings, or plant parts are grown.
Raised bed	A flat-topped soil bed built 15–30 cm above surrounding ground level to improve drainage and soil aeration.
Sunken bed	A shallow basin-like bed dug 10–20 cm below ground level to catch and conserve water in dry periods.
Flat bed	A tilled plot prepared level with the surrounding ground surface.
Ridge and furrow	Long parallel banks of heaped soil (ridges) separated by drainage channels (furrows).
Mound	A rounded heap of loose topsoil constructed especially for root and tuber crops.

Clear Study Notes

1. Why Farmers Shape Soil into Different Seed Beds

In Basic 7 (B7.2.3.1), you studied plant nutrient sources and soil types. Once soil is cleared and loosened, farmers shape it into different seed beds so crops receive the right balance of water, air (oxygen for roots), and root support for the local rainfall and soil texture.

2. Comparing Seed Bed Types and Suitable Crops

Five major seed beds are widely used across Ghanaian farms and school gardens:

Seed Bed Type	Structure & Water Behaviour	Best Conditions	Suitable Crops & Stages
Flat Bed	Level with surrounding ground; moderate drainage	Well-drained loamy soils with moderate rainfall	Maize, cowpea, sorghum, millet, okra
Raised Bed	Built 15–30 cm above ground; drains excess water well; loose & well-aerated	Rainy season, heavy clay soils, or low-lying wet areas	Vegetable nurseries & transplanted crops: tomato, pepper, cabbage, lettuce, carrot, onion
Sunken Bed	Excavated 10–20 cm below ground with raised edges; collects & holds moisture	Dry season or sandy/low-rainfall areas (e.g., savanna zones)	Dry-season vegetables (okra, ayoyo/corchorus, pepper, spinach, seedlings needing steady moisture)
Ridges & Furrows	Long parallel raised soil banks; water drains along furrows	Sloping or rainy plots; protects roots from waterlogging	Sweet potato, groundnut, cassava, maize (on ridges), irrigated crops (watered via furrows)
Mounds	Large circular heaps of loose topsoil; deep aerated root zone	Areas with shallow topsoil or heavy tuber crops	Yam, cassava, cocoyam

3. Similarities and Differences Among Seed Beds

Similarities: Every seed bed requires clearing weeds, loosening/tilling soil so roots can penetrate, removing stones/stumps, and mixing in organic matter (compost or manure) where needed.

Differences: They differ mainly in height relative to ground level, moisture handling (raised beds and ridges shed excess water; sunken beds trap and hold water), and crop rooting depth (mounds give deepest loose soil for large tubers like yam).

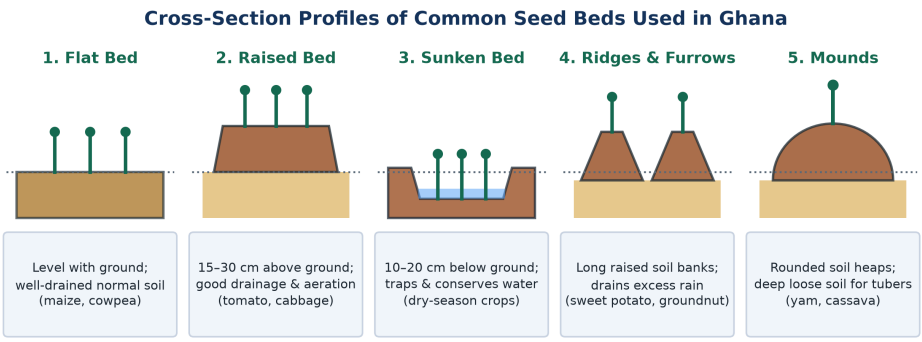


Figure 2.4: Cross-section profiles of flat beds, raised beds, sunken beds, ridges, and mounds.

Guided Decision Task: Choosing the Right Seed Bed for Four School Garden Situations

Aim: To match seed bed types accurately to soil conditions, seasons, and crop stages.
Materials: Seed-bed comparison table and Figure 2.4.
Procedure / Steps:

1. Situation 1: Sowing tomato seeds for a nursery during heavy June rains on clayey school soil.
2. Situation 2: Growing leafy okra and ayoyo during the dry Harmattan season using hand watering.
3. Situation 3: Planting yam setts that need deep, loose, well-drained soil to expand underground.
4. Situation 4: Planting cowpea on level, well-drained loamy soil during moderate rains.

Key Observations / Findings:

- Situation 1 → Raised bed (prevents flooding/damping-off of delicate nursery seedlings).
- Situation 2 → Sunken bed (keeps irrigation water right around the roots and slows evaporation).
- Situation 3 → Mounds (deep, loose topsoil lets heavy yam tubers grow without resistance or rotting).
- Situation 4 → Flat bed (fast to prepare on well-drained level loam).

Scientific Conclusion: No single seed bed is best everywhere—farmers match the bed shape to water supply, soil drainage, and crop growth habit.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

- Mistake: Using sunken beds in waterlogged clay soil during heavy rains. Correction: Sunken beds collect water; in heavy rains they will flood and rot crop roots—use raised beds or ridges instead!
- Mistake: Confusing ridges and mounds. Correction: Ridges are long continuous banks of soil with furrows between them, whereas mounds are separate, rounded heaps.

Lesson Summary — Key Points to Remember

✓ Common seed beds include flat beds, raised beds, sunken beds, ridges, and mounds.
✓ All seed beds share basic soil preparation (clearing, tilling, removing debris, adding organic matter), but differ in elevation, drainage, and moisture retention.
✓ Raised beds/ridges suit wet soils and shallow vegetables; sunken beds conserve water in dry conditions; mounds provide deep loose soil for tubers.

Practice Exercise 7 (B8.2.3.1.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which seed bed is BEST for growing dry-season vegetables in an area with scarce water?
A. High mound B. Sunken bed C. Steep ridge D. Unprepared hardpan

2. Why do gardeners prepare raised beds for tomato and cabbage nurseries during the rainy season?
A. To flood the roots continuously B. To allow excess rainwater to drain away and improve soil aeration C. To stop sunlight from reaching seedlings D. To make harvesting yam tubers easier

3. Which crop is traditionally planted on large rounded soil mounds in Ghana?
A. Lettuce B. Cabbage C. Yam D. Rice

4. What is one key similarity among flat beds, raised beds, and sunken beds?
A. They all sit 20 cm below ground level B. They all require clearing weeds and loosening topsoil before planting C. They are all built exclusively for yam tubers D. They all drain water away rapidly

Part B: Short Structured Questions (Write clear scientific answers)

5. List five types of seed beds used in Ghanaian communities.

6. Compare a raised bed and a sunken bed under two headings: (a) height relative to ground level, (b) water behaviour.

7. Match each crop to a suitable seed bed: (a) tomato seedlings in rainy season, (b) sweet potato vines, (c) yam setts, (d) dry-season okra.

8. Why should farmers avoid walking directly on top of prepared raised seed beds after shaping them?

LESSON 8 # Planting Different Crop Parts on Seed Beds **B8.2.3.1.2 • Weeks 12–13**

Target Indicator: Plant different types of crops on different seed beds.
Scheme Topic: Crop Production | **Resources:** Seeds, seedlings, stem cuttings, leaves, roots/tubers, garden soil, watering can, mulch

Learning Outcomes — By the end of this lesson, you should be able to:
• Describe the step-by-step practice of preparing seed beds and planting crops properly.
• Identify the different plant parts used as planting (propagating) materials: seeds, seedlings, stem cuttings, leaves, and roots/underground stems.
• Demonstrate correct planting techniques for different plant parts on appropriate seed beds.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Propagation	Producing new plants from seeds (sexual propagation) or vegetative plant parts like cuttings, leaves, and tubers (asexual propagation).
Seedling	A young plant grown from seed in a nursery, ready to be transplanted to its permanent seed bed.
Stem cutting	A healthy section of stem bearing buds/nodes that grows roots and shoots when planted (e.g., cassava stick, sweet potato vine).
Transplanting	Carefully moving young seedlings from a nursery bed to a prepared field bed, usually in the cool late afternoon.
Mulching	Covering the soil surface around planted crops with dry grass or leaves to conserve moisture, keep soil cool, and suppress weeds.

Clear Study Notes

1. Preparing a Seed Bed and Planting Correctly

Good crop growth begins with careful bed preparation and planting practice:

1. Clear weeds and stones, then dig and loosen the topsoil to a fine tilth (adding well-rotted compost or poultry manure).
2. Shape the bed (flat, raised, sunken, ridge, or mound) according to season and crop.
3. Water the bed lightly before planting so soil is moist, not muddy.
4. Space planting holes evenly so leaves will not crowd each other for sunlight and air.
5. Cover planted materials gently at the proper depth, firm the soil slightly around them, water gently, and apply light mulch where appropriate.

2. Selecting and Planting Different Plant Parts

Not every crop is planted from seed! Crops can be established on seed beds using five kinds of plant parts:

Plant Part Used	Examples of Crops	Suitable Seed Bed	Key Planting Technique
Seeds	Maize, cowpea, okra, groundnut, tomato (nursery)	Flat bed, raised bed (nursery), ridges	Sow healthy, whole seeds at 2–3× seed thickness deep; avoid burying small seeds too deeply.
Seedlings (Transplants)	Tomato, garden egg, pepper, cabbage, lettuce	Raised bed (wet) or sunken bed (dry)	Water nursery before lifting; keep soil ball around roots; transplant in late afternoon to avoid wilting.
Stem cuttings	Cassava (sticks), sweet potato (vines), sugarcane	Ridges, mounds, or flat bed	Cut mature, disease-free stems (20–25 cm long with 3–4 nodes/buds); plant slanted or horizontally with buds facing upward.
Leaves	Bryophyllum (life plant / 'never die'), African violet	Raised or flat nursery box/bed	Place mature fleshy leaf flat on moist loamy soil (or insert petiole slightly); new plantlets sprout from leaf notches.
Roots & Underground Organs	Yam (tubers/setts), cocoyam (corms), ginger (rhizomes), sweet potato root slips	Mounds or deep ridges	Plant healthy disease-free setts/corms with buds covered 8–12 cm deep; mulch top of yam mound against sun scorching.

Guided School Garden Practical: Establishing Two Different Plant Parts on Two Seed Beds

Aim: To prepare a raised bed and a ridge/sunken bed and properly plant (a) tomato seedlings and (b) cassava or sweet potato stem cuttings.
Materials: Garden soil plots, compost, healthy tomato seedlings, sweet potato vines or cassava cuttings (with visible nodes), ruler, watering can, dry grass mulch.
Procedure / Steps:
1. Plot A (Raised bed, 1 m × 2 m, 20 cm high): Loosen soil, mix in compost, level top with a rake, and water lightly. In late afternoon, transplant tomato seedlings 45 cm apart with soil intact around roots.

2. Plot B (Ridge or Sunken bed): Prepare a ridge 25 cm high (or sunken bed in dry season). Insert 25 cm healthy sweet potato vine cuttings at a gentle slant so at least 2 nodes sit below soil and 1–2 nodes remain above.
3. Label both beds clearly with wooden stakes showing Crop, Plant Part Used, and Date Planted; apply thin dry-grass mulch and water gently.

Key Observations / Findings:
• Transplanted seedlings stand upright without severe wilting when planted in late afternoon and watered; buried nodes on stem cuttings sprout new roots and shoots within 1–2 weeks.

Scientific Conclusion: Each plant part—seed, seedling, stem cutting, leaf, or root/tuber—requires an appropriate planting depth, orientation, and seed bed to establish successfully.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Planting a cassava stem cutting upside down. Correction: Buds (leaf scars) must point upward/outward; planting upside down delays sprouting and weakens yield.
• Mistake: Burying tiny vegetable seeds like tomato or cabbage 10 cm deep. Correction: Tiny seeds lack enough stored food to push through deep soil—sow them shallowly (about 0.5–1 cm) and keep topsoil moist.

Lesson Summary — Key Points to Remember

✓ Proper seed-bed planting includes clearing, tilling, adding organic matter, shaping the bed, spacing, planting at proper depth, watering, and mulching.
✓ Crops are planted from five main plant parts: seeds, seedlings, stem cuttings, leaves, and roots/underground storage organs.
✓ Transplanting seedlings in the cool late afternoon with intact root soil reduces transplant shock.

Practice Exercise 8 (B8.2.3.1.2)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which crop is commonly established on seed beds or ridges using stem cuttings?
A. Maize B. Cassava C. Cowpea D. Okra

2. Why is it best to transplant tomato seedlings in the late afternoon rather than at midday?
A. Insects sleep at night B. Cooler temperatures and lower sunlight reduce water loss and wilting shock C. Soil becomes deeper in the evening D. Seeds germinate faster in the dark

3. Which plant can produce complete new plantlets from notches along the edge of a detached leaf placed on moist soil?
A. Maize B. Yam C. Bryophyllum D. Groundnut

4. What is the main purpose of covering a newly planted yam mound with a cap of dry grass (mulching)?
A. To stop yam shoots from growing upward B. To protect the tuber sett from scorching sun heat and conserve soil moisture C. To attract termites to eat the mound D. To make the mound completely flat

Part B: Short Structured Questions (Write clear scientific answers)

5. Name the five plant parts that can be selected and planted on seed beds, giving one crop example for each.
6. Give two precautions a student must take when planting stem cuttings such as cassava sticks.
7. Why should very small seeds (like pepper or cabbage) NOT be buried as deeply as maize or cowpea seeds?
8. State two benefits of watering the nursery bed before lifting seedlings for transplanting.

Content Standard B8.2.3.2: Demonstrate understanding of the differences in height, size, and flowering of crops grown in different seed beds.

LESSON 9

Comparing Crop Growth, Size, and Flowering Across Seed Beds

B8.2.3.2.1 • Weeks 12–13

Target Indicator: Compare and contrast the differences in height, size, and flowering of crops grown in different seed beds.
Scheme Topic: Crop Production | **Resources:** Ruler/metre rule, string, graph paper, sample growth table & bar chart, crops on different seed beds

Learning Outcomes — By the end of this lesson, you should be able to:

• Measure crop growth indicators accurately: plant height, leaf/stem size, number of flowers, and number of fruits.
• Organise crop measurement data from different seed beds into tables and bar charts/graphs.
• Explain scientific reasons for differences in growth, flowering, and fruiting across flat, raised, and sunken seed beds.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Growth parameter	A measurable feature showing how well a plant is growing—such as stem height, leaf width, stem girth, flower count, or fruit count.
Waterlogging	When soil air spaces fill completely with excess water, starving plant roots of oxygen.

Key Term	Clear Scientific Meaning
Soil aeration	The presence of oxygen-filled air pockets in soil pores needed for healthy root respiration.
Yield	The usable harvest produced by a crop, such as number or mass of fruits, grains, or tubers.

Clear Study Notes

1. How Scientists Measure Crop Performance on Different Seed Beds

After planting the same crop variety (e.g., tomato or okra) at the same spacing on three seed beds (Flat, Raised, Sunken) and watering equally, students track four measurements every week:

• Plant Height (cm): Measure vertically with a ruler from the soil surface at the stem base to the highest growing tip.

• Plant / Leaf Size: Measure the length and width of the largest mature leaf or stem circumference using a piece of string placed against a ruler.

• Number of Flowers: Count open flowers per tagged plant at Week 5–6.

• Number of Fruits: Count developing fruits per tagged plant at Week 7–8.

2. Using Tables and Graphs to Compare Growth

Tables organise numbers cleanly, while bar charts make differences immediately visible at a glance. Notice in the illustrative school garden data below how the same tomato crop performs differently depending on the season!

Seed Bed (Illustrative Classroom Sample)	Mean Height Wk 6 (Rainy, cm)	Mean Height Wk 6 (Dry, cm)	Mean Flowers / Plant (Rainy)	Mean Fruits / Plant (Rainy)
Flat Bed	36 cm	28 cm	10	6
Raised Bed	48 cm	25 cm	16	12
Sunken Bed	24 cm	44 cm	5	2

3. Scientific Reasons Behind the Differences

Why did the Raised Bed produce the tallest plants (48 cm), most flowers (16), and most fruits (12) in the rainy season, whereas the Sunken Bed did best (44 cm) in the dry season?

• Oxygen & Drainage in Rainy Season: In heavy rains, sunken beds collect excess water and become waterlogged. Roots cannot respire properly without oxygen, turn brown, and absorb fewer minerals—stunting height, leaf size, flowering, and fruiting. Raised beds drain extra water away and stay loose and well aerated, giving strong roots, tall stems, and abundant flowers.

• Water Conservation in Dry Season: During dry weather, water runs off raised beds quickly and evaporates from exposed bed sides, causing moisture stress and fewer flowers. Sunken beds trap irrigation water and shelter soil from drying winds, keeping roots hydrated so crops grow taller and fuller.

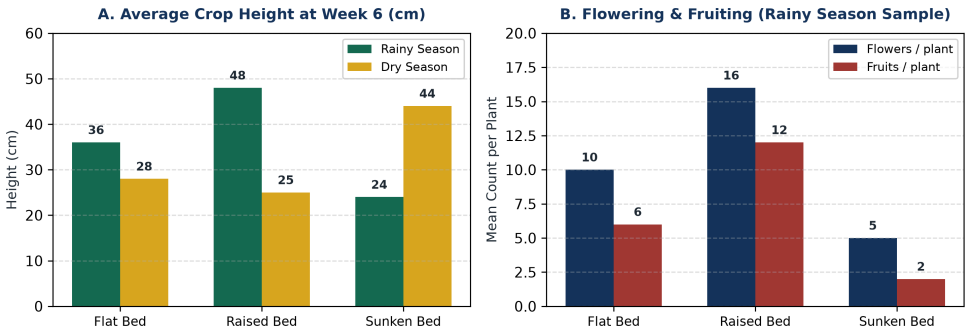


Figure 2.5: Illustrative bar charts comparing crop height (rainy vs. dry season) and flowering/fruiting across three seed beds.

Guided Data Analysis: Interpreting a Seed-Bed Growth Experiment

Aim: To read measurement tables and bar charts and explain why crop performance differs across seed beds.

Materials: Illustrative school garden table, Figure 2.5, calculator/notebook.

Procedure / Steps:

1. Compare the Rainy Season mean height on the Raised Bed (48 cm) against the Sunken Bed (24 cm). Calculate the height difference (48 − 24 = 24 cm; twice as tall!).

2. Examine flowering and fruiting in Graph B: compare flower-to-fruit ratios across all three beds.

3. Explain why the pattern flips in Graph A during the Dry Season (Sunken Bed 44 cm vs Raised Bed 25 cm).

Key Observations / Findings:

• In the rainy season, Raised Bed > Flat Bed > Sunken Bed across height, flowers, and fruits.

• In the dry season, Sunken Bed > Flat Bed > Raised Bed for crop height.

Scientific Conclusion: Root aeration (drainage) drives faster growth and flowering in wet seasons, while soil moisture conservation drives superior growth in dry seasons.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Measuring plant height from the bottom of a furrow rather than from the soil surface around the plant's own stem. Correction: Always start the zero mark of your ruler right at the soil surface touching the base of the plant.
• Mistake: Assuming raised beds ALWAYS produce better yields in every weather. Correction: In dry weather with limited water, raised beds dry out faster than sunken beds!

Lesson Summary — Key Points to Remember

✓ Crop performance on different seed beds is compared by measuring plant height, leaf/stem size, flower count, and fruit count.
✓ Tables and bar graphs reveal patterns clearly across rainy and dry conditions.
✓ Differences in growth, flowering, and yield result from how each seed bed manages soil drainage, root oxygen (aeration), moisture retention, and nutrient uptake.

Practice Exercise 9 (B8.2.3.2.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. In the rainy-season school garden data (Figure 2.5), which seed bed produced the highest mean number of flowers per plant?
A. Sunken bed B. Flat bed C. Raised bed D. All had equal flowers

2. Why do tomato plants on a sunken bed often look yellow and stunted during heavy continuous rains?
A. They receive too much oxygen in the soil B. Waterlogging drives air out of soil pores so roots lack oxygen to respire C. Their stems cannot see the sun D. Sunken beds have no soil at all

3. When comparing three seed beds in a fair scientific test, which factor should be kept the SAME across all three beds?
A. The elevation of the bed B. The crop variety and planting spacing C. The drainage speed of the bed D. The final height of the plants

4. Using Figure 2.5A, how much taller were the dry-season plants on the Sunken Bed than on the Raised Bed?
A. 12 cm B. 16 cm C. 19 cm D. 24 cm

Part B: Short Structured Questions (Write clear scientific answers)

5. Name four plant measurements a student should record when comparing crop growth across different seed beds.

6. Give two reasons why a raised bed outperforms a flat bed for growing vegetables on clay soil during the rainy season.

7. Why can low flowering on a poorly drained seed bed reduce the final number of fruits harvested?

8. Suppose you are presenting findings from a 6-week seed-bed experiment to your class. Why is including both a table and a bar chart helpful?

Sub-Strand 4: Animal Production

Content Standard B8.2.4.1: Recognise the different types of feed for different types of animals.

LESSON 10 **Types of Animal Feed, Nutrients, and Feed Labels** **B8.2.4.1.1 • Week 13**

Target Indicator: Compare and contrast the different types of feed for different types of animals.
Scheme Topic: Animal Production | **Resources:** Pictures and charts of livestock feed, sample commercial feed bag labels

Learning Outcomes — By the end of this lesson, you should be able to:

• Classify common farm animals (ruminants vs. monogastrics/poultry) and match each group to suitable feed types.
• Compare roughages, concentrates, succulents, and mineral/vitamin supplements.
• Identify the major nutrients found in animal feed and their local sources.
• Read and interpret commercial animal feed package labels to select the right feed for an animal's stage of growth.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Roughage (Forage)	Bulky, high-fibre plant feed such as fresh pasture grass, hay, silage, or maize stover, suited to ruminants and rabbits.
Concentrate	Low-fibre, nutrient-dense feed rich in energy (e.g., maize, wheat bran) or protein (e.g., soybean meal, fishmeal, groundnut cake).
Ruminants	Four-chambered stomach herbivores (cattle, sheep, goats) that chew cud and digest fibrous roughage efficiently.
Monogastrics (Non-ruminants)	Single-stomach animals (poultry, pigs, dogs) that cannot digest large amounts of roughage and rely mainly on concentrates.
Feed supplement / Premix	Concentrated vitamins, minerals (e.g., calcium, salt licks, oyster shell), or amino acids added in small amounts to balance rations.

Clear Study Notes

1. Matching Feed Types to Different Animals

In Basic 7 (B7.2.4.1), you learned about domestic animals and their uses. Different farm animals have different digestive systems, so they cannot all eat the exact same diet:

Feed Category	Characteristics	Common Feed Examples	Best Suited Animals
Roughages / Forages	High crude fibre (>18%), bulky, lower energy per kg	Elephant grass, stylo, leucaena leaves, hay, silage, groundnut haulms, maize stover	Ruminants (cattle, sheep, goats) & rabbits
Energy Concentrates	Low fibre, rich in carbohydrates or fats for heat and activity	Crushed maize, sorghum, rice bran, wheat bran, dried cassava chips, palm kernel cake	Poultry (broilers, layers), pigs, dairy cattle (supplement)
Protein Concentrates	Rich in amino acids for rapid muscle, egg & milk growth	Soybean meal, groundnut cake, fishmeal, blood meal, cottonseed cake	Growing chicks, broilers, laying hens, piglets, calves
Succulents & Supplements	High moisture greens or concentrated mineral/vitamin boosters	Fresh cabbage/plantain peels (for goats/pigs), oyster shell grit, bone meal, salt licks, vitamin premix	All livestock (e.g., oyster shell calcium for laying hens; salt licks for goats/cattle)

2. Types of Nutrients and Sources in Animal Feed

A balanced animal ration contains six essential nutrient classes: Carbohydrates (energy: maize, cassava peels, wheat bran), Proteins (tissue building, eggs, milk: soybean meal, fishmeal, groundnut cake, leguminous leaves), Fats/Oils (concentrated energy: palm kernel cake, oilseeds), Vitamins (disease protection: fresh green leaves, vitamin premixes), Minerals (bones, teeth, eggshells: bone meal, oyster shells, dicalcium phosphate, common salt), and Water.

3. Selecting Feed Using Package Labels

Commercial poultry and pig feeds come with guaranteed nutrient analysis labels on the sack. Look closely at Crude Protein %, Energy, Calcium %, and Target Age/Purpose before buying:

• Broiler Starter / Chick Mash (~20–23% crude protein): high protein for fast muscle and feather growth in young chicks (Weeks 1–4).

• Grower Mash (~15–17% crude protein): moderate protein for steady frame development in growing pullets.

• Layer Mash (~16–18% crude protein + HIGH calcium ~3.5–4.0%): extra calcium and phosphorus so hens form strong eggshells without weakening their own bones!

Guided Label Reading: Selecting the Right Commercial Feed Bag

Aim: To read nutrient percentages on sample commercial feed labels and select the correct bag for different farm animals.

Materials: Three illustrative commercial feed label specifications below.

Procedure / Steps:

1. Bag X: Crude Protein 22%, Crude Fibre 4.0%, Calcium 1.0%, Energy: High (3,000 kcal/kg).
2. Bag Y: Crude Protein 17%, Crude Fibre 5.0%, Calcium 3.8%, Energy: Medium (2,700 kcal/kg).
3. Bag Z: Crude Fibre 26%, Crude Protein 11%, Dry rough forage pellets.
4. Match Bag X, Bag Y, and Bag Z to: (i) 25-week-old laying hens producing eggs, (ii) 2-week-old broiler chicks, and (iii) Mature goats.

Key Observations / Findings:

- Bag X (22% protein, low fibre, high energy) → 2-week-old broiler chicks needing fast body growth.
- Bag Y (17% protein, 3.8% calcium) → Laying hens needing high calcium for hard eggshells.
- Bag Z (26% fibre forage pellets) → Mature goats (ruminants equipped to digest high-fibre roughage).

Scientific Conclusion: Inspecting crude protein, fibre, and mineral percentages on feed labels prevents stunted growth, soft eggshells, and wasted money.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Feeding chick starter mash or pig feed to goats as their sole diet, or feeding only dry grass to broiler chickens. Correction: Chickens have a simple monogastric digestive tract and cannot extract enough nutrients from fibrous grass; goats need roughage/fibre to keep their rumen working properly.

• Mistake: Giving laying hens grower mash with low calcium. Correction: Without adequate calcium (3.5–4%) in layer mash or oyster shell grit, hens lay thin-shelled or soft-shelled eggs.

Lesson Summary — Key Points to Remember

✓ Animal feeds fall into roughages (high fibre, for ruminants/rabbits), concentrates (high energy/protein, for poultry/pigs and supplements), succulents, and mineral/vitamin additives.

✓ Nutrients in feed include carbohydrates, proteins, fats, vitamins, minerals, and water, each sourced from specific crops or animal by-products.

✓ Always check crude protein, crude fibre, energy, and calcium proportions on commercial feed labels to match the animal species and production stage.

Practice Exercise 10 (B8.2.4.1.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which group of farm animals can digest high-fibre roughages like elephant grass most efficiently?
A. Broiler chickens B. Pigs C. Ruminants (cattle, sheep, goats) D. Catfish

2. Which feed ingredient is primarily a source of PROTEIN in poultry rations?
A. Crushed yellow maize B. Soybean meal / fishmeal C. Cassava peels D. Rice husk

3. Why does Layer Mash contain a significantly higher percentage of Calcium than Chick Starter Mash?
A. To make feathers turn white B. To supply calcium needed to build strong eggshells C. To replace drinking water D. To increase dietary fibre

4. Maize grain, wheat bran, and cassava chips are mainly classified as:
A. Mineral supplements B. Energy concentrates C. Roughages D. Vitamin premixes

Part B: Short Structured Questions (Write clear scientific answers)

5. State two differences between a roughage and a concentrate.

6. Match each animal with the most appropriate feed: (a) dairy cow, (b) 3-week-old broiler chick, (c) laying hen, (d) rabbit.

7. List four nutrients found in livestock feed and give one local feed source for each.

8. A poultry farmer sees label A (Crude Protein 21%, Calcium 1.0%) and label B (Crude Protein 16.5%, Calcium 3.8%). Which feed should she buy for actively laying hens, and why?

Content Standard B8.2.4.2: Demonstrate understanding of the importance of water and animal feed to the growth of animals.

LESSON 11

Importance of Water and Feed for Animal Growth and Reproduction

B8.2.4.2.1 • Week 13

Target Indicator: Explain the importance of water and animal feed to the growth of animals.
Scheme Topic: Animal Production | **Resources:** Pictures and charts of livestock watering and feeding systems

Learning Outcomes — By the end of this lesson, you should be able to:
- List and explain the roles of balanced animal feed in growth, production, health, and reproduction.
- List and explain the vital physiological roles of clean drinking water in farm animals.
- Predict the physical signs and harmful consequences when animals are deprived of adequate water or feed.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Dehydration	A dangerous condition caused by losing or lacking enough body water, leading to poor digestion, heat stress, and organ failure.
Feed conversion	How efficiently an animal turns eaten feed into body weight, meat, milk, or eggs—which drops sharply without water.
Reproduction	Breeding, pregnancy/egg-laying, and raising healthy young—which require adequate energy, protein, vitamins, minerals, and clean water.
Thermoregulation	Keeping body temperature within a safe range (e.g., through panting or sweating using body water).

Clear Study Notes

1. Why Animal Feed Is Essential for Growth and Reproduction

Feed provides the building blocks and fuel every farm animal needs throughout its life:

• Growth and Body Repair: Proteins and minerals build muscles, bones, skin, feathers, and blood so young animals grow rapidly.

• Energy for Activity and Warmth: Carbohydrates and fats fuel heartbeat, breathing, walking/grazing, and maintaining body heat.

• Production (Meat, Milk, Eggs): A lactating cow or laying hen needs extra nutrients above maintenance simply to make milk or eggs.

• Reproduction and Immunity: Well-fed females reach breeding maturity on time, conceive readily, carry healthy pregnancies or hatch strong chicks, and produce rich milk for newborns; vitamins and proteins also build strong immunity against diseases.

2. Why Clean Water Is Essential for Growth and Survival

Water makes up roughly 60–75% of a farm animal's body weight. An animal can survive far longer without solid food than without water! Clean, cool drinking water performs five irreplaceable jobs:

1. Digestion and Nutrient Transport: Softens dry feed, enables digestive enzymes to work, and forms blood plasma that carries nutrients to body cells.

2. Temperature Regulation: Water absorbs body heat and cools animals via evaporation when cattle sweat or chickens/dogs pant in hot Ghanaian weather.

3. Waste Removal: Flushes toxic metabolic wastes such as urea out of the kidneys in urine.

4. Lubrication and Cushioning: Lubricates joints and eyeballs and protects developing young inside the womb.

5. Milk and Egg Production: Milk is about 87% water and a whole egg is about 65–75% water—no water means an immediate drop in milk and egg output!

3. Predicting What Happens Without Adequate Water

If a farmer forgets to refill drinkers in a poultry coop or keeps goats tied in the sun without water, predictable warning signs appear quickly: first, animals stop eating dry feed (feed intake drops because dry feed cannot be swallowed or digested easily), causing stunted growth and weight loss. Next follow intense thirst, dry/sticky gums, sunken eyes, dry ruffled feathers or dull coat, constipation, rapid breathing/heat stress, sharp fall in egg or milk yield, reproductive failure (abortions or weak offspring), and eventually kidney failure and death.

Guided Prediction Scenario: Comparing Two Poultry Pens

Aim: To predict growth, feed intake, and health outcomes when water supply is restricted versus adequate.

Materials: Notebook, Case scenario.

Procedure / Steps:

1. Pen 1 has clean, cool water available continuously alongside balanced grower mash.

2. Pen 2 receives the exact same balanced mash, but its drinkers run dry for 18 hours on a hot afternoon.

3. Predict what happens in Pen 2 regarding: (a) amount of mash eaten, (b) body temperature/panting, (c) weekly weight gain, and (d) laying performance if hens are mature.

Key Observations / Findings:

• In Pen 2, chickens eat much less mash even though feeders are full, pant heavily with spread wings (heat stress), show slow weight gain, and laying hens lay fewer or smaller eggs.

Scientific Conclusion: Adequate clean water is essential before feed nutrients can even be digested or converted into growth and animal products.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking dirty puddle water is fine for livestock because animals are 'tough'. Correction: Contaminated water carries parasites and bacteria that cause diarrhoea, slowing growth and spreading disease across the herd or flock.

• Mistake: Thinking dry feed alone determines how fast an animal grows. Correction: Even the most expensive feed cannot support growth without sufficient clean water—water deprivation immediately halts eating and growth.

Lesson Summary — Key Points to Remember

✓ Balanced animal feed supplies energy, protein, minerals, and vitamins needed for tissue growth, maintenance, milk/egg production, immunity, and reproduction.

✓ Water makes up most of an animal's body and is vital for digestion, blood transport, cooling, waste excretion, and milk/egg formation.

✓ Without adequate water, animals suffer dehydration, stop eating, grow slowly, produce fewer eggs/milk, fail to reproduce well, and may die from heat stress or organ failure.

Practice Exercise 11 (B8.2.4.2.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which statement best explains why laying hens stop laying quickly when their water drinkers run dry?
A. Water makes up the majority of an egg and is needed to digest feed B. Chickens only sleep when water is absent C. Water turns directly into feathers D. Feed becomes toxic only in the dark

2. During a hot afternoon in Ghana, how does body water help protect cattle and chickens from overheating?
A. By stopping all blood circulation B. By cooling the body through evaporation during sweating or panting C. By freezing the stomach D. By hardening the bones

3. What is usually the FIRST behavioural effect seen when farm animals lack adequate drinking water?
A. They eat twice as much dry mash B. Their appetite for dry feed drops sharply C. Their wool or feathers grow thicker D. They lay double-sized eggs

4. Which nutrient role is MOST directly linked to building muscle tissue in growing animals and repairing body cells?
A. Water only B. Proteins in feed C. Crude fibre D. Sand grit

Part B: Short Structured Questions (Write clear scientific answers)

5. List four roles of clean water in the body of a farm animal.

6. Explain two ways good feeding supports animal reproduction.

7. Predict four signs you would observe in goats or poultry that have been deprived of adequate water for two days.

8. Why should livestock water troughs be scrubbed clean and kept in the shade?

BASIC 8 SCIENCE • FIRST TERM

STRAND 3: SYSTEMS

A system is a collection of parts working together to perform a larger function. Building on your Basic 7 study of human digestion, First Term Basic 8 zooms in on the human body's mechanical digestion tools—the mammalian tooth, the four functional types of human teeth, and how to prevent tooth decay and gum disease.

Sub-Strand 1: The Human Body Systems

Content Standard B8.3.1.1: Demonstrate knowledge of parts of mammalian tooth and the functions of the different types of teeth in relation to feeding in man.

LESSON 12

External and Internal Parts of a Mammalian Tooth

B8.3.1.1.1 • Week 5

Target Indicator: Identify parts of a mammalian tooth.
Scheme Topic: Mammalian Tooth | **Resources:** Pictures, models, and vertical-section charts of a mammalian tooth

Learning Outcomes — By the end of this lesson, you should be able to:
• Identify and label the three main external regions of a mammalian tooth: crown, neck, and root.
• Identify and label the internal structures in a vertical section: enamel, dentine, pulp cavity (with blood vessels and nerves), cementum, gum (gingiva), and jawbone.
• Explain the specific function of each external and internal part of the human tooth.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Crown	The visible white upper part of the tooth projecting above the gumline.
Neck	The slightly narrowed region where the crown meets the root right at the gumline.
Root	The lower part of the tooth embedded firmly inside a socket of the jawbone.
Enamel	The shiny, mineral-rich, hardest substance in the human body that forms the protective outer cap of the crown.
Dentine	The yellowish bone-like tissue beneath the enamel that forms the main bulk of the tooth and absorbs chewing shocks.
Pulp cavity	The living inner chamber containing blood vessels (to nourish the tooth) and nerves (to sense heat, cold, and pain).
Cementum	A tough bone-like layer covering the root that anchors the tooth inside the jawbone socket via periodontal fibres.

Clear Study Notes

1. Three External Regions of a Mammalian Tooth

In Basic 7 (B7.3.1.1), you learned that digestion starts in the mouth, where teeth mechanically break food into small pieces so digestive enzymes can act faster. Every mammalian tooth has three external regions from top to bottom:
• Crown: The exposed upper part above the gum that bites, tears, crushes, or grinds food.
• Neck: The narrow waist right at the gumline separating the crown from the root.
• Root: One or more tapered projections buried beneath the gum inside a bony socket in the upper or lower jaw.

2. Internal Parts and Their Functions

Cutting a tooth vertically reveals specialised living and non-living layers working together:

Part of Tooth	Location & Nature	Main Function
Enamel	Outer covering of the crown; hardest tissue in the human body (non-living mineral crystals)	Protects the softer inner layers from wear, biting force, and bacterial acids.
Dentine	Main bulk beneath enamel & cementum; softer bone-like living tissue	Supports the enamel and cushions impact forces during chewing.
Pulp cavity	Hollow central core extending down the root canal	Houses living nerves and blood vessels to keep the tooth alive.
Blood vessels (capillaries)	Inside the pulp cavity	Supply oxygen and nutrients (like calcium and phosphate) to living dentine cells and remove waste.
Nerve endings	Inside the pulp cavity	Detect temperature (very hot/cold), pressure, and pain warnings if decay reaches deep inside.

Part of Tooth	Location & Nature	Main Function
Cementum	Outer covering of the root	Cements/anchors the tooth root via fibrous ligaments into the jawbone socket.
Gum (Gingiva) & Jawbone	Pink soft tissue collar & bony socket surrounding root	Gum seals out bacteria around the neck; jawbone holds the roots firmly during heavy biting.

Vertical (Longitudinal) Section of a Mammalian Tooth

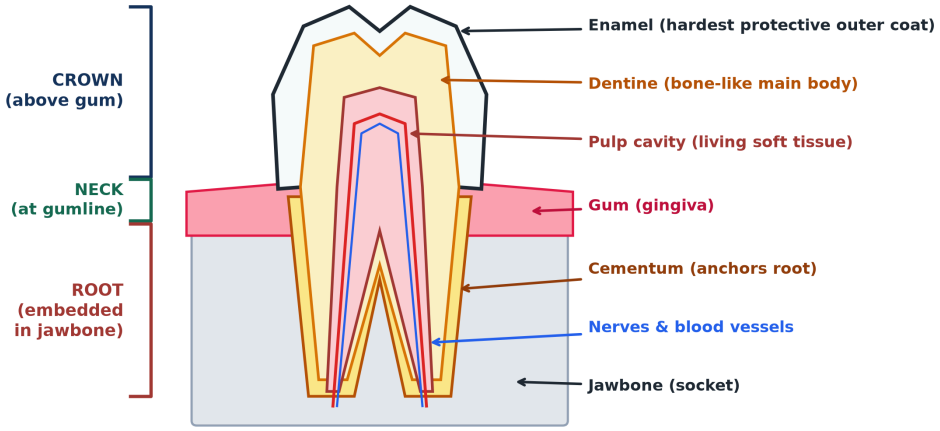


Figure 3.1: Labelled vertical (longitudinal) section of a mammalian tooth showing external regions and internal tissues.

Guided Drawing & Labelling Task: Vertical Section of a Mammalian Molar

Aim: To sketch and accurately label all external and internal structures of a mammalian tooth.

Materials: Plain paper, sharp pencil, eraser, ruler, Figure 3.1.

Procedure / Steps:

1. Draw the outer outline of a two-rooted molar crown, neck, and roots.
2. Draw the inner dentine boundary and central pulp cavity branching into both roots.
3. Draw the horizontal gum collar and surrounding jawbone block.
4. Use a ruler to draw neat horizontal pointer lines (without crossing lines) labelling: Crown, Neck, Root, Enamel, Dentine, Pulp cavity, Blood vessels & nerves, Gum, Cementum, and Jawbone.

Key Observations / Findings:

• Enamel covers ONLY the crown above the neck, while cementum covers ONLY the root below the neck; dentine and pulp run through both crown and root.

Scientific Conclusion: A mammalian tooth combines an ultra-hard protective outer coat (enamel) with a shock-absorbing body (dentine) and a living sensory core (pulp).

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking enamel covers the entire tooth down to the tip of the root. Correction: Enamel covers ONLY the crown; the root is covered by cementum.

• Mistake: Thinking a tooth is completely dead rock like a pebble. Correction: Although enamel is non-living mineral, the dentine and pulp cavity inside are living tissues with active blood supply and sensitive nerves!

Lesson Summary — Key Points to Remember

✓ Externally, a mammalian tooth consists of the crown (above gum), neck (at gumline), and root (inside jawbone).

✓ Internally, the crown is protected by hard enamel over dentine, while the root is covered by cementum anchored in the jawbone.

✓ The central pulp cavity contains blood vessels (nourishment) and nerves (sensation).

Practice Exercise 12 (B8.3.1.1.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which part of a mammalian tooth is the hardest substance in the entire human body?
A. Dentine B. Cementum C. Enamel D. Pulp cavity

2. The region of the tooth situated right at the gumline between the crown and root is called the:
A. Enamel B. Neck C. Socket D. Pulp

3. Why does biting ice-cold water feel painful when a deep cavity exposes the pulp cavity?
A. Because enamel contains strong muscles B. Because the pulp cavity houses sensitive nerve endings C. Because cementum freezes instantly D. Because gums turn into bone

4. Which layer covers the root of a tooth and helps anchor it firmly in the jawbone?
A. Enamel B. Cementum C. Saliva D. Plaque

Part B: Short Structured Questions (Write clear scientific answers)

5. Name the three main external regions of a mammalian tooth and briefly describe where each is found.
6. State the function of (a) enamel, (b) dentine, and (c) blood vessels in the pulp cavity.
7. Explain why a healthy gum (gingiva) is important for protecting the root of a tooth.
8. Compare the location of enamel and cementum on a single mammalian tooth.

LESSON 13 ## Types of Human Teeth and Their Functions **B8.3.1.1.2 • Week 5**

Target Indicator: Discuss the functions of the different types of teeth such as incisors, canines, premolars, and molars.
Scheme Topic: Mammalian Tooth | **Resources:** Pictures, dental models, and charts of human teeth

Learning Outcomes — By the end of this lesson, you should be able to:
- Identify the four types of human teeth: incisors, canines, premolars, and molars.
- Describe the shape, position in the jaw, number of roots, and adult count of each tooth type.
- Relate the physical structure of each tooth type directly to its feeding function.
- Draw and label each of the four types of teeth accurately.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Dentition	The type, number, and arrangement of teeth in an animal's jaws.
Incisors	Flat, chisel-shaped front teeth used for cutting and biting food.
Canines	Pointed, conical teeth beside the incisors used for tearing and gripping tough food.
Premolars	Broad teeth behind canines with two cusps (bicuspid) and 1–2 roots used for crushing and grinding.
Molars	Large, broad back teeth with 4–5 cusps and 2–3 roots used for heavy chewing and fine grinding.
Cusp	A raised point or ridge on the biting surface of a premolar or molar.

Clear Study Notes

1. Human Sets of Teeth (Milk Teeth vs. Permanent Teeth)

Humans are omnivores—we eat both plant foods (kenkey, banku, fruits, yam) and animal foods (fish, meat, eggs). Because we eat varied foods, we have four specialised types of teeth rather than teeth that all look alike!

• Milk (Temporary / Deciduous) Teeth: Babies begin cutting 20 milk teeth around 6 months of age (8 incisors, 4 canines, 8 molars—no premolars). These shed gradually between ages 6 and 12.

• Permanent Teeth: Adults normally have 32 permanent teeth—16 in the upper jaw and 16 in the lower jaw (arranged symmetrically: 8 per half-jaw, written in dental formula as I 2/2, C 1/1, PM 2/2, M 3/3).

2. Structure and Function of Each Type of Tooth

Notice how each tooth's shape matches its job—a core idea of structure-function in Biology:

Type of Tooth	Position & Adult Count	Structure (Crown & Roots)	Feeding Function & Food Example
Incisors	Front of jaws (8 total: 4 upper, 4 lower)	Flat, sharp, chisel-shaped crown; 1 single root	Biting and cutting pieces off food (e.g., biting into a mango or roasted plantain).
Canines	Corners of mouth beside incisors (4 total: 2 upper, 2 lower)	Conical, pointed, strong crown; 1 very long root	Gripping, piercing, and tearing tough food (e.g., tearing meat or sugarcane rind).
Premolars	Behind canines (8 total: 4 upper, 4 lower)	Broad crown with 2 pointed cusps; 1–2 roots	Crushing and grinding food into smaller pieces (e.g., groundnuts, fried yam).
Molars	Back of jaws (12 total: 6 upper, 6 lower, including 4 wisdom teeth)	Largest, widest flat surface with 4–5 cusps & ridges; 2–3 strong roots	Heavy chewing and fine grinding of food mixed with saliva before swallowing.

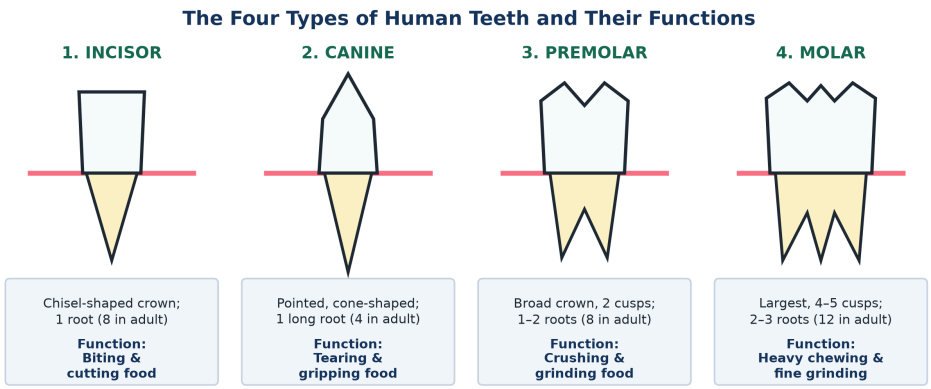


Figure 3.2: Distinguishing shapes, roots, and feeding functions of the four types of human teeth.

Guided Mirror Observation & Drawing: Mapping Your Own Permanent Teeth

Aim: To observe the four tooth shapes in the human mouth and draw each type accurately.

Materials: Clean pocket mirror, clean washed hands (if gently checking with clean tongue/mirror), drawing book, pencil, Figure 3.2.

Procedure / Steps:

1. Stand in good daylight with a clean mirror and examine your upper and lower front teeth (4 flat chisel-edged incisors in the middle of each jaw).

2. Look just beside the incisors for your pointed canines, followed further back by premolars (2 cusps) and broad multi-cusped molars.

3. Draw neat side-by-side sketches of an incisor, a canine, a premolar, and a molar—paying attention to crown edge/cusps and root count (1, 1 long, 1–2, 2–3).

Key Observations / Findings:

• Biting an apple or pawpaw uses front incisors; chewing groundnuts moves automatically back to premolars and molars.

Scientific Conclusion: Each tooth type has a distinct crown and root design tailored to cutting, tearing, crushing, or grinding food during mechanical digestion.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking children's milk teeth include premolars. Correction: The 20 milk teeth consist of 8 incisors, 4 canines, and 8 milk molars—premolars appear ONLY in the 32-tooth permanent set.

• Mistake: Confusing premolars and molars when counting roots or cusps. Correction: Premolars have 2 cusps and 1–2 roots; molars are larger with 4–5 cusps and 2–3 roots (2 in lower molars, 3 in upper molars).

Lesson Summary — Key Points to Remember

✓ Adult humans normally have 32 permanent teeth across four functional types: 8 incisors, 4 canines, 8 premolars, and 12 molars.

✓ Incisors have chisel edges for biting/cutting; canines are pointed for tearing; premolars have 2 cusps for crushing/grinding; molars have 4–5 cusps and 2–3 roots for heavy grinding.

✓ Chewing increases food surface area and mixes food with saliva to aid swallowing and chemical digestion.

Practice Exercise 13 (B8.3.1.1.2)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. How many permanent teeth does a healthy adult human normally possess in total?
A. 20 B. 24 C. 28 D. 32

2. Which tooth type is chisel-shaped with a single root and specialised for cutting and biting food?
A. Molar B. Premolar C. Canine D. Incisor

3. Why do molars have 2 or 3 roots and a broad surface with 4–5 cusps?
A. To pierce fruit skin only B. To withstand heavy forces while grinding and crushing food finely C. To help humans whistle D. Because they sit in front of incisors

4. Which type of tooth is absent from a toddler's set of 20 milk teeth?
A. Incisors B. Canines C. Premolars D. Milk molars

Part B: Short Structured Questions (Write clear scientific answers)

5. Name the four types of teeth in humans and give the exact total number of each in an adult.

6. State two structural differences between a canine and a molar.

7. Explain why thorough chewing by premolars and molars helps digestion in the stomach and small intestine.

8. Identify which tooth type does most of the work when: (a) biting into a slice of pineapple, (b) tearing tough dried meat, (c) grinding roasted maize.

LESSON 14

Causes and Prevention of Tooth Decay and Gum Disease

B8.3.1.1.3 • Week 6

Target Indicator: Explain the causes and prevention of tooth and gum decay.

Scheme Topic: Tooth And Gum Decay | **Resources:** Pictures and charts of healthy vs. decayed teeth, toothbrush, fluoride toothpaste, chewing stick

Learning Outcomes — By the end of this lesson, you should be able to:

- Explain how plaque, tartar, tooth decay (dental caries), and gum disease develop.
- Identify habits and dietary factors that accelerate tooth and gum damage.
- Describe and demonstrate proper daily oral hygiene methods and preventive practices for healthy teeth and gums.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Plaque	A sticky, colourless film of bacteria, food sugars, and saliva that constantly coats unbrushed teeth.
Tartar (Calculus)	Hardened mineralised plaque that builds up along the gumline and irritates gums.
Tooth decay (Dental caries)	Destruction of tooth enamel and dentine by acids produced when mouth bacteria feed on leftover sugars.
Gum disease (Gingivitis)	Red, swollen, tender gums that bleed easily during brushing, usually triggered by plaque and tartar at the gumline.
Fluoride	A protective mineral in toothpaste and water that strengthens tooth enamel against acid attack.

Clear Study Notes

1. How Tooth Decay (Dental Caries) Happens Step by Step

Healthy enamel is extremely hard—yet weak acid slowly dissolves its calcium minerals! Tooth decay happens in a clear chain reaction:

1. Sticky sweets, biscuits, toffees, sugary drinks, and starchy food fragments linger in tooth grooves and between teeth after eating.
2. Natural bacteria living in the mouth mix with saliva and food debris to form a sticky biofilm called plaque.
3. Plaque bacteria feed on the leftover sugars and produce weak acids (such as lactic acid) as waste.
4. Trapped against the tooth by plaque, the acid slowly dissolves the outer enamel, forming a tiny hole (cavity). At this early stage there is no pain yet!
5. Untreated, decay spreads faster through softer dentine and reaches the living pulp cavity—causing throbbing toothache, bacterial abscess/swelling, and even loss of the tooth.

2. Causes of Gum Disease and Physical Tooth Damage

When plaque stays along the gumline for days, minerals in saliva harden it into rough, yellow-brown deposits called tartar (calculus). Tartar traps even more bacteria beneath the gum margin, causing gingivitis (red, puffy gums that bleed when brushing) and bad breath. If ignored, infection damages the periodontal fibres and jawbone (periodontitis), causing healthy-looking teeth to loosen and fall out!

Mechanical habits also crack enamel: never use your teeth to open soda bottle tops, crack palm kernels/bones, bite pens, or chew ice cubes.

3. Proper Cleaning and Prevention of Tooth and Gum Decay

Tooth decay and gum disease are almost 100% preventable through consistent daily care:

Preventive Practice	How to Do It Properly & Why It Works
Brush twice daily (morning & last thing before bed)	Use a pea-sized amount of FLUORIDE toothpaste on a soft/medium toothbrush for 2 full minutes. Night brushing is critical because saliva flow drops during sleep!
Use proper brushing technique	Angle bristles gently at 45° toward the gumline and brush with small circular or sweeping strokes (away from the gum), covering outer, inner, and chewing surfaces + tongue. Do NOT scrub sideways violently.
Clean between teeth / rinse	Floss or rinse mouth thoroughly with clean water right after sugary snacks or meals when brushing is not possible. Traditional Ghanaian chewing sticks (tweapea), kept clean and frayed gently at the tip, also remove plaque mechanically.
Healthy tooth-friendly diet	Limit sticky sweets, toffees, and sugary fizzy drinks between meals. Eat calcium- and vitamin-rich foods (milk, small fish eaten with bones, green leafy vegetables, oranges, crunchy fruits).
Protect enamel & visit dentist	Never crack bottle caps, bones, or hard nuts with teeth. Replace frayed toothbrushes every 3 months. Visit a dental clinic every 6–12 months for check-ups and safe scaling of tartar.

Guided Demonstration: Proper Two-Minute Tooth-Brushing Technique (Using Model or Chart)

Aim: To demonstrate systematic brushing strokes that remove plaque along the gumline without injuring enamel or gums.

Materials: Clean toothbrush (or jaw model + brush), cup of clean water, fluoride toothpaste.

Procedure / Steps:

1. Apply a pea-sized drop of fluoride toothpaste onto a soft/medium brush.

2. Place bristles at a 45° angle where teeth meet gums; use gentle short circular motions on outer surfaces of upper and lower teeth.

3. Sweep downward on upper teeth ('from pink gum to white tooth') and upward on lower teeth on the inside surfaces; tilt brush vertically behind front teeth.

4. Scrub the grooved chewing surfaces of premolars and molars back-and-forth gently, then brush the tongue lightly to remove odour-causing bacteria. Spit out foam (avoid excessive immediate rinsing so a trace of fluoride stays protecting enamel).

Key Observations / Findings:

• Gentle angled circular brushing reaches plaque tucked in the gum crevice and molar grooves without wearing away gums or enamel.

Scientific Conclusion: Regular gentle brushing with fluoride toothpaste—especially before bedtime—prevents plaque accumulation, acid decay, and gum bleeding.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking sugar itself directly burns holes in teeth like acid. Correction: Sugar feeds plaque bacteria, and it is the ACID produced by those bacteria that dissolves tooth enamel!

• Mistake: Brushing very hard with stiff bristles makes teeth cleaner. Correction: Hard scrubbing wears away enamel near the neck and pushes gums back, causing tooth sensitivity—gentle thorough technique with a soft/medium brush works much better.

Lesson Summary — Key Points to Remember

✓ Plaque is a sticky bacterial film; when bacteria digest sugary/starchy food leftovers, they release acid that dissolves enamel (tooth decay / dental caries).

✓ Hardened plaque (tartar) along the gumline causes swollen, bleeding gums (gum disease).

✓ Prevent decay by brushing twice daily (especially at night) with fluoride toothpaste and proper strokes, rinsing after sweets, avoiding cracking hard objects, eating balanced calcium/vitamin-rich foods, and visiting a dentist regularly.

Practice Exercise 14 (B8.3.1.1.3)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. What directly dissolves the hard enamel of a tooth during tooth decay?

A. Saliva alone B. Acids produced by plaque bacteria feeding on leftover sugars C. Cold drinking water D. Vitamin C in fruits

2. Why is brushing your teeth BEFORE going to sleep at night especially important?

A. Bacteria only live at night B. Saliva production slows down during sleep, so leftover sugars and acids stay unwashed on teeth for hours C. Toothpaste only works in darkness D. Enamel turns soft after sunset

3. What is tartar (calculus), and what problem does it commonly cause?

A. Soft chewing gum that cleans molars B. Hardened plaque along the gumline that irritates gums and causes bleeding/gum disease C. A type of fluoride mineral inside enamel D. A broken canine root

4. Which daily practice helps strengthen tooth enamel against acid decay?

A. Opening soda bottles with molars B. Brushing with fluoride toothpaste twice daily C. Sipping sweet carbonated drinks every hour D. Using a hard wire brush

Part B: Short Structured Questions (Write clear scientific answers)

5. Explain step by step how eating sugary biscuits without brushing leads to a painful toothache.

6. Distinguish between plaque and tartar.

7. State four practical habits that keep teeth and gums healthy.

8. Why does a dentist advise brushing gently away from the gumline with a soft/medium brush instead of scrubbing vigorously side-to-side?

BASIC 8 SCIENCE • FIRST TERM

STRAND 4: FORCES AND ENERGY

Building on your Basic 7 foundation on forms of energy and heat transfer, First Term Basic 8 explores how energy converts from one form into another (including hydroelectric dams in Ghana), compares renewable and non-renewable energy sources and how to manage renewable energy sustainably, and clarifies the scientific difference and relationship between heat and temperature.

Sub-Strand 1: Energy

Content Standard B8.4.1.1: Demonstrate the skill to evaluate the conversion of energy from one form to another.

LESSON 15

Energy Conversion and Everyday Transformation Chains

B8.4.1.1.1 • Week 7

Target Indicator: Describe energy conversion.
Scheme Topic: Energy Conversion | **Resources:** Pictures, charts, torchlight, battery, bell/radio, solar calculator

Learning Outcomes — By the end of this lesson, you should be able to:

- Recall common forms of energy (chemical, electrical, light, heat/thermal, sound, kinetic, potential) and state the Law of Conservation of Energy.
- Define energy conversion and explain how energy changes from one form to another in everyday devices and living organisms.
- Trace multi-step energy conversion chains accurately using arrow diagrams.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Energy conversion (transformation)	The change of energy from one form (input) into one or more other forms (output).
Law of Conservation of Energy	Energy can neither be created nor destroyed; it can only be converted from one form to another.
Potential energy	Stored energy due to position (gravitational), stretch (elastic), or chemical bonds (chemical energy in food, fuel, and batteries).
Kinetic energy	Energy possessed by any moving object (running athlete, spinning fan, flowing river, wind).
Wasted (dissipated) energy	Unwanted energy output—commonly heat or noise—produced alongside the useful energy during conversion.

Clear Study Notes

1. What Is Energy Conversion?

In Basic 7 (B7.4.1.1), you learned the forms of energy: chemical, electrical, heat (thermal), light, sound, kinetic (motion), and potential energy. Energy rarely stays in one form! Whenever you switch on a torchlight, cook over charcoal, kick a football, or charge a phone, energy is changing form—a process called energy conversion (or energy transformation).

Every energy conversion obeys the Law of Conservation of Energy: energy cannot be created out of nothing or destroyed; the total amount of energy stays constant, though some often spreads out into surroundings as unwanted heat or sound.

2. Common Energy Conversion Chains in Daily Life

Scientists represent energy conversions with arrow chains showing Input Energy → Intermediate Energy → Output Energy(ies):

Device / Process	Starting (Input) Energy	Energy Conversion Chain
Torchlight (Dry cell)	Chemical energy (in battery)	Chemical → Electrical → Light (+ Heat)
Electric iron / Kettle	Electrical energy	Electrical → Heat (Thermal)
Ceiling fan / Blender	Electrical energy	Electrical → Kinetic (Mechanical) (+ Sound + Heat)
Radio / Loudspeaker	Electrical energy	Electrical → Sound (+ Heat)
Green plant (Photosynthesis)	Light (Solar) energy	Light → Chemical energy (stored in glucose/starch)
Human running / Cycling	Chemical energy (in food)	Chemical → Kinetic (motion) + Heat
Charcoal stove / Firewood	Chemical energy (in wood/charcoal)	Chemical → Heat + Light
Car / Motorcycle engine	Chemical energy (petrol/diesel)	Chemical → Heat → Kinetic (+ Sound)
Mango falling from a tree	Gravitational potential energy	Potential → Kinetic → Sound + Heat (on hitting ground)

Device / Process	Starting (Input) Energy	Energy Conversion Chain
Solar calculator / Panel	Light (Solar) energy	Light → Electrical energy

Guided Investigation: Tracing Energy Chains in Four Classroom Devices

Aim: To observe real classroom/home devices and map their exact energy conversion chains.

Materials: Dry-cell torchlight, wind-up or battery clock/bell, solar calculator, small hand-clapping/pendulum demonstration.

Procedure / Steps:

1. Switch on the torchlight for 30 seconds and hold your palm close to the bulb—feel warmth in addition to seeing light.
2. Hold the solar calculator under bright classroom light (then cover the solar strip with a finger to watch the display fade).
3. Lift a stone or eraser 50 cm above a desk (stored gravitational potential energy) and drop it onto the desk.
4. Write the full arrow energy chain for each observation, identifying BOTH useful output and unwanted energy.

Key Observations / Findings:

- Torchlight: Chemical (battery) → Electrical → Light (useful) + Heat (felt on hand).
- Solar calculator: Light (Sun/room) → Electrical → Light/display.
- Dropping eraser: Gravitational Potential → Kinetic (falling) → Sound ('thud') + tiny Heat on impact.

Scientific Conclusion: Every real-world action involves energy converting from one form to another, often producing secondary heat or sound alongside the desired output.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

- Mistake: Saying a battery 'uses up electricity until energy vanishes'. Correction: A dry-cell battery stores CHEMICAL energy; as it converts chemical energy into electrical energy, the chemicals get used up (transformed), not destroyed!
- Mistake: Skipping intermediate steps in a battery-powered torch or car. Correction: In a torchlight, chemical energy first becomes ELECTRICAL energy in the wires before turning into LIGHT and HEAT at the bulb.

Lesson Summary — Key Points to Remember

✓ Energy conversion is the changing of energy from one form to another.
✓ According to the Law of Conservation of Energy, energy cannot be created or destroyed—only transformed.
✓ Examples include Chemical → Electrical → Light + Heat (torchlight), Light → Chemical (photosynthesis), and Chemical → Kinetic + Heat (muscular movement).

Practice Exercise 15 (B8.4.1.1.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which energy conversion chain correctly describes a battery-operated radio playing music?
A. Sound → Chemical → Light B. Chemical → Electrical → Sound (+ Heat) C. Heat → Kinetic → Electrical D. Light → Potential → Sound

2. During photosynthesis, green plants convert:
A. Chemical energy into light energy B. Light energy into chemical energy C. Electrical energy into sound energy D. Kinetic energy into nuclear energy

3. When a ripe pawpaw falls from a high branch toward the ground, its gravitational potential energy converts mainly into:
A. Chemical energy B. Electrical energy C. Kinetic energy D. Nuclear energy

4. Which statement accurately expresses the Law of Conservation of Energy?
A. Energy is destroyed every time a lamp glows B. Energy can be created inside electric wires C. Energy cannot be created or destroyed, only converted from one form to another D. Only heat energy can convert into other forms

Part B: Short Structured Questions (Write clear scientific answers)

5. Define energy conversion and give one everyday example from a Ghanaian kitchen.

6. Write the complete energy conversion chain for: (a) an electric pressing iron, (b) a bicycle rider eating food and pedalling uphill, (c) a solar calculator.

7. When you switch on a classroom ceiling fan, why does the motor housing feel warm after an hour?

8. Explain why a pendulum eventually slows down and stops swinging in air—does this violate the Law of Conservation of Energy?

LESSON 16

Importance of Energy Conversion and Hydroelectric Power

B8.4.1.1.2 • Week 7

Target Indicator: Discuss the importance of conversion of energy.

Scheme Topic: Energy Conversion | **Resources:** Pictures and diagrams of Akosombo / Bui hydroelectric dams, turbines, solar panels, windmills

Learning Outcomes — By the end of this lesson, you should be able to:

- Explain step by step how a dammed river produces electricity (hydroelectric power generation).
- Describe how natural forms of energy (falling water, sunlight, wind, biomass) are harnessed and converted into useful energy.
- Discuss why energy conversion is vital to homes, schools, healthcare, agriculture, transport, and industry.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Hydroelectric power (Hydro)	Electricity generated by converting the energy of dammed or flowing river water through turbines and generators.
Penstock	A large enclosed pipe that channels stored water downward from a dam reservoir to spin the turbine.
Turbine	A wheel fitted with blades/buckets that spins rapidly when pushed by rushing water, steam, or wind.
Generator (Dynamo)	A machine spun by a turbine that converts mechanical/kinetic energy into electrical energy using magnets and wire coils.

Clear Study Notes

1. How a Dammed River Produces Electricity (e.g., Akosombo & Bui Dams)

Most raw energy in nature—like water sitting in the Volta River—cannot directly light a bulb or power a hospital X-ray machine until we convert it! Ghana's Akosombo and Bui Hydroelectric Dams demonstrate a magnificent four-stage energy conversion chain:

1. Water Storage (Potential Energy): A strong concrete dam wall holds back river water in a deep elevated reservoir. Because the water sits high above the powerhouse, it holds huge gravitational potential energy.

2. Falling Water in Penstocks (Kinetic Energy): When intake gates open, water rushes downward through huge sloped steel pipes called penstocks, converting gravitational potential energy into high-speed kinetic energy.

3. Spinning the Turbine (Mechanical Kinetic Energy): The high-pressure jet of water strikes the blades of a giant water turbine, making the turbine and its vertical steel shaft spin fast.

4. Generating Electricity (Electrical Energy): The spinning shaft turns a generator (magnets rotating inside copper coils), converting kinetic energy into electrical energy! Transformers boost the voltage so power lines can carry electricity across Ghana, where it finally converts into light, heat, sound, and motion inside homes, schools, and factories.

2. Harnessing Natural Energy Forms into Useful Forms

Beyond hydro dams, human technology harnesses other natural energy sources by converting them into usable electricity, heat, or mechanical work:

Natural Energy Source	Harnessing Device	Useful Conversion Performed
Sunlight (Solar)	Solar photovoltaic (PV) panel / Solar water heater	Light → Electrical energy (for lights, boreholes, phones); Solar radiation → Heat (for drying cocoa/fish or heating water)
Moving Air (Wind)	Wind turbine / Windmill	Kinetic (wind) → Mechanical rotation → Electrical energy (or pumping groundwater)
Organic Matter / Waste (Biomass)	Biogas digester / Improved cookstove	Chemical energy (dung, crop waste, wood) → Methane gas → Heat & Electricity
Moving River Water	Hydroelectric dam (turbine + generator)	Gravitational Potential → Kinetic → Electrical energy

3. Why Energy Conversion Matters in Everyday Life

Without energy conversion, stored energy in batteries, dams, fuel, and sunshine would be useless to us! Converting energy makes modern life possible: electrical energy converts into cool air in fridges to preserve vaccines and food; fuel chemical energy converts into motion in trotro buses and tractors to transport harvests; solar energy converts into electricity for rural schools and clinics; and food chemical energy converts into body heat and muscular movement so we can live and learn.

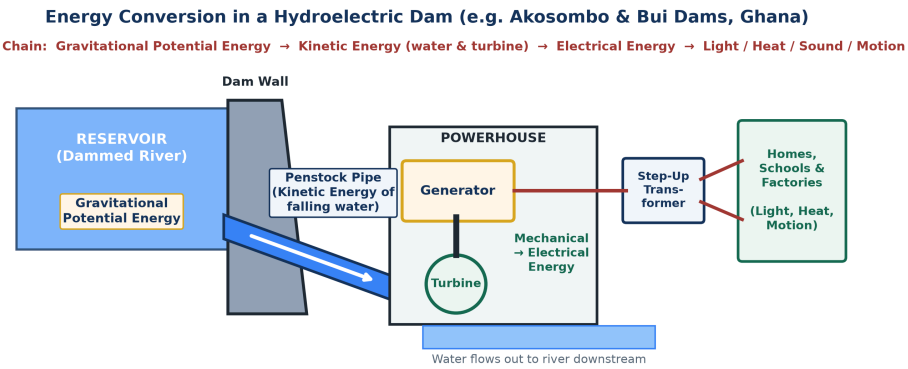


Figure 4.1: Stages of energy conversion inside a hydroelectric dam such as Akosombo or Bui Dam.

Guided Flow-Tracing Task: Following Energy From Raincloud to a Classroom Bulb

Aim: To trace every conversion stage from a dammed river to an electrical appliance in class.
Materials: Figure 4.1, notebook.
Procedure / Steps:
1. Trace Stage 1 (Reservoir behind Akosombo Dam) → Stage 2 (Water rushing down penstock) → Stage 3 (Spinning turbine blades) → Stage 4 (Generator coil) → Stage 5 (Classroom LED light bulb & ceiling fan).
2. Write the exact form of energy at each stage and explain what would happen to electricity output during a severe drought when reservoir water level falls very low.
Key Observations / Findings:
• Energy chain: Gravitational Potential (reservoir) → Kinetic (falling water + spinning turbine) → Electrical (generator & grid) → Light/Heat (bulb) or Kinetic (fan).
• Lower reservoir height means less gravitational potential energy per litre of water, reducing electricity generation.
Scientific Conclusion: Hydroelectric dams convert the natural potential energy of elevated river water into versatile electrical energy that powers nearly every sector of society.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking the dam itself makes electricity by mixing water with wires. Correction: Water NEVER touches the electrical coils—water pushes the mechanical TURBINE blades, and the spinning shaft turns magnets inside the GENERATOR above the water.
• Mistake: Thinking water is destroyed inside a hydroelectric dam. Correction: The same river water flows out through the tailrace downstream to continue the river and water cycle—only its height (potential energy) was used!

Lesson Summary — Key Points to Remember

✓ A dammed river produces electricity through the chain: Gravitational Potential Energy (reservoir) → Kinetic Energy (penstock & turbine) → Electrical Energy (generator).
✓ Solar panels, wind turbines, and biogas digesters similarly harness natural energy into useful electrical, heat, and mechanical forms.
✓ Energy conversion enables lighting, cooking, food preservation, healthcare, transport, farming, and industry.

Practice Exercise 16 (B8.4.1.1.2)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. In a hydroelectric power station, what is the role of the turbine?
A. To store rainwater in clouds B. To spin when hit by rushing water and turn the generator shaft C. To filter mud out of drinking water D. To step up electrical voltage

2. Which sequence gives the correct order of energy conversions inside a hydroelectric dam?
A. Electrical → Kinetic → Potential B. Gravitational Potential → Kinetic → Electrical C. Chemical → Light → Potential D. Thermal → Sound → Electrical

3. Which device harnesses natural wind energy to generate electricity?
A. Solar heater B. Wind turbine C. Penstock D. Dry cell

4. Why does a severe drought reduce electricity supply from hydroelectric dams like Akosombo?
A. Sun dries out copper wires B. Lower water volume and height in the reservoir store less gravitational potential energy C. Fish block all transformers D. Generators melt in dry air

Part B: Short Structured Questions (Write clear scientific answers)

5. Describe in four clear steps how water held behind a river dam generates electricity for homes.
6. State two ways sunlight can be harnessed and converted into useful energy in Ghana.
7. State three reasons why energy conversion is essential in daily life in your community.
8. Why is electricity regarded as one of the most useful intermediate forms of energy after conversion?

Content Standard B8.4.1.2: Show an understanding of the sources of renewable energy and how to manage these sources in a sustainable manner.

LESSON 17 Renewable and Non-Renewable Energy Sources

B8.4.1.2.1 • Week 8

Target Indicator: Describe renewable and non-renewable forms of energy.

Scheme Topic: Renewable & Non-Renewable Energy | **Resources:** Pictures, charts of solar panels, wind turbines, hydro dams, coal, crude oil, natural gas, biogas digesters

Learning Outcomes — By the end of this lesson, you should be able to:

- Explain the meaning of renewable and non-renewable energy sources.
- Classify energy sources accurately—including wind, coal, hydro, crude oil, natural gas, solar, and biogas.
- Describe step by step how useful energy is produced from a renewable energy source such as biogas or solar photovoltaic panels.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Renewable energy source	An energy source that is continuously or rapidly replenished by natural processes within a human lifetime and will not run out if managed properly.
Non-renewable energy source	A finite energy source formed over millions of years that cannot be quickly replaced once used up.
Fossil fuels	Non-renewable fuels—coal, crude oil (petroleum), and natural gas—formed from ancient buried plants and marine organisms.
Biogas	A renewable fuel gas (mainly methane, CH_4, and CO_2) produced when bacteria break down organic waste in an oxygen-free digester.

Clear Study Notes

1. Renewable vs. Non-Renewable Sources of Energy

All energy sources we rely on fall into two major categories based on whether nature replaces them fast enough for continuous human use:

Feature	Renewable Energy Sources	Non-Renewable Energy Sources
Replenishment	Continuously renewed by sun, weather, rivers, or organic cycles	Finite underground reserves; take millions of years to form
Exhaustible?	Will not run out if used sustainably	Will eventually deplete as reserves are burned
Environmental impact	Usually clean during operation; low CO_2 emissions	Burning emits greenhouse gases (CO_2), smoke, and air pollutants
Curriculum Examples	Solar (sunlight), Wind, Hydro (moving water), Biogas / Biomass (wood/crop waste if replanted)	Coal, Crude oil (petrol, diesel, kerosene, LPG), Natural gas

2. Classifying Specific Energy Sources (From Curriculum Exemplar 2)

Be able to classify every source listed in the NaCCA curriculum accurately:

- Renewable: Wind, Hydro (river dams), Solar (sunlight), Biogas (fermented organic waste).
- Non-renewable: Coal (solid fossil fuel), Crude oil (liquid fossil fuel refined into petrol/diesel/kerosene), Natural gas (gaseous fossil fuel).

3. How to Produce Energy from a Renewable Source: Biogas & Solar PV

Production from Biogas: Farm animal manure (cow/pig/poultry dung), kitchen food scraps, and water are mixed in an inlet tank and fed into an airtight underground chamber called a biogas digester. Naturally occurring anaerobic bacteria digest the waste without oxygen, releasing biogas (mostly methane, CH_4) that rises into a dome and flows through a pipe to fuel cooking stoves, gas lamps, or small electricity generators! Meanwhile, the leftover nutrient-rich slurry flows out for use as organic farm fertiliser.

Production from Solar Energy: Sunlight strikes silicon semiconductor cells on a Solar Photovoltaic (PV) panel, knocking electrons loose to generate direct current (DC) electricity. A charge controller stores this energy safely in deep-cycle batteries for night use, and an inverter converts DC to alternating current (AC) for school lights, computers, and refrigerators.

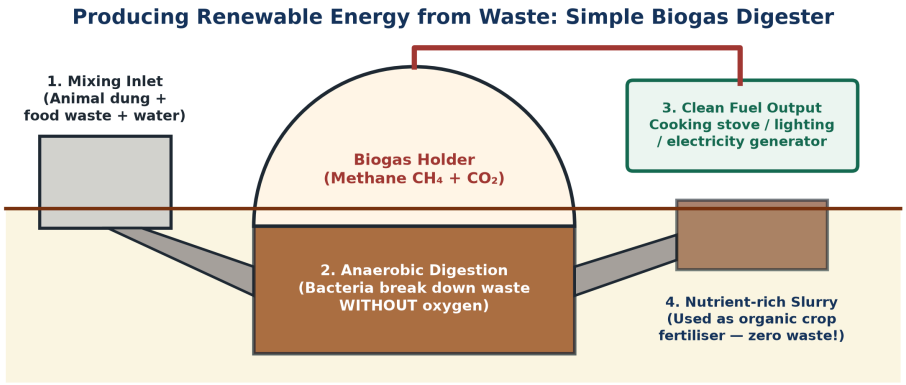


Figure 4.2: Producing clean renewable methane biogas and organic fertiliser from farm and food waste.

Guided Classification & Process Design: School Biogas vs. Diesel Generator

Aim: To classify seven energy sources and trace how organic canteen waste can produce clean renewable energy.

Materials: Notebook, Figure 4.2.

Procedure / Steps:

1. Sort these seven sources into Renewable and Non-renewable: wind, coal, hydro, crude oil, natural gas, solar, biogas.

2. Describe the four stages inside a school biogas digester using food peels and livestock droppings: (1) Mixing inlet, (2) Oxygen-free bacterial digestion, (3) Gas collection & piping to kitchen burner, (4) Slurry fertiliser collection.

Key Observations / Findings:

- Renewable (4): wind, hydro, solar, biogas. Non-renewable (3): coal, crude oil, natural gas.
- A biogas digester simultaneously solves two school problems: waste disposal and clean cooking fuel!

Scientific Conclusion: Renewable technologies such as biogas digesters and solar PV panels turn abundant local resources into clean, endlessly replaceable energy.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

- Mistake: Classifying Liquefied Petroleum Gas (LPG) cooking gas or natural gas as 'renewable' just because it burns cleanly. Correction: Natural gas and LPG (from crude oil) are FOSSIL FUELS—once burned, they cannot be replaced quickly, so they are NON-RENEWABLE! Biogas (from fresh dung/waste) is renewable.
- Mistake: Assuming biomass (firewood/charcoal) automatically stays renewable even when trees are clear-cut without replanting. Correction: Wood is only renewable if trees are continuously replanted at or above the rate they are harvested.

Lesson Summary — Key Points to Remember

✓ Renewable energy sources (solar, wind, hydro, biogas) are naturally replenished quickly and will not run out when properly managed.

✓ Non-renewable energy sources (coal, crude oil, natural gas) are finite fossil fuels formed over millions of years.

✓ Renewable energy can be produced locally using solar PV panels (light → electricity) or biogas digesters (anaerobic bacterial breakdown of organic waste → methane gas).

Practice Exercise 17 (B8.4.1.2.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which list contains ONLY renewable sources of energy?
A. Solar, coal, wind B. Hydro, wind, biogas C. Natural gas, crude oil, solar D. Coal, natural gas, hydro

2. What type of gas makes up the main fuel component of biogas produced inside a biodigester?
A. Oxygen B. Nitrogen C. Methane D. Helium

3. Why are coal, crude oil, and natural gas described as non-renewable energy sources?
A. They cannot produce heat when burned B. They are finite reserves that take millions of years to form and deplete as used C. They are found only in rivers D. They come directly from today's sunlight

4. Inside a biogas digester, bacteria break down dung and food scraps in the ABSENCE of:
A. Water B. Carbon C. Oxygen D. Organic matter

Part B: Short Structured Questions (Write clear scientific answers)

5. Define 'renewable energy source' and 'non-renewable energy source', giving two examples of each.

6. Classify each of the following as renewable or non-renewable: (a) wind, (b) crude oil, (c) biogas, (d) coal, (e) hydro, (f) natural gas, (g) solar.

7. Describe briefly how energy is produced using a simple biogas digester.

8. State two advantages of using solar PV panels instead of a petrol generator at a rural clinic.

LESSON 18

Sustainable Management of Renewable Energy Sources

B8.4.1.2.2 • Week 8

Target Indicator: Demonstrate how to manage sources of renewable energy sustainably.

Scheme Topic: Renewable & Non-Renewable Energy | **Resources:** Pictures, charts, case studies on managing solar, hydro, wind, and biomass/biogas

Learning Outcomes — By the end of this lesson, you should be able to:

- Explain what sustainable management of renewable energy means.
- Outline the key stages involved in managing renewable energy systems sustainably over their lifespan.
- Construct a structured table identifying practical challenges in managing different renewable sources (solar, hydro, wind, biomass/biogas) and appropriate solutions.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Sustainable management	Using, maintaining, and renewing energy resources wisely so they meet today's needs without damaging the environment or depriving future generations.
Siltation	Build-up of mud, sand, and debris washed into a hydroelectric dam reservoir, reducing its water-holding capacity.
Catchment protection	Planting and conserving trees and vegetation along riverbanks to regulate stream flow and stop soil erosion into dams.
Intermittency	Natural ups and downs in energy supply—for example, solar panels produce no power at night and wind slows during calm days.

Clear Study Notes

1. Why Even Renewable Energy Needs Sustainable Management

Many learners assume 'renewable' means 'automatic and trouble-free forever'. In reality, a poorly maintained solar system fails when batteries die early, an unprotected hydro reservoir fills with mud (siltation), and over-harvested firewood destroys forests faster than trees regrow! Sustainable management ensures clean energy remains reliable, affordable, and environmentally safe year after year.

2. Stages in Managing Renewable Energy Sources Sustainably

Sustainable renewable energy management follows five logical stages:

1. Assessment & Smart Siting: Choosing the best site for each technology (e.g., sunny unshaded roofs for solar, windy coastal/highland ridges for wind, permanent rivers for hydro).

2. Sustainable Harvesting & Catchment Care: Harvesting firewood/biomass only alongside active tree-replanting (woodlots), and planting buffer trees along riverbanks to protect hydro dams from siltation.

3. Routine Inspection & Preventive Maintenance: Wiping dust/Harmattan grit off solar panels, topping up or replacing worn batteries, checking biogas pipes for gas leaks, and servicing turbine bearings.

4. Efficient Storage & Energy Conservation: Using battery banks, smart controllers, energy-saving LED bulbs, and avoiding wasteful appliance use so supply stretches reliably.

5. Safe End-of-Life Recycling & Disposal: Recycling lead-acid/lithium solar batteries safely at licensed centres instead of dumping toxic battery acid or heavy metals into soil and water.

3. Challenges in Managing Renewable Energy Sources (Table)

Notice how every renewable source brings specific technical and environmental management challenges—and clear practical solutions:

Renewable Source	Key Management Challenges	Sustainable Management Solution
Solar Energy	• No sunlight at night / cloudy rainy days (intermittency) • Harmattan dust & bird droppings shade panels • High upfront cost & limited battery lifespan (3–5 yrs) • Improper disposal of old batteries	• Pair panels with properly sized batteries & charge controllers • Wipe panels regularly with clean water • Use energy-efficient LEDs/DC appliances • Recycle used batteries responsibly
Hydroelectric (Water)	• Lower river levels during severe dry seasons/drought • Siltation (soil eroded from deforested banks fills dam) • High dam construction cost & flooding of river communities	• Protect river-basin forests (catchment reforestation) • Desilt intakes and maintain dam gates • Combine hydro with solar backup during dry months
Wind Energy	• Wind speed fluctuates daily/ seasonally • Requires open unobstructed windy sites • Mechanical wear on blades/gears & turbine noise	• Survey wind patterns before installation • Service moving parts & lubrication on schedule • Site turbines away from homes and bird migration corridors

Renewable Source	Key Management Challenges	Sustainable Management Solution
Biomass & Biogas	• Over-cutting trees for firewood/ charcoal causes deforestation • Smoke pollution from traditional open three-stone fires • Biogas digesters fail if fed plastics/ chemicals or left leaking	• Replant fast-growing woodlots & use improved cookstoves • Feed digesters ONLY biodegradable waste + proper water ratio • Inspect gas valves & slurry pipes routinely

Guided Problem-Solving Case: Rescuing a Failing School Solar & Woodlot Project

Aim: To diagnose sustainability failures and apply the five management stages.

Materials: Table 4.3, notebook.

Procedure / Steps:

1. Case A: A school installed roof solar panels two years ago, but now the lights go dim after just 30 minutes at night, Harmattan dust coats the glass, and someone left an old cracked battery behind the science block.

2. Case B: The school kitchen cuts down mango and neem trees around the fence faster than seedlings are planted.

3. Identify the management stage neglected in each case and write two corrective steps.

Key Observations / Findings:

• Case A neglected Stage 3 (Maintenance: cleaning dust off panels), Stage 4 (checking battery health & switching to low-wattage LEDs), and Stage 5 (safe recycling of spent battery).

• Case B neglected Stage 2 (sustainable harvesting + replanting) and efficiency (switching to an improved cookstove or biogas from canteen scraps).

Scientific Conclusion: Renewable energy stays truly sustainable only when paired with maintenance, conservation, ecosystem care, and safe battery recycling.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking solar panels require zero cleaning or upkeep after installation. Correction: Thick Harmattan dust or shade from growing tree branches can cut solar power output dramatically unless panels are cleaned and kept unshaded.

• Mistake: Thinking used solar batteries can be thrown into ordinary school rubbish pits. Correction: Old lead-acid or lithium batteries contain toxic metals and acid that poison groundwater—they must be sent for proper recycling.

Lesson Summary — Key Points to Remember

✓ Managing renewable energy sustainably involves five stages: smart siting, sustainable harvesting/catchment protection, routine maintenance, efficient storage/use, and safe end-of-life recycling.

✓ Each source faces distinct challenges—intermittency and dust (solar), drought and siltation (hydro), wind variability (wind), and deforestation or gas leaks (biomass/biogas)—that require planned practical solutions.

Practice Exercise 18 (B8.4.1.2.2)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Why should trees be planted and protected along rivers feeding a hydroelectric dam?

A. To stop water from flowing to the turbines B. To prevent soil erosion and siltation that shallow the dam reservoir C. To block sunlight from hitting the river D. To convert river water into crude oil

2. During the Harmattan season in Ghana, what simple maintenance task helps solar panels keep producing maximum electricity?

A. Painting the panels black B. Wiping dust gently off the glass surface C. Covering the panels with plastic sheets D. Disconnecting the batteries permanently

3. Which practice makes harvesting firewood/biomass UNSUSTAINABLE?

A. Establishing fast-growing community woodlots B. Using improved fuel-efficient cookstoves C. Felling forest trees continuously without replanting seedlings D. Turning kitchen food peels into biogas

4. Why must worn-out solar storage batteries NOT be dumped in open refuse pits?

A. They dissolve into clean rainwater B. They contain toxic lead/chemicals that contaminate soil and groundwater C. They attract bees to flowers D. They regenerate energy underground

Part B: Short Structured Questions (Write clear scientific answers)

5. Outline four stages involved in managing renewable energy sources sustainably.

6. State two challenges in managing SOLAR energy and give one practical solution for each.

7. State two challenges in managing HYDROELECTRIC power and explain how communities can help solve one of them.

8. Explain why feeding plastic sachets into a biogas digester damages sustainable biogas production.

Content Standard B8.4.1.3: Demonstrate an understanding of the relationship between heat and temperature.

LESSON 19

Heat and Temperature: Differences and Relationship

B8.4.1.3.1 • Week 9

Target Indicator: Discuss the differences and the relationship between heat and temperature in the environment.
Scheme Topic: Heat And Temperature | **Resources:** Pictures and charts, laboratory thermometer, clinical thermometer, beakers of water

Learning Outcomes — By the end of this lesson, you should be able to:
• Define heat and temperature accurately in scientific terms.
• Create and explain a comparison table showing the distinguishing features of heat and temperature (meaning, SI unit, instrument, mass dependence, direction of flow).
• Explain the relationship between heat and temperature using everyday environmental examples.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Heat (Thermal energy)	A form of energy transferred between objects because of a temperature difference (total kinetic energy of moving particles in a body).
Temperature	A measure of how hot or cold a body is (determined by the average kinetic energy of its particles).
Joule (J)	The standard SI unit for measuring all forms of energy, including heat.
Kelvin (K) & Degree Celsius (°C)	Kelvin is the SI unit of temperature; degree Celsius (°C) is the common everyday scale (0 °C = ice melting, 100 °C = pure water boiling).
Thermometer	An instrument used to measure temperature (e.g., laboratory mercury/alcohol thermometer or digital clinical thermometer).
Calorimeter	An insulated laboratory vessel used in experiments to measure heat energy transferred.

Clear Study Notes

1. Distinguishing Features of Heat and Temperature

In everyday speech, people often say 'the heat today is 34 degrees', mixing up two very different scientific ideas! In Basic 7 (B7.4.1.2), you learned how heat travels by conduction, convection, and radiation. Now compare Heat and Temperature side by side:

Feature	Heat	Temperature
Scientific meaning	Form of energy (total thermal energy transferred due to temperature difference)	Degree of hotness or coldness of a body (average kinetic energy per particle)
SI Unit (and common unit)	Joule (J) [also kilojoule kJ, or calorie]	Kelvin (K) [commonly degree Celsius, °C]
Measuring instrument	Calorimeter (calculated from mass, specific heat & temp change)	Thermometer (laboratory, clinical, or digital)
Depends on mass / size of object?	YES — depends on mass, material, and temperature (a full bucket of 80 °C water holds far more heat than a cup of 80 °C water!)	NO — independent of mass (a cup and a pot of boiling water are both at 100 °C)
Flow / Direction	Always flows naturally from a hotter body to a colder body until temperatures equalise	Indicates the direction heat will flow (from higher °C to lower °C)

2. The Relationship Between Heat and Temperature in the Environment

Although heat and temperature are NOT the same thing, they are closely linked:

• Cause and Effect: When heat energy enters an object without changing its state, its particles move faster and its temperature RISES. When an object loses heat energy to cooler surroundings, its particles slow down and its temperature DROPS. For example, sunlight supplies radiant heat to soil, roofs, and rivers during the afternoon, raising their temperature; at night they lose heat into the cooler night air and their temperature falls.

• Mass Matters! Think of a lit matchstick spark versus a big pot of warm bathwater (45 °C). The tiny spark has a very high temperature (hundreds of °C), so it feels sharp if it touches skin—yet it holds very little total heat energy because its mass is tiny. Meanwhile, the large pot of 45 °C water holds far more total heat energy because it contains trillions more water molecules!

• Heat Flow & Thermal Equilibrium: Leave a hot cup of tea (80 °C) on a classroom table at 28 °C. Heat flows from the hotter tea into the cooler surrounding air until both reach 28 °C (thermal equilibrium).

Guided Thought & Thermometer Experiment: Comparing a Cup vs. a Bucket of Hot Water

Aim: To distinguish temperature from total heat content and observe heat flow direction.
Materials: Laboratory thermometer (0–110 °C), one small cup (200 mL) and one larger beaker/bowl (1,000 mL) of hot water at the same initial temperature (~60 °C), two equal ice cubes (or cool water baths).
Procedure / Steps:

1. Measure the initial temperature of both the 200 mL cup and the 1,000 mL beaker of hot water using the thermometer—verify both read ~60 °C.
2. Place one identical ice cube into each container and observe which melts faster and which water stays warmer after 3 minutes.
3. Explain the result using the concepts of Heat (Joules) vs. Temperature (°C).

Key Observations / Findings:
• Both containers started at the exact same temperature (60 °C), yet the larger 1,000 mL beaker melts its ice cube completely while remaining noticeably warmer than the small cup!

Scientific Conclusion: Temperature does NOT depend on mass, but Heat energy DOES—greater mass at the same temperature holds greater total thermal energy.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Saying 'heat is measured in degrees Celsius (°C)'. Correction: Temperature is measured in Kelvin (SI) or degrees Celsius (°C); Heat is a form of energy measured in Joules (J)!
• Mistake: Thinking two objects at the same temperature must contain the same amount of heat. Correction: A large pot of water at 70 °C contains much more heat energy than a single spoonful of water at 70 °C because heat depends on mass.

Lesson Summary — Key Points to Remember

✓ Heat is energy transferred between objects due to a temperature difference (SI unit: Joule, J; depends on mass).
✓ Temperature is the degree of hotness or coldness of a body (SI unit: Kelvin, K, commonly °C; measured by a thermometer; does not depend on mass).
✓ Heat always flows spontaneously from a body at higher temperature to one at lower temperature; adding heat generally raises an object's temperature.

Practice Exercise 19 (B8.4.1.3.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. What are the correct SI units for Heat and Temperature respectively?
A. Degree Celsius and Joule B. Joule and Kelvin C. Thermometer and Calorimeter D. Watt and Kilogram

2. A small cup of water and a large drum of water are both heated to 65 °C. Which statement is TRUE?
A. Both contain the exact same amount of heat energy B. The small cup contains more heat energy C. Both have the same temperature, but the drum contains far more heat energy D. Neither contains any thermal energy

3. When a hot metal spoon at 90 °C is placed into a bowl of tap water at 25 °C, what happens?
A. Heat flows from the tap water into the spoon B. Heat flows from the hot spoon into the tap water until both reach the same temperature C. Cold flows out of the spoon D. Both temperatures stay permanently unchanged

4. Which instrument is used in a clinic or science lab to measure temperature directly?
A. Barometer B. Thermometer C. Ammeter D. Spring balance

Part B: Short Structured Questions (Write clear scientific answers)

5. Create a comparison table listing three clear differences between Heat and Temperature.

6. Explain the relationship between heat and temperature when sunshine warms a metal rooftop in the afternoon.

7. Why can a tiny red-hot spark from a grinding stone have a much higher temperature than a pot of boiling water, yet cause less severe burns if it drops onto dry cloth?

8. A student leaves hot sobolo at 75 °C on a kitchen counter in a room at 29 °C. Describe what happens over the next hour and name the final state reached.

BASIC 8 SCIENCE • FIRST TERM

STRAND 5: HUMANS AND THE ENVIRONMENT

Human health depends directly on how responsibly we manage our surroundings. Building on your Basic 7 foundation on waste management (reduce, reuse, recycle) and infectious diseases, First Term Basic 8 examines sustainable community waste management systems (including upcycling and composting), asks you to design and evaluate a local waste plan, and equips you with scientific knowledge of common communicable diseases, their symptoms, risk factors, and prevention.

Sub-Strand 1: Waste Management

Content Standard B8.5.1.1: Demonstrate knowledge of waste management systems and apply it in an environment.

LESSON 20

Sustainable Waste Management Practices (Reduce, Reuse, Recycle, Compost)

B8.5.1.1.1 • Week 10

Target Indicator: Explain sustainable waste management practices.

Scheme Topic: Waste Management | **Resources:** Pictures and charts of waste hierarchy, sorted waste bins, compost heap, recycled/upcycled items

Learning Outcomes — By the end of this lesson, you should be able to:

- Explain the meaning of sustainable waste management and why open dumping and burning harm the environment.
- Sort community waste into biodegradable, recyclable, reusable, and hazardous streams.
- Explain and apply the Waste Management Hierarchy: Reduce, Reuse, Recycle/Upcycle, Compost/Recover, and Safe Disposal.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Sustainable waste management	Handling waste from generation to final treatment in ways that conserve resources, protect human health, and leave a clean environment for future generations.
Reduce (Source reduction)	Generating less waste in the first place (e.g., carrying a reusable shopping bag instead of taking many single-use polythene bags).
Reuse	Using an intact item again for the same or another practical purpose without industrial reprocessing (e.g., refilling glass bottles or water flasks).
Recycle	Collecting and processing waste materials (plastic bottles, scrap aluminium/iron, paper) into raw material for new products.
Upcycle	Creatively transforming waste items into higher-value useful products (e.g., turning plastic bottles into school tree guards or pencil holders).
Composting	Allowing microbes and earthworms to decompose sorted organic kitchen/garden waste into rich dark humus fertiliser.

Clear Study Notes

1. Why Unsustainable Disposal Hurts Ghanaian Communities

In Basic 7 (B7.5.1.1), you learned basic waste types and ways to manage them. Whenever people dump rubbish into open gutters or burn mixed plastics on school compounds, serious harms follow: blocked drains trigger floods and breed Anopheles mosquitoes (malaria); rotting refuse spreads cholera and typhoid via houseflies; and burning plastic sachets and tyres releases toxic, cancer-causing fumes that harm lungs and worsen air pollution.

2. Waste Segregation: First Step to Sustainable Management

Mixing wet food peels with dry paper, broken glass, and used batteries turns valuable resources into filthy garbage! Sustainable systems separate waste right at the source into distinct colour-coded or labelled bins:

- Organic / Biodegradable Bin (Green): Food scraps, cassava/plantain peels, dry leaves, grass clippings → ideal for composting or biogas digesters!
- Recyclables Bin (Blue / Yellow): Clean plastic water sachets & PET bottles, cardboard, office paper, aluminium cans, clean glass bottles → sent to recyclers or upcycled in school projects.
- General Residual / Hazardous Waste (Black / Red): Broken syringes/clinic waste, expired batteries, old paint tins, electronic e-waste → kept strictly out of compost and handed to licensed hazardous-waste handlers or sanitary landfills.

3. The Sustainable Waste Hierarchy (From Most to Least Preferred)

Follow the inverted waste pyramid from top (best) to bottom (last resort):

Priority Level	Practice	Practical Ghanaian School & Home Examples
1st (Best)	REDUCE (Prevent waste)	Use a refillable drinking bottle instead of buying daily sachet water; buy bulk grains rather than tiny plastic wraps; use both sides of exercise sheets.
2nd	REUSE	Wash glass jam jars to store spices; mend torn uniforms or bags; carry reusable woven cloth bags to the market; use empty gallon containers for watering plants.
3rd	RECYCLE & UPCYCLE	Collect clean water sachets and plastic bottles for certified plastic-recycling aggregators; upcycle old tyres into flower planters or plastic bottles into nursery pots.
4th	COMPOST & RECOVER ENERGY	Turn canteen food scraps and swept leaves into school-garden compost manure or feed a biogas digester for cooking gas.
5th (Last Resort)	SAFE DISPOSAL	Dispose of truly unusable residual waste in covered sanitary bins collected for engineered landfills—NEVER burn plastics or dump in gutters.

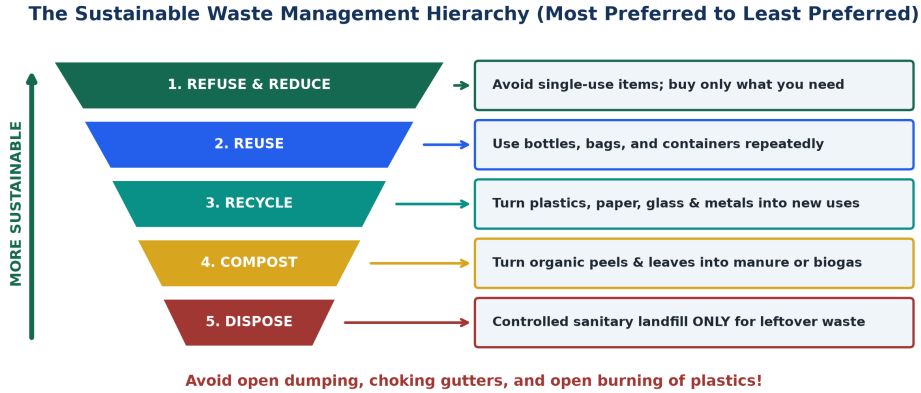


Figure 5.1: The Sustainable Waste Management Hierarchy showing priority practices from Reduce to Safe Disposal.

Guided Waste-Sorting Audit: Turning Classroom Rubbish into Resources

Aim: To classify typical school-day waste items and apply the best sustainable practice to each.

Materials: Sample clean/safe classroom waste inventory list, Figure 5.1.

Procedure / Steps:

1. Examine six common school waste items: (1) Empty plastic water sachets, (2) Banana & orange peels, (3) Used torn exercise paper, (4) Broken fluorescent tube/dead dry-cell battery, (5) Intact clean jam jar, (6) Swept dry mango leaves.

2. Match each item to its proper bin and best sustainable action (Reduce, Reuse, Recycle, Compost, or Hazardous Safe Disposal).

Key Observations / Findings:

• 1) Plastic sachets → Reduce via reusable bottle / Rinse & send for Recycling; 2) Fruit peels → Compost or Biogas; 3) Exercise paper → Use both sides first (Reduce), then Recycle/Compost shreddings; 4) Dead battery/tube → Hazardous Safe Disposal (NEVER burn or compost!); 5) Clean jar → Reuse for storage; 6) Dry leaves → Compost/Mulch for school garden.

Scientific Conclusion: Once sorted at the source, more than three-quarters of ordinary school waste turns out to be compostable, reusable, or recyclable rather than useless rubbish!

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking burning plastic waste in a pit makes waste 'disappear safely'. Correction: Burning plastics releases toxic dioxins, heavy smoke, and greenhouse gases (CO_2) into the air you breathe, while toxic ash poisons soil and groundwater.

• Mistake: Confusing REUSE and RECYCLE. Correction: REUSE means using the item again as it is (e.g., refilling a bottle); RECYCLE means breaking down the material industrially (melting plastic or pulping paper) to manufacture a brand-new product.

Lesson Summary — Key Points to Remember

✓ Sustainable waste management protects health and conserves resources by separating waste at source and following the waste hierarchy.

✓ From highest to lowest priority: Reduce → Reuse → Recycle/Upcycle → Compost/Recover energy → Safe Disposal.

✓ Avoid open dumping in drains and open burning of plastics; sort biodegradable waste into compost/biogas and keep hazardous e-waste/batteries separate.

Practice Exercise 20 (B8.5.1.1.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which action sits at the VERY TOP (highest priority) of the sustainable waste management hierarchy?
A. Safe landfill disposal B. Open burning C. Reducing waste generation at source D. Recycling scrap metal

2. Washing an empty plastic ice-cream tub and using it to store chalk or dry beans is an example of:
A. Composting B. Recycling C. Reusing D. Incinerating

3. Why should used dry-cell batteries NEVER be thrown onto a school compost heap?
A. They make compost dry out B. They are non-biodegradable and leak toxic heavy metals that poison compost and crops C. They attract earthworms D. They convert into methane too fast

4. Which waste items are BEST suited for making garden compost manure?
A. Broken glass, nylon sachets, and nails B. Plantain peels, leftover food greens, and dry swept leaves C. Used motor oil and old tyres D. PVC pipes and tin cans

Part B: Short Structured Questions (Write clear scientific answers)

5. List the five priority levels of the sustainable waste hierarchy from most preferred to least preferred.

6. Distinguish between 'reusing' and 'recycling' a glass bottle, giving one clear example of each.

7. State two environmental or health dangers of burning mixed plastic waste on a school compound.

8. Explain why sorting waste into separate bins BEFORE collection helps a school save money and protect its garden.

LESSON 21

Designing and Evaluating a Community Waste Management Plan

B8.5.1.1.2 • Week 10

Target Indicator: Apply knowledge of waste management practices to manage waste in a community.
Scheme Topic: Waste Management | **Resources:** Observation checklist, waste-audit tally chart, school/community map

Learning Outcomes — By the end of this lesson, you should be able to:

- Conduct a simple waste audit to identify major waste types and litter hotspots in a school or local community.
- Design a practical, step-by-step waste management action plan assigning bins, duties, composting/recycling pathways, and awareness steps.
- Identify clear criteria and indicators to evaluate whether a community waste management project is succeeding and improve weak areas.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Waste audit	A systematic survey weighing, counting, or estimating the types and amounts of waste generated in a place.
Litter hotspot	A specific area (e.g., canteen, gutter corner, market stall) where rubbish repeatedly piles up.
Action plan	A structured schedule naming clear tasks, tools needed, responsible persons, timelines, and collection routines.
Evaluation criteria	Measurable standards (such as cleaner gutters, compost produced, or reduction in sachet litter) used to judge a plan's success.

Clear Study Notes

1. Five Steps to Design a School or Community Waste Plan

Knowing the 3Rs is only helpful when put into action! Follow these five engineering-style steps to apply sustainable waste practices in your school or neighbourhood:

Step	Action in Community / School	Key Questions Answered
Step 1: Survey & Audit	Walk the compound/street to map litter hotspots and inspect what waste dominates (e.g., water sachets, food wraps, dry leaves, tins).	What kinds of waste are generated most, and where does litter accumulate?
Step 2: Set Clear Goals	Set realistic targets—e.g., 'Eliminate sachet burning and cut playground litter by 80% within one term.'	What measurable improvement do we want to achieve?
Step 3: Design Bin Stations & Pathways	Place clearly labelled 3-bin sets (Organic / Plastic Recyclables / General) at canteen, classrooms, and gates; build a fenced compost pit behind the garden.	Where will sorted bins go, and who collects recyclables/ compost?
Step 4: Assign Duties & Educate	Draw a weekly Sanitation Club roster; place colourful posters in English/ local languages; encourage students to bring reusable water flasks.	Who empties bins daily, and how will everyone remember the rules?

Step	Action in Community / School	Key Questions Answered
Step 5: Monitor & Evaluate	Inspect weekly using a cleanliness scorecard, track compost bags harvested, identify recurring problems (like mixed bins), and refine the plan.	Is the plan actually working, and what needs fixing?

2. Evaluating a Waste Management Plan

A true scientist evaluates results honestly rather than stopping after placing bins! Good evaluation checks four measurable signs:

1. Segregation Accuracy: Are students putting food peels into the Organic bin and sachets into the Plastic bin—or tossing everything together?
2. Hotspot Reduction: Are gutters, sports fields, and canteen corners noticeably cleaner with zero smoke from rubbish burning?
3. Resource Recovery: How many buckets of mature compost went to the school garden, or how many sacks of clean plastics were collected for recycling?
4. Community Participation: Do learners, food vendors, and neighbours actively follow the system even without teachers watching?

Guided Project Blueprint: School Canteen & Playground Waste Plan

Aim: To design and establish an evaluation checklist for a Basic 8 class waste management project.

Materials: Exercise book, ruler, compound sketch.

Procedure / Steps:

1. Hotspot Identified: Behind the school canteen, leftover kenkey leaves, fruit peels, and empty water sachets spill over a single broken bin and blow into the gutter.

2. Design Solution: (a) Install three covered, colour-labelled drums (Green: Food/Leaves, Blue: Clean Plastics, Black: General Waste) under canteen shade; (b) Ask water/drink vendors to keep a small rinsing bucket & sachet sack; (c) Empty Green bin daily into the school garden compost bay; (d) Rotate 4-student Sanitation Monitors weekly.

3. Evaluation Checklist (Checked every Friday for 4 weeks): Score 0–5 on (i) Gutter cleanliness, (ii) Correct bin sorting %, (iii) Zero burning episodes, (iv) Compost moisture/odour condition.

Key Observations / Findings:

• When bins are covered, clearly labelled with pictures + words, and emptied on schedule before overflowing, sorting accuracy rises sharply and gutter litter drops.

Scientific Conclusion: Successful community waste management combines infrastructure (sorted covered bins), clear routines, community education, and regular evaluation.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking simply buying dustbins solves a community waste problem. Correction: Without clear labelling, convenient placement, regular emptying schedules, composting/recycling outlets, and user education, bins quickly overflow or fill with mixed unsorted trash.

• Mistake: Designing a waste plan without an evaluation step. Correction: Without weekly inspection and feedback, broken bins, overflowing points, or mixed sorting go unnoticed until the project collapses.

Lesson Summary — Key Points to Remember

✓ Applying waste management in a community follows five stages: Survey/Audit → Set Goals → Set Up Sorted Infrastructure → Assign Duties & Educate → Monitor & Evaluate.

✓ Evaluation checks sorting accuracy, cleanliness of former hotspots, volume of compost/recyclables recovered, and ongoing participation.

Practice Exercise 21 (B8.5.1.1.2)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. What should a Basic 8 Environmental Club do FIRST before placing new waste bins around a market or school?
A. Burn existing rubbish immediately B. Conduct a waste audit to identify types of waste and litter hotspots C. Dig a deep hole anywhere on the pitch D. Fine every student blindly

2. Which detail is MOST important when labelling waste sorting bins for both young pupils and community elders?
A. Writing long paragraphs in tiny font B. Using clear colour codes plus simple pictures and short labels of allowed items C. Leaving bins unmarked so people guess D. Painting all bins the exact same black colour

3. How can food vendors at a school gate actively help reduce mixed plastic litter?
A. By throwing wrappers into gutters B. By keeping a dedicated sack for clean water sachets/bottles and encouraging reusable bowls/cups C. By hiding dustbins indoors D. By burning waste behind stalls at noon

4. Which indicator provides direct evidence that a school waste plan is succeeding after six weeks?
A. Bins are overflowing onto the grass B. Previously choked drains stay clear, sorting errors drop, and garden compost is being produced C. Smoke rises daily from the rubbish pit D. Students ignore bin labels

Part B: Short Structured Questions (Write clear scientific answers)

5. Outline the five steps you would follow to design a waste management plan for your school compound.

6. Suppose your Friday evaluation shows students keep throwing plastic water sachets into the Organic Food Waste bin. Suggest two practical improvements.

7. State three measurable evaluation criteria a committee can use to judge a community waste management project.
8. Why should waste bins in rainy Ghanaian communities always have fitted lids and drainage/raised stands?

Sub-Strand 2: Human Health

Content Standard B8.5.2.1: Demonstrate knowledge of common communicable diseases, such as Hepatitis, of humans, causes, symptoms, effects and their prevention.

LESSON 22

Common Communicable Diseases: Hepatitis, HIV/AIDS, Cholera & Tuberculosis

B8.5.2.1.1 • Weeks 11–12

Target Indicator: Explain the symptoms, effects and prevention of common communicable diseases.
Scheme Topic: Communicable Diseases | **Resources:** Health posters, Ghana Health Service charts, sample data tally table

Learning Outcomes — By the end of this lesson, you should be able to:
• Define communicable (infectious) diseases and group them by how they spread (water/food, air, blood/body fluids, vectors).
• Compile and interpret simple survey data on common communicable diseases in Ghanaian communities.
• Describe the causes, key symptoms, health effects, and prevention of common communicable diseases—with special focus on Hepatitis (especially Hepatitis B), plus HIV/AIDS, Cholera, Tuberculosis, and Malaria.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Communicable disease	An infectious illness caused by microorganisms (viruses, bacteria, protozoa, fungi) that can pass from an infected person, animal, or contaminated source to a healthy person.
Pathogen	A microscopic disease-causing organism (virus, bacterium, protozoan, or fungus).
Hepatitis	Inflammation/damage of the liver caused most commonly by Hepatitis viruses (A, B, C), leading to jaundice, fatigue, and potential liver cirrhosis or cancer.
Jaundice	Yellowish discolouration of the whites of the eyes (sclera) and skin caused by bilirubin build-up when the liver is damaged.
Vaccination (Immunisation)	Giving a safe vaccine so the immune system builds protective antibodies before encountering the real pathogen.

Clear Study Notes

1. What Are Communicable Diseases and How Do They Spread?

In Basic 7 (B7.5.2.1–2), you studied water-borne and food-borne diseases like cholera and typhoid. A communicable disease is any illness caused by a pathogen (virus, bacterium, protozoan, or fungus) that spreads from person to person through:

• Contaminated food and water (faecal-oral route): Hepatitis A, Cholera, Typhoid fever.
• Blood and body fluids / unprotected sex / contaminated sharp tools: Hepatitis B, Hepatitis C, HIV/AIDS.
• Airborne droplets from coughing or sneezing: Tuberculosis (TB), Measles, Common flu/COVID-19.
• Insect vectors: Malaria (female Anopheles mosquito).

2. Focus Study: Viral Hepatitis (Especially Hepatitis B)

Hepatitis means inflammation of the liver—the vital organ that detoxifies blood, stores energy, and makes bile. Different viruses cause different forms:

• Hepatitis A (HAV): Spread via contaminated food or drinking water (faecal-oral); usually short-term acute illness prevented by handwashing, safe food/water, and vaccination.
• Hepatitis B (HBV) & Hepatitis C (HCV): Serious blood- and body-fluid-borne viral infections common in West Africa. HBV spreads through unscreened blood transfusions, sharing unsterilised razor blades, manicure/pedicure tools, needles, or tattoo instruments, from infected mother to baby during childbirth, and unprotected sexual contact.
• Symptoms of Hepatitis: Yellowing of the whites of the eyes and skin (jaundice), dark tea-coloured urine, pale stools, nausea/vomiting, loss of appetite, pain in the upper-right abdomen (over the liver), and extreme tiredness. (Note: Many people with chronic Hepatitis B show NO early symptoms for years!)
• Serious Effects: Untreated chronic Hepatitis B or C can permanently scar the liver (liver cirrhosis), cause liver failure, or lead to liver cancer.
• Prevention: Get tested and complete the 3-dose Hepatitis B vaccine series (now routine in Ghana's infant immunisation schedule and recommended for unprotected youths/adults); NEVER share blades, clippers, syringes, or toothbrushes; insist on screened blood transfusions; practise safe behaviour.

3. Summary Profile of Common Communicable Diseases

Compare five common communicable diseases affecting Ghanaian communities:

Disease & Pathogen	Mode of Spread	Key Symptoms & Effects	Prevention Methods
Hepatitis B (Virus - HBV)	Infected blood, unsterile blades/needles, mother-to-child, unprotected sex	Yellow eyes (jaundice), dark urine, upper-right belly pain, fatigue; causes liver cirrhosis/cancer	Hepatitis B vaccination; never share blades/needles/ toothbrushes; screened blood; safe sex
HIV/AIDS (Virus - HIV)	Unprotected sex, infected blood, shared sharp needles/ blades, mother-to-child	Weakens immune white blood cells, weight loss, chronic fever/ diarrhoea, opportunistic infections	Abstinence/faithfulness/condoms; avoid sharing sharps; screened blood; PMTCT drugs
Tuberculosis / TB (Bacterium)	Airborne droplets when infected person coughs, sneezes, or speaks closely	Persistent cough (>2 weeks), chest pain, blood-stained sputum, night sweats, weight loss	BCG vaccine in infancy; cover mouth when coughing; airy sunlit rooms; complete full 6-month antibiotic cure
Cholera (Bacterium - Vibrio cholerae)	Drinking/eating water or food contaminated with infected faeces	Sudden severe watery ('rice-water') diarrhoea, vomiting, rapid fatal dehydration	Boil/chlorinate water; wash hands with soap; eat hot covered food; stop open defecation
Malaria (Protozoan - Plasmodium)	Bite of infected female Anopheles mosquito	High fever, chills/shivering, headache, body pains, anaemia	Sleep under ITNs; drain stagnant water; indoor spraying; prompt RDT testing & ACT treatment

Guided Data Tally (Illustrative Classroom Sample Data): Local Clinic Awareness Survey

Aim: To organise community health survey counts into a table and link top illnesses to preventive actions.

Materials: Notebook, pen, Illustrative Classroom Sample Data below.

Procedure / Steps:

1. A Basic 8 class interviewed 40 adults in an Illustrative Classroom Sample Data survey asking which communicable diseases family members had experienced or sought testing for over the past year:

2. Sample Tally Results (out of 40 respondents): Malaria = 22; Acute Diarrhoea/Cholera-like = 8; Hepatitis B screening/ cases = 5; Tuberculosis = 3; Measles = 2.

3. Rank the top three diseases in this sample data and suggest one targeted community prevention measure for each.

Key Observations / Findings:

• Top 3 in Illustrative Sample: 1st Malaria (22/40 = 55%), 2nd Water-borne Diarrhoea/Cholera (8/40 = 20%), 3rd Hepatitis B (5/40 = 12.5%).

• Note: Always collect real figures from a local clinic or CHPS compound when doing your actual school fieldwork!

Scientific Conclusion: Organising health observations in a clear tally helps communities target scarce resources toward sleeping under ITNs, safe water/handwashing, and Hepatitis B vaccination.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking Hepatitis B spreads by shaking hands, hugging, or sharing a school bench. Correction: Hepatitis B is NOT spread by casual touch or coughing—it spreads via infected blood, body fluids, shared unsterilised sharps/ razors, childbirth, or unprotected sex.

• Mistake: Thinking someone who looks healthy cannot carry Hepatitis B or HIV. Correction: Both infections can remain silent without obvious symptoms for years while still damaging organs or spreading to others unless tested.

Lesson Summary — Key Points to Remember

✓ Communicable diseases are illnesses caused by pathogens (viruses, bacteria, protozoa) that transmit between hosts via contaminated food/water, air, blood/body fluids, or vectors.

✓ Hepatitis inflames the liver, causing jaundice (yellow eyes), dark urine, fatigue, and—if chronic like Hepatitis B—possible liver cirrhosis or liver cancer.

✓ Vaccination (e.g., Hepatitis B and BCG), personal hygiene, clean water, safe handling of sharp instruments, ITNs, and prompt full medical treatment prevent spread.

Practice Exercise 22 (B8.5.2.1.1)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which internal human organ is primarily attacked and inflamed by Hepatitis viruses?
A. Heart B. Lungs C. Liver D. Kidneys

2. Yellowing of the whites of the eyes and skin seen in many Hepatitis patients is known scientifically as:
A. Anaemia B. Jaundice C. Dehydration D. Gingivitis

3. How is Hepatitis B MOST commonly transmitted in communities?
A. Bites of houseflies on mangoes B. Contact with infected blood/body fluids (e.g., sharing blades/needles, mother-to-child, unprotected sex) C. Shaking hands or sitting next to someone in class D. Breathing dusty Harmattan air

4. Which bacterium causes sudden severe watery diarrhoea that can lead to fatal dehydration within hours unless treated?
A. Plasmodium B. Vibrio cholerae (Cholera) C. Hepatitis A virus D. HIV

Part B: Short Structured Questions (Write clear scientific answers)

5. Define 'communicable disease' and give one example spread by (a) airborne droplets, (b) contaminated water, and (c) infected blood.

6. List three symptoms and two long-term serious effects of viral Hepatitis (such as Hepatitis B).

7. State three specific actions a teenager can take to protect themselves against Hepatitis B infection.

8. Why must a Tuberculosis patient complete the full 6-month course of prescribed medicines even after their cough stops in Week 4?

LESSON 23

Analysing Risk Factors of Communicable Diseases

B8.5.2.1.2 • Weeks 11–12

Target Indicator: Analyse the risk factors of communicable diseases.

Scheme Topic: Communicable Diseases | **Resources:** Health risk-assessment charts, community sanitation pictures, soap/water handwashing station

Learning Outcomes — By the end of this lesson, you should be able to:

- Explain what a 'risk factor' means in public health and disease prevention.
- Identify environmental, behavioural, biological/nutritional, and healthcare risk factors that increase vulnerability to communicable diseases.
- Analyse how changing modifiable risk factors breaks disease transmission chains in Ghanaian homes and schools.

Key Terms & Definitions

Key Term	Clear Scientific Meaning
Risk factor	Any environmental condition, personal behaviour, or biological state that increases a person's likelihood of catching or spreading a disease.
Immunity	The body's natural defence ability to fight off pathogens using white blood cells and antibodies.
Sanitation	Conditions and systems for safe disposal of human excreta, wastewater, and solid rubbish to keep environments clean.
Malnutrition	Lacking sufficient balanced nutrients (protein, vitamins, minerals), which weakens body immunity against infections.

Clear Study Notes

1. What Is a Disease Risk Factor?

Have you noticed why two people living in the same town do not face equal chances of falling sick? While the pathogen (virus, bacterium, or parasite) is the direct CAUSE of an infection, a risk factor is anything that makes it easier for that pathogen to enter your body or overcome your immune system! Almost all major risk factors for communicable diseases in Ghana can be controlled or reduced.

2. Analysing the Four Groups of Communicable Disease Risk Factors

Scientists and public health officers group communicable disease risk factors into four practical categories:

Risk Factor Category	Specific High-Risk Conditions	Diseases Triggered & Why	How to Reduce the Risk
1. Environmental & Sanitation Risks	Stagnant gutters/puddles, open defecation, overflowing refuse, unventilated crowded bedrooms	Malaria (mosquito breeding), Cholera/Typhoid/Hepatitis A (flies & faecal water contamination), TB (airborne spread in stuffy rooms)	Desilt drains, use clean toilets, cover rubbish, open windows for cross-ventilation
2. Personal & Behavioural Risks	Not washing hands with soap after toilet/before meals; sharing blades/clippers/ needles; unprotected sex; refusing ITNs	Cholera/Hepatitis A (dirty hands), Hepatitis B & HIV (blood/body fluids), Malaria (night mosquito bites)	Wash hands with soap under running water; use personal sterilised blades; sleep under ITN
3. Biological & Immune Risks	Missed vaccinations; malnutrition (lack of protein, Vitamin A/C/Zinc); infancy, old age, pregnancy, or untreated HIV	Severe Measles, Hepatitis B, TB, severe Malaria (weakened antibodies & white blood cells cannot fight germs quickly)	Complete immunisation schedule (BCG, HepB, Measles); eat balanced local meals (beans, fish, greens, fruits)
4. Healthcare & Economic Risks	Self-medicating or buying expired/incomplete drugs; delaying clinic testing; poverty/overcrowding; unscreened blood	Drug-resistant TB/Malaria; ongoing spread from untreated sick people; HBV/HIV via unsafe procedures	Visit certified CHPS/clinics promptly; finish full prescriptions; use screened blood & sterile single-use syringes

3. From Risk Analysis to Community Action

Analysing risk factors empowers communities to fix root causes rather than only buying medicine after falling ill. For instance, if a boarding school sees recurring diarrhoea and cough outbreaks, risk analysis checks three points immediately: (1) Are Veronica buckets with soap functioning outside toilets and dining halls? (2) Are drinking water tanks clean and covered? (3) Are dormitory windows kept open for fresh airflow rather than sealed shut?

Guided Risk-Audit Case Study: Investigating Outbreaks in Town X

Aim: To identify hidden environmental, behavioural, and immune risk factors in a scenario and recommend targeted prevention.

Materials: Case vignette below, notebook.

Procedure / Steps:

1. Read Scenario: In Town X, three linked health complaints increased last month: (a) Teenagers at an unhygienic barbering corner developed Hepatitis B; (b) Children drinking from an uncovered shallow well near a pit latrine got cholera; (c) Learners sleeping without bed nets beside choked gutters suffered frequent malaria.

2. For (a), (b), and (c), identify the specific risk factor category and exact risky practice, then prescribe two risk-reduction changes for each.

Key Observations / Findings:

• (a) Behavioural/healthcare risk: reusing unsterilised razor blades/clippers → Switch to single-use new blade per customer, sterilise clippers with spirit/UV steriliser, and get Hepatitis B vaccination.

• (b) Environmental/sanitation risk: faecal seepage from pit latrine into uncovered well → Boil/chlorinate drinking water, seal well apron, relocate/upgrade toilets safely, and enforce soap handwashing.

• (c) Environmental + behavioural risk: choked stagnant gutters + sleeping without ITNs → Community drain clearing + nightly ITN use.

Scientific Conclusion: Identifying and eliminating underlying risk factors stops multiple communicable diseases before anyone gets infected.

Common Misconceptions & Corrections (Avoid These Exam Traps!)

• Mistake: Thinking a 'risk factor' guarantees you will automatically catch a disease—or that having no risk factor makes risky habits harmless. Correction: Risk factors change PROBABILITY; wearing a helmet or washing hands lowers risk sharply, while sharing blades raises infection risk sharply.

• Mistake: Blaming illnesses strictly on 'bad luck' or witches instead of modifiable risk factors. Correction: Communicable diseases spread through observable physical, biological, and behavioural risk factors that individuals and communities can actively change.

Lesson Summary — Key Points to Remember

✓ A risk factor is any environmental, behavioural, biological, or healthcare condition that increases susceptibility to or transmission of disease.

✓ Key risk factors include poor sanitation/stagnant water, crowded poorly ventilated rooms, unwashed hands, sharing unsterilised sharp objects, missed vaccines, malnutrition, and delayed/incomplete medical treatment.

✓ Reducing modifiable risk factors through clean water, handwashing, sterilised sharps, vaccination, balanced nutrition, ventilation, and ITNs prevents outbreaks effectively.

Practice Exercise 23 (B8.5.2.1.2)

Part A: Multiple-Choice Questions (Choose the letter A–D of the best option)

1. Which of the following best defines a 'risk factor' for a communicable disease?
A. The exact drug used to cure an illness B. Any condition or behaviour that increases the chance of catching or spreading the disease C. An antibody made after vaccination D. A type of microscope used in hospitals

2. Crowded bedrooms with tightly shut windows significantly increase the risk of catching:
A. Tuberculosis (TB) B. Hepatitis B C. Cholera D. Tooth decay

3. How does childhood malnutrition act as a biological risk factor for communicable diseases?
A. It strengthens tooth enamel B. It weakens the body's immune system, reducing antibodies and white blood cell defences C. It kills malaria parasites directly D. It cleans stagnant gutters

4. Using a brand-new disposable razor blade for every client at a barbering salon directly removes a risk factor for:
A. Cholera and typhoid B. Hepatitis B and HIV C. Malaria and asthma D. Tuberculosis and cholera

Part B: Short Structured Questions (Write clear scientific answers)

5. Explain the difference between a 'pathogen' and a 'risk factor', using Malaria or Cholera as an example.

6. List one specific risk factor under each heading: (a) Environmental, (b) Behavioural, (c) Biological/Immune, (d) Healthcare.

7. Analyse why heavy flooding during the rainy season in a town with pit latrines increases the risk of BOTH Cholera and Malaria.

8. Suggest three practical steps a Basic 8 student can take personally this term to lower their risk of communicable diseases.

Complete Answer Key & Explanations

Solutions for All 23 Lessons (92 Multiple-Choice + 92 Short Structured = 184 Questions)

Remember: Attempt every question independently before checking your answers here. Use explanations to understand WHY each correct option works.

Lesson 1 (B8.1.1.1.1): Types of Mixtures and Their Characteristics

Part A (MCQs):

1. C — Sugar dissolves completely in water to form a uniform solution throughout.

2. B — Water is the dissolving medium present in larger quantity, so it is the solvent.

3. C — Groundnut paste does not dissolve completely; its visible particles remain suspended briefly and settle out on standing.

4. B — Colloidal particles are large enough to scatter light (Tyndall effect) without settling out.

Part B (Short Structured Model Answers):

5. Solids: gari, bottle tops, pebbles, sugar, powder. Liquids: oil, vinegar, water.

6. 1) A solute is the substance that dissolves (e.g., salt), whereas a solvent is the substance that dissolves the solute (e.g., water). 2) The solute is usually present in a smaller proportion and changes state into solution, whereas the solvent is usually in a larger proportion and retains its physical state.

7. In sand and sugar, the brown/grey sand grains and white sugar crystals remain visibly distinct and unevenly distributed (heterogeneous). In a salt solution, salt particles spread evenly at the molecular/ionic level throughout the water so every drop looks uniform (homogeneous).

8. In a darkened room, shine a narrow beam of torchlight through the side of each glass. The beam passes through the sugar solution invisibly (no Tyndall effect), whereas in the dilute milk colloid the light beam becomes clearly visible inside the liquid because the colloidal particles scatter light.

Lesson 2 (B8.1.1.1.2): Scientific Methods of Separating Mixtures

Part A (MCQs):

1. B — Water dissolves the salt while sand remains insoluble; filtering removes the sand, and evaporating the filtrate recovers the salt.

2. D — The solid trapped by the filter paper is the residue; the clear liquid passing through is the filtrate.

3. B — Oil and water are immiscible liquids with different densities; a separating funnel lets the denser bottom water layer drain out first.

4. C — Cold water flowing through the condenser jacket cools hot water vapour so it condenses into liquid distillate.

Part B (Short Structured Model Answers):

5. (a) Sieving (or handpicking for large stones), (b) Filtration (or decantation after settling), (c) Magnetic separation, (d) Simple distillation.

6. Because dissolved sugar breaks down into tiny molecules that pass straight through the pores of filter paper together with the water.

7. Entering from the bottom fills the condenser jacket completely and ensures cool water flows upward against the descending hot vapour (counter-current cooling), giving maximum condensation efficiency.

8. 1. Aim (purpose of experiment), 2. Apparatus / Materials, 3. Method / Procedure (with a labelled diagram), 4. Observations / Results, and 5. Conclusion.

Lesson 3 (B8.2.1.1.1): The Process of the Carbon Cycle

Part A (MCQs):

1. C — Green plants absorb atmospheric CO_2 during photosynthesis to manufacture glucose and plant tissue.

2. B — Decomposers break down dead organic matter and release CO_2 through their own respiration.

3. B — Respiration, burning (combustion), and decay all release CO_2 back into the air.

4. C — Carbon is conserved and recycled continuously through living organisms, soils, fuels, and the atmosphere.

Part B (Short Structured Model Answers):

5. Photosynthesis, respiration, burning (combustion), and decay (decomposition).

6. (a) In photosynthesis, green plants absorb CO_2 from the air and combine it with water using sunlight to form carbon-rich food (glucose/starch). (b) In combustion (burning), fuels like firewood, charcoal, coal, or petrol react with oxygen and release their stored carbon back into the air as CO_2.

7. Atmospheric CO_2 --[Photosynthesis]--> Grass/Maize leaves --[Feeding]--> Goat --[Respiration]--> Atmospheric CO_2.

8. Decomposers break down dead plants, dead animals, and animal wastes; without them, carbon would remain locked up in dead organic matter instead of returning to the air as CO_2 and to the soil.

Lesson 4 (B8.2.1.1.2): Environmental Role of the Carbon Cycle

Part A (MCQs):

1. B — Photosynthesis consumes CO_2 and releases O_2, while respiration consumes O_2 and releases CO_2.

2. C — Only producers (green plants and algae) trap atmospheric CO_2 during photosynthesis to build carbohydrates.

3. B — Carbon (IV) Oxide (CO_2) and methane (CH_4) both contain carbon and trap heat in the atmosphere.

4. B — Combustion adds large volumes of CO_2 to the air and removes plants needed for photosynthesis.

Part B (Short Structured Model Answers):

5. During daylight, green plants absorb CO_2 and release O_2 through photosynthesis. At the same time (day and night), plants carry out respiration, taking in O_2 and releasing CO_2.

6. Air (CO_2) → Maize plant (producer fixes carbon via photosynthesis) → Grasshopper (primary consumer eats maize) → Weaver bird / Guinea fowl (secondary consumer eats grasshopper) → Decomposers (break down dead bird/droppings and respire CO_2 back into the air).

7. Greenhouse gases such as CO_2 and methane are part of the carbon cycle. Normally, photosynthesis and oceans absorb roughly as much CO_2 as respiration and natural decay release, keeping Earth's heat balance stable. When burning fossil fuels and deforestation add extra CO_2 faster than it is absorbed, greenhouse gases build up and trap excess heat.

8. 1) Plant and care for trees and gardens on the school compound to increase CO_2 absorption. 2) Compost dry leaves and organic food scraps instead of burning them, avoiding unnecessary CO_2 emissions.

Lesson 5 (B8.2.2.1.1): Life Cycle and Economic Importance of the Anopheles Mosquito

Part A (MCQs):

1. C — Eggs hatch into feeding larvae, which become non-feeding pupae before emerging as winged adults.

2. B — Because Anopheles larvae lack a long breathing siphon, they lie parallel to the surface film.

3. B — The pupa ('tumbler') does not feed while its body reorganises into an adult.

4. B — Female mosquitoes require proteins from blood meals to mature their eggs.

Part B (Short Structured Model Answers):

5. The four stages are Egg, Larva, Pupa, and Adult. The Egg, Larva, and Pupa stages live in water (aquatic), while the Adult lives on land/air.

6. 1) Larval stage: Anopheles larvae lack a long siphon and lie parallel to the water surface, whereas Culex larvae hang at an angle via a breathing siphon. 2) Adult stage: Anopheles adults rest at an inclined ~45° angle with head down, whereas Culex adults rest with bodies parallel to the surface.

7. Negative economic impacts: (1) High family and national healthcare expenditure on malaria diagnosis, medicine, and control; (2) Loss of school attendance and worker/farmer productivity due to illness. Ecological role: Mosquito larvae serve as food for small fish and aquatic insects in ponds.

8. To prevent newly emerged adult mosquitoes from escaping into the classroom and biting learners or spreading malaria.

Lesson 6 (B8.2.2.1.2): Impact of the Anopheles Mosquito and Malaria Control in Ghana

Part A (MCQs):

1. B — Plasmodium is the protozoan parasite carried by the female Anopheles mosquito vector.

2. B — Larvae and pupae must surface to breathe air; an oil film seals the surface and suffocates them.

3. B — ITNs physically block night-biting female Anopheles mosquitoes and kill those that land on the netting.

4. A — Larvivorous fish eat aquatic mosquito larvae and pupae before they can mature into adults.

Part B (Short Structured Model Answers):

5. A vector is an organism that transmits a disease-causing pathogen from one host to another. For malaria, the vector is the female Anopheles mosquito and the causative parasite is Plasmodium.

6. High fever, chills/shivering, severe headache, body/joint pains, nausea/vomiting, extreme tiredness, and anaemia (any four).

7. Anopheles eggs, larvae, and pupae must live in still water to develop. Removing stagnant water breaks the life cycle at the aquatic stages so fewer new adult mosquitoes are produced.

8. 1) Environmental: Organise regular communal labour to desilt gutters, drain puddles, cover water barrels, and remove discarded tins/tyres. 2) Personal protection: Encourage every household to screen windows and sleep under ITNs. 3) Health-seeking: Encourage prompt clinic testing (RDT) and completing prescribed ACT medication.

Lesson 7 (B8.2.3.1.1): Types of Seed Beds and Matching Crops

Part A (MCQs):

1. B — A sunken bed sits below ground level so it captures and retains watering moisture around plant roots.

2. B — Raised beds prevent waterlogging in wet seasons and keep root zones well aerated.

3. C — Yam requires deep, loose, well-drained soil provided by mounds so large underground tubers can develop.

4. B — All seed beds start with clearing debris and tilling soil so roots can grow and absorb nutrients.

Part B (Short Structured Model Answers):

5. Flat bed, raised bed, sunken bed, ridges (and furrows), and mounds.

6. (a) A raised bed is built 15–30 cm above ground level, whereas a sunken bed is dug 10–20 cm below ground level. (b) A raised bed drains excess water quickly to prevent waterlogging, whereas a sunken bed traps and conserves moisture for dry periods.

7. (a) Raised bed, (b) Ridges, (c) Mounds, (d) Sunken bed.

8. Stepping on the bed compacts the soil, closing air spaces (reducing aeration), slowing water infiltration, and making it harder for young roots to penetrate.

Lesson 8 (B8.2.3.1.2): Planting Different Crop Parts on Seed Beds

Part A (MCQs):

1. B — Cassava is propagated using mature stem cuttings ('sticks') bearing healthy buds/nodes.
2. B — Transplanting in cool late afternoon prevents rapid transpiration and helps damaged root hairs recover overnight.
3. C — Bryophyllum (life plant) reproduces vegetatively via tiny buds formed along its leaf margins.
4. B — Capping/mulching shades the soil, keeps moisture inside, and prevents heat damage to the sprouting yam sett.

Part B (Short Structured Model Answers):

5. 1) Seeds (e.g., maize/okra), 2) Seedlings (e.g., tomato/pepper), 3) Stem cuttings (e.g., cassava/sweet potato), 4) Leaves (e.g., Bryophyllum), 5) Roots/underground storage organs (e.g., yam/cocoyam/ginger).
6. 1) Ensure the buds/nodes point upward (right-side up) when inserting into the soil. 2) Select healthy, disease-free mature stems with at least 3–4 nodes and bury 2–3 nodes in moist soil.
7. Small seeds contain very little stored food (endosperm/cotyledon reserves); if planted too deeply, the young shoot runs out of food before reaching sunlight above the soil.
8. 1) Softens the soil so delicate root hairs are not torn off when lifted. 2) Keeps a moist ball of soil clinging to the roots to protect them during transfer.

Lesson 9 (B8.2.3.2.1): Comparing Crop Growth, Size, and Flowering Across Seed Beds

Part A (MCQs):

1. C — The raised bed averaged 16 flowers per plant, compared with 10 on the flat bed and 5 on the sunken bed.
2. B — Waterlogging starves roots of oxygen, preventing active mineral uptake and stunting growth.
3. B — To test the effect of seed bed type fairly, crop variety, initial seedling size, and spacing must be kept constant.
4. C — 44 cm (Sunken Bed) − 25 cm (Raised Bed) = 19 cm.

Part B (Short Structured Model Answers):

5. 1) Plant height (cm), 2) Leaf or stem size, 3) Number of flowers per plant, and 4) Number of fruits per plant.
6. 1) Excess rainwater drains off quickly, preventing waterlogging and root rot. 2) Loose, raised soil holds more air pockets (oxygen), helping roots respire and absorb nutrients efficiently.
7. Fruits develop from pollinated flowers; fewer healthy flowers mean fewer opportunities for pollination and fruit set.
8. A table displays the exact measured values for every bed and week, while a bar chart lets classmates visually compare heights, flowers, and yields across beds immediately.

Lesson 10 (B8.2.4.1.1): Types of Animal Feed, Nutrients, and Feed Labels

Part A (MCQs):

1. C — Ruminants possess a four-chambered stomach containing microbes that break down fibrous cellulose in roughages.
2. B — Soybean meal and fishmeal supply high levels of protein and essential amino acids for growth.
3. B — Eggshells are largely calcium carbonate, so laying hens require high dietary calcium.
4. B — They are rich in digestible starch/carbohydrates and low in fibre compared with grass.

Part B (Short Structured Model Answers):

5. 1) Roughages are high in crude fibre and bulky (e.g., grass, hay), whereas concentrates are low in fibre and dense in energy or protein (e.g., maize, soybean meal). 2) Roughages are digested mainly by ruminants, whereas concentrates are key feeds for monogastrics like poultry and pigs.
6. (a) Fresh pasture grass/hay plus dairy concentrate, (b) Broiler starter mash (~20–23% protein), (c) Layer mash with oyster shell/high calcium, (d) Fresh/wilted greens (forage) plus pellets.
7. Carbohydrates (maize/cassava peels), Proteins (groundnut cake/soybean meal/fishmeal), Minerals (bone meal/oyster shell/salt lick), Vitamins (fresh green leaves/vitamin premix).
8. She should buy Label B because laying hens need high calcium (~3.8%) to produce strong eggshells and moderate protein (~16.5%); excess protein in Label A wastes money and lacks enough calcium for eggs.

Lesson 11 (B8.2.4.2.1): Importance of Water and Feed for Animal Growth and Reproduction

Part A (MCQs):

1. A — Eggs are roughly two-thirds to three-quarters water, and water is necessary for feed digestion and nutrient transport.
2. B — Evaporating water from skin (sweat) or respiratory surfaces (panting) removes excess heat.
3. B — Animals reduce feed intake immediately because dry food requires water for swallowing and digestion.
4. B — Proteins supply amino acids that build new muscle, blood, organs, and milk.

Part B (Short Structured Model Answers):

5. 1) Aids digestion and softens feed, 2) Transports dissolved nutrients in blood, 3) Regulates body temperature (panting/sweating), 4) Removes toxic wastes in urine (and forms major part of milk and eggs).
6. 1) Balanced energy, proteins, and minerals help young animals reach breeding maturity normally and conceive successfully. 2) Adequate nutrients allow pregnant females to nourish growing embryos/foetuses and produce sufficient milk after birth.
7. 1) Reduced appetite / refusal to eat dry feed, 2) Sunken eyes and dry mouth/skin, 3) Intense panting or restlessness followed by extreme weakness, 4) Constipation, concentrated dark urine, stunted growth, and sharp drop in milk/egg output.
8. Scrubbing removes algae, droppings, and harmful microbes that cause disease, while shading keeps the water cool so animals drink willingly without heat stress.

Lesson 12 (B8.3.1.1.1): External and Internal Parts of a Mammalian Tooth

Part A (MCQs):

1. C — Enamel is densely packed with calcium minerals, making it harder than bone or dentine so it resists chewing wear.

2. B — The neck is the narrowed transition zone where crown meets root at the gum margin.

3. B — The pulp cavity contains living nerves that sense temperature and pain.

4. B — Cementum surrounds the root and attaches periodontal ligaments to the jawbone socket.

Part B (Short Structured Model Answers):

5. 1) Crown: the visible upper portion above the gum. 2) Neck: the narrow waist right at the gumline. 3) Root: the lower portion embedded inside the jawbone socket.

6. (a) Enamel protects the crown against wear, biting forces, and weak acids. (b) Dentine forms the main body of the tooth and absorbs chewing shocks. (c) Blood vessels supply oxygen and nutrients to keep the living tissues of the tooth healthy.

7. The gum forms a tight protective collar around the neck of the tooth, sealing out food debris and harmful bacteria from reaching the cementum, periodontal fibres, and jawbone.

8. Enamel forms the outer covering of the crown (above the gum), whereas cementum forms the outer covering of the root (below the gum inside the jawbone).

Lesson 13 (B8.3.1.1.2): Types of Human Teeth and Their Functions

Part A (MCQs):

1. D — An adult has 8 incisors + 4 canines + 8 premolars + 12 molars = 32 permanent teeth.

2. D — Incisors sit at the front of the jaw with flat sharp edges ideal for biting and cutting.

3. B — Multiple roots anchor molars firmly under strong jaw pressure while wide cusped crowns grind food effectively.

4. C — Milk dentition consists of 8 incisors, 4 canines, and 8 molars—no premolars.

Part B (Short Structured Model Answers):

5. Incisors (8), Canines (4), Premolars (8), and Molars (12).

6. 1) A canine has a narrow, conical pointed crown, whereas a molar has a wide, flat crown with 4–5 cusps. 2) A canine has 1 long single root, whereas a molar has 2–3 roots.

7. Grinding food into tiny pieces greatly increases its total surface area, allowing digestive enzymes in saliva, stomach juice, and intestinal juice to break down nutrients faster and more completely.

8. (a) Incisors, (b) Canines, (c) Premolars and Molars.

Lesson 14 (B8.3.1.1.3): Causes and Prevention of Tooth Decay and Gum Disease

Part A (MCQs):

1. B — Mouth bacteria in plaque ferment sugars into weak acids that demineralise tooth enamel.

2. B — During the day saliva helps wash and neutralise acids; at night low saliva lets plaque acid attack enamel continuously.

3. B — Unremoved plaque mineralises into hard tartar, harbouring bacteria that inflame the gums.

4. B — Fluoride reacts with enamel minerals to make the surface harder and more acid-resistant.

Part B (Short Structured Model Answers):

5. 1) Sugar sticks to teeth and mixes with bacteria to form plaque. 2) Bacteria digest the sugar and release weak acids. 3) Acids dissolve the enamel to create a cavity. 4) Decay spreads through dentine into the pulp cavity, irritating living nerves and causing toothache.

6. Plaque is a soft, sticky colourless film of bacteria and food debris that can be brushed away daily; tartar is plaque that has hardened with saliva minerals over time and usually requires a dentist to scrape/scale off.

7. 1) Brush twice daily (morning and bedtime) for 2 minutes with fluoride toothpaste. 2) Rinse or clean between teeth after sugary snacks and limit sweets between meals. 3) Never use teeth to crack hard objects like bottle tops or bones. 4) Eat calcium- and vitamin-rich foods and visit a dentist regularly.

8. Vigorous horizontal scrubbing wears thin the enamel at the neck of the tooth and injures/recedes delicate gum tissue, whereas gentle circular or sweeping strokes clear plaque safely from the gum crevice.

Lesson 15 (B8.4.1.1.1): Energy Conversion and Everyday Transformation Chains

Part A (MCQs):

1. B — Chemical energy in the battery produces electrical energy in the circuits, which the speaker converts into sound.

2. B — Plants capture solar light energy and store it as chemical energy inside glucose and starch.

3. C — As the fruit drops and speeds up, stored potential energy turns into kinetic energy of motion.

4. C — Total energy in any closed system remains constant; it only changes forms.

Part B (Short Structured Model Answers):

5. Energy conversion is the change of energy from one form into another. Kitchen example: burning charcoal converts chemical energy in charcoal into heat (for cooking) and light energy (glowing embers).

6. (a) Electrical → Heat; (b) Chemical (food) → Kinetic (pedalling) + Gravitational Potential (climbing hill) + Heat (body warmth); (c) Light (solar) → Electrical.

7. Although the fan's useful conversion is Electrical → Kinetic energy, some electrical energy is also converted into unwanted Heat (thermal) energy due to friction and electrical resistance.

8. No. Air resistance and pivot friction gradually convert the pendulum's mechanical energy into thermal (heat) energy in the surrounding air and pivot; no energy is destroyed.

Lesson 16 (B8.4.1.1.2): Importance of Energy Conversion and Hydroelectric Power

Part A (MCQs):

1. B — Falling water strikes the turbine blades, converting water kinetic energy into rotational mechanical energy to drive the generator.

2. B — Elevated reservoir water (potential) rushes down to spin the turbine (kinetic), driving the generator (electrical).

3. B — A wind turbine uses rotating blades turned by wind to drive an electrical generator.

4. B — Less water at a lower head stores less gravitational potential energy to drive the turbines.

Part B (Short Structured Model Answers):

5. 1) Water stored high behind the dam holds gravitational potential energy. 2) Water flows down penstock pipes, converting potential energy into kinetic energy. 3) Rushing water spins the turbine blades (mechanical kinetic energy). 4) The turbine turns a generator to produce electrical energy, which travels via transformers and power lines to homes.

6. 1) Using solar photovoltaic (PV) panels to convert sunlight into electrical energy for homes, streetlights, and water pumps. 2) Using solar dryers or solar water heaters to convert solar radiation into heat energy for drying cocoa/fish or warming water.

7. 1) Converts chemical/electrical energy into heat for cooking and preserving food (refrigeration). 2) Converts electrical energy into light for evening study and safety. 3) Converts fuel/electrical energy into kinetic energy for transport, milling grain, and pumping water.

8. Because electricity can be transmitted quickly over long distances through cables and converted cleanly at the user's end into light, heat, sound, cooling, or mechanical motion at the flick of a switch.

Lesson 17 (B8.4.1.2.1): Renewable and Non-Renewable Energy Sources

Part A (MCQs):

1. B — Hydro, wind, and biogas are naturally replenished; coal, natural gas, and crude oil are finite fossil fuels.

2. C — Anaerobic bacteria convert organic waste into methane (CH_4) mixed with some CO_2.

3. B — Fossil fuels form over geological timescales, so human consumption rapidly exhausts existing deposits.

4. C — Biogas formation is an anaerobic process occurring inside a sealed oxygen-free chamber.

Part B (Short Structured Model Answers):

5. A renewable energy source is naturally replenished quickly within a human lifetime (e.g., solar, wind, hydro, biogas). A non-renewable energy source is finite and takes millions of years to form, depleting as used (e.g., coal, crude oil, natural gas).

6. Renewable: wind, biogas, hydro, solar. Non-renewable: crude oil, coal, natural gas.

7. Organic waste (animal dung/food scraps) is mixed with water in an inlet tank and fed into a sealed underground digester without oxygen. Anaerobic bacteria break down the waste, producing methane biogas that collects in the dome and is piped to stoves or generators for cooking and electricity, while digested slurry becomes farm fertiliser.

8. 1) Sunlight is free and renewable with no ongoing petrol cost or supply shortages. 2) Solar panels operate silently without polluting exhaust smoke or carbon monoxide.

Lesson 18 (B8.4.1.2.2): Sustainable Management of Renewable Energy Sources

Part A (MCQs):

1. B — Plant roots hold soil in place, preventing mud (silt) from washing into and filling the reservoir.

2. B — Removing accumulated Harmattan dust lets full sunlight reach the photovoltaic cells.

3. C — Cutting trees without replanting depletes the forest resource faster than it can regenerate.

4. B — Batteries contain hazardous heavy metals and corrosive acids that must be safely recycled.

Part B (Short Structured Model Answers):

5. 1) Smart assessment and siting; 2) Sustainable harvesting and ecosystem/catchment protection (e.g., replanting trees); 3) Routine inspection and preventive maintenance (cleaning solar panels, checking biogas pipes); 4) Efficient storage/ conservation and safe recycling of spent components like batteries.

6. Challenge 1: Intermittency (no generation at night/cloudy days) → Solution: Use properly sized battery storage with charge controllers. Challenge 2: Dust build-up reducing output (or high battery replacement cost) → Solution: Clean panels regularly and use energy-saving LEDs to extend battery life.

7. Challenges: (1) Drought lowering river water levels; (2) Siltation (mud filling the reservoir due to soil erosion). Community solution: Plant and protect trees along riverbanks (avoiding bush burning and farming right on river edges) to reduce soil erosion and siltation.

8. Plastics are non-biodegradable, so anaerobic bacteria cannot break them down; they clog inlet/outlet pipes and reduce the digester's usable volume for organic waste.

Lesson 19 (B8.4.1.3.1): Heat and Temperature: Differences and Relationship

Part A (MCQs):

1. B — Heat is energy (SI unit: Joule, J) and Temperature has SI unit Kelvin (K), though °C is commonly used.

2. C — Temperature (65 °C) is independent of mass, whereas total heat energy increases with mass.

3. B — Heat flows spontaneously from a hotter body (90 °C) to a cooler body (25 °C).

4. B — A thermometer measures temperature; a calorimeter is used in heat measurements.

Part B (Short Structured Model Answers):

5. 1) Meaning: Heat is thermal energy in transit, whereas Temperature is the degree of hotness/coldness. 2) SI Unit: Heat is measured in Joules (J), whereas Temperature is measured in Kelvin (K) [or °C]. 3) Dependence on mass: Heat depends on the mass of the substance, whereas Temperature is independent of mass (or Measuring instrument: Calorimeter vs. Thermometer).

6. As the metal roof absorbs radiant heat energy from the sun, its particles vibrate faster, causing its temperature to rise noticeably.

7. Because the spark's mass is minuscule, its total heat energy (in Joules) is very small despite its high temperature, so it cools almost instantly without transferring much heat—unlike a large mass of boiling water.

8. Heat flows continuously from the hotter sobolo to the cooler surrounding air and counter until both reach the same room temperature (~29 °C), a state called thermal equilibrium.

Lesson 20 (B8.5.1.1.1): Sustainable Waste Management Practices (Reduce, Reuse, Recycle, Compost)

Part A (MCQs):

1. C — Preventing waste from being created in the first place (Reduce) saves the most energy, materials, and cost.

2. C — Reusing keeps the original intact container in service without melting or reprocessing it.

3. B — Dry-cell batteries contain hazardous chemicals/metals that contaminate soil and food crops.

4. B — Only biodegradable organic matter decomposes naturally via microbes into nutrient-rich compost.

Part B (Short Structured Model Answers):

5. 1) Reduce (prevent waste), 2) Reuse, 3) Recycle / Upcycle, 4) Compost / Recover energy (e.g., biogas), 5) Safe disposal in engineered facilities.

6. Reusing keeps the bottle intact for another use (e.g., washing it to store cooking oil or palm kernel oil). Recycling means crushing and melting waste glass in a factory to mould brand-new glass bottles or jars.

7. 1) Releases toxic fumes and dioxins that cause respiratory problems (asthma, coughing) and air pollution. 2) Releases Carbon (IV) Oxide greenhouse gas and leaves toxic melted plastic residue in the soil.

8. Sorting keeps wet food/leaf waste clean for free compost manure (saving fertiliser costs) and keeps clean plastic bottles/ paper unsoiled so recyclers can collect or buy them.

Lesson 21 (B8.5.1.1.2): Designing and Evaluating a Community Waste Management Plan

Part A (MCQs):

1. B — A waste audit reveals where litter builds up and what waste streams exist so bins and solutions match actual needs.

2. B — Distinct colours with clear icons/pictures let everyone tell instantly which bin takes food waste versus plastics.

3. B — Dedicated vendor collection bags and reusable bowls keep plastics clean for recycling and reduce waste generation.

4. B — Clear drains, high sorting accuracy, and active recovery (compost/recyclables) measure tangible success.

Part B (Short Structured Model Answers):

5. 1) Survey/Audit waste types and litter hotspots; 2) Set clear measurable goals; 3) Set up sorted bin stations, compost bay, and recycling collection; 4) Assign student/vendor duties and run awareness education; 5) Monitor weekly and evaluate results to fix weak spots.

6. 1) Place a large blue Plastic Bin right beside the Organic bin (bins separated far apart invite wrong-bin dumping) with large picture stickers of sachets. 2) Announce quick assembly reminders and assign prefects/monitors near the canteen during break for two weeks to guide sorting.

7. 1) Percentage of litter-free drains and hotspots; 2) Sorting accuracy inside colour-coded bins; 3) Mass/bags of recyclables collected or compost produced (and absence of open rubbish burning).

8. Lids keep rainwater out (preventing foul wet smells and mosquito breeding) and stop wind, goats, or dogs from scattering rubbish, while raised stands prevent bottom rusting.

Lesson 22 (B8.5.2.1.1): Common Communicable Diseases: Hepatitis, HIV/AIDS, Cholera & Tuberculosis

Part A (MCQs):

1. C — Hepatitis causes inflammation and cellular damage in the liver, producing jaundice.

2. B — Damaged liver cells fail to clear bilirubin pigment, causing yellow discolouration called jaundice.

3. B — HBV is a blood- and body-fluid-borne virus; it is not transmitted through casual touch.

4. B — Cholera bacteria produce a toxin causing extreme watery diarrhoea and rapid dehydration.

Part B (Short Structured Model Answers):

5. A communicable disease is an infectious illness caused by a pathogen that can spread from one organism to another. Examples: (a) Tuberculosis, (b) Cholera / Hepatitis A, (c) Hepatitis B / HIV.

6. Symptoms: jaundice (yellow eyes/skin), dark urine, extreme fatigue, nausea/loss of appetite, upper-right abdominal pain (any three). Long-term effects: liver scarring (cirrhosis), liver failure, and liver cancer.

7. 1) Get tested and receive the full Hepatitis B vaccination series. 2) Never share razor blades, barber clippers (unless sterilised), needles, earrings/piercing tools, or toothbrushes. 3) Avoid unsafe blood exposure and practise sexual abstinence/safe practices.

8. Stopping antibiotics early leaves hardy TB bacteria alive in the lungs, causing the disease to return in a dangerous drug-resistant form that is much harder to cure.

Lesson 23 (B8.5.2.1.2): Analysing Risk Factors of Communicable Diseases

Part A (MCQs):

1. B — Risk factors are conditions (such as dirty water, overcrowding, or missed vaccines) that raise infection probability.

2. A — TB spreads via airborne droplets; stagnant indoor air in crowded rooms concentrates infectious droplets.

3. B — Protein and micronutrient deficiencies impair antibody production and immune cell response.

4. B — Fresh single-use blades eliminate blood-to-blood transmission of HBV and HIV.

Part B (Short Structured Model Answers):

5. A pathogen is the living germ that directly causes the disease (e.g., Vibrio cholerae bacterium causes cholera). A risk factor is a condition or habit that helps the pathogen infect people (e.g., drinking untreated water or failing to wash hands with soap after using the toilet).

6. (a) Stagnant choked gutters / open defecation; (b) Sharing unsterilised razor blades / skipping handwashing; (c) Missed vaccinations / malnutrition; (d) Stopping antibiotics early / using unscreened blood.

7. Floodwater overflows pit latrines and washes faecal bacteria (Vibrio cholerae) into drinking wells and food sources (increasing cholera risk), while remaining puddles after floods create abundant stagnant breeding pools for Anopheles mosquitoes (increasing malaria risk).

8. 1) Wash hands thoroughly with soap under running water after toilet use and before eating; 2) Sleep every night under an Insecticide-Treated Net (ITN); 3) Never share razor blades, toothbrushes, or needles, and verify Hepatitis B vaccination status.

Alphabetical Glossary of First Term Scientific Terms

90 Core Terms & Definitions Aligned to Lessons 1–23

Scientific Term	Definition	Reference
Air composition balance	Maintaining stable proportions of atmospheric Oxygen (~21%) and Carbon (IV) Oxide (~0.04%) via photosynthesis and respiration.	Lesson 4
Alloy	A solid homogeneous mixture of two or more metals (or a metal and non-metal) melted together, such as brass or steel.	Lesson 1
Anaerobic digestion	Breakdown of organic waste by bacteria in the absence of oxygen inside a biogas digester to release methane.	Lesson 17
Biogas	A renewable fuel gas (primarily methane, CH_4, plus CO_2) produced by anaerobic microbial digestion of dung and food waste.	Lesson 17
Calorimeter	An insulated container used in science experiments to measure heat energy transferred.	Lesson 19
Canines	Pointed single-rooted teeth positioned beside the incisors for gripping and tearing food.	Lesson 13
Carbon cycle	The continuous natural circulation of carbon between air (CO_2), living organisms, soils, oceans, and fossil fuels.	Lesson 3
Catchment protection	Conserving vegetation along riverbanks to prevent soil erosion and siltation of reservoirs.	Lesson 18
Cementum	Hard bone-like tissue covering the root of a mammalian tooth to anchor it inside the jawbone socket.	Lesson 12
Combustion (Burning)	Rapid reaction of carbon-containing fuel with oxygen releasing heat, light, and Carbon (IV) Oxide.	Lesson 3
Communicable disease	An infectious disease caused by a pathogen that spreads from person to person, animal to person, or via contaminated food/water.	Lesson 22
Complete metamorphosis	A four-stage insect life cycle: Egg -> Larva -> Pupa -> Adult (as in Anopheles mosquito).	Lesson 5
Composting	Controlled biological decomposition of sorted organic peels and dry leaves into dark nutrient-rich humus fertiliser.	Lesson 20
Concentrate (Feed)	Low-fibre, nutrient-dense livestock feed rich in energy (maize, wheat bran) or protein (soybean meal, fishmeal).	Lesson 10
Crown	The exposed upper part of a tooth covered in enamel above the gumline.	Lesson 12
Decay (Decomposition)	Breakdown of dead organisms and waste by bacteria and fungi, returning CO_2 to the air and nutrients to soil.	Lesson 3
Dehydration	Harmful lack of body water causing reduced feed intake, weakness, heat stress, or death.	Lesson 11
Dentine	Yellowish bone-like living tissue forming the main bulk of a tooth beneath enamel and cementum.	Lesson 12
Distillate	Pure liquid collected after boiling and re-condensing vapour during distillation.	Lesson 2
Emulsion	A cloudy liquid-in-liquid mixture of tiny droplets of one immiscible liquid dispersed in another (e.g., oil shaken with water).	Lesson 1
Enamel	The non-living, calcium-rich outermost covering of the tooth crown—the hardest substance in the human body.	Lesson 12
Energy conversion	The transformation of energy from one form into one or more different forms.	Lesson 15
Evaporation (Separation)	Heating a solution so the solvent vaporises, leaving behind the dissolved solid solute.	Lesson 2
Filtrate	The clear liquid that passes through filter paper during filtration.	Lesson 2
Flat bed	A prepared garden planting plot levelled even with the surrounding ground surface.	Lesson 7
Fluoride	A protective mineral in toothpaste that strengthens tooth enamel against bacterial acids.	Lesson 14
Food chain	A linear pathway tracing how carbon compounds and food energy pass from producers through consumers.	Lesson 4
Fossil fuels	Non-renewable carbon-rich fuels (coal, crude oil, natural gas) formed over millions of years from buried ancient remains.	Lessons 3 & 17
Generator	A machine spun by a turbine that converts kinetic energy into electrical energy using magnets and coils.	Lesson 16
Greenhouse gas	An atmospheric gas (e.g., CO_2, methane) that absorbs outgoing heat from Earth, keeping the planet warm.	Lesson 4

Scientific Term	Definition	Reference
Gum disease (Gingivitis)	Swollen, tender, bleeding gums caused by plaque and tartar build-up along the gumline.	Lesson 14
Heat	Thermal energy transferred between bodies due to a temperature difference; SI unit is Joule (J).	Lesson 19
Hepatitis	Inflammation/damage of the liver caused chiefly by Hepatitis viruses (A, B, C), marked by jaundice and fatigue.	Lesson 22
Heterogeneous mixture	A non-uniform mixture whose separate parts remain visibly distinct (e.g., gari and sand, oil and water).	Lesson 1
Homogeneous mixture	A uniform mixture whose components are evenly mixed and appear as one phase throughout (e.g., salt solution).	Lesson 1
Hydroelectric power	Electricity generated by using elevated river water stored behind a dam to spin turbines and generators.	Lesson 16
Immiscible liquids	Two liquids (such as palm oil and water) that do not dissolve in each other and separate into layers.	Lessons 1 & 2
Immunity	The body's natural ability to recognise and defend itself against disease-causing pathogens.	Lesson 23
Incisors	Flat, chisel-edged front teeth used for biting and cutting pieces of food.	Lesson 13
Insecticide-Treated Net (ITN)	A mosquito bed net treated with safe insecticide to block and kill night-biting mosquitoes.	Lesson 6
Jaundice	Yellowish colouration of the whites of the eyes and skin resulting from liver dysfunction.	Lesson 22
Kinetic energy	Energy possessed by an object or water/air because it is moving.	Lessons 15 & 16
Larva (Wriggler)	The active, feeding aquatic stage of the mosquito after hatching; Anopheles larvae lie parallel to the water surface.	Lesson 5
Law of Conservation of Energy	Energy can neither be created nor destroyed; it only changes from one form to another.	Lesson 15
Malnutrition	Poor nutrition (deficiency in proteins, vitamins, or minerals) that weakens resistance to infections.	Lesson 23
Miscible liquids	Two liquids (such as ethanol and water) that mix evenly in all proportions to form one uniform liquid.	Lessons 1 & 2
Mixture	Two or more substances physically combined in any proportion, each retaining its own properties.	Lesson 1
Molars	Broad back teeth with 4–5 cusps and 2–3 roots specialised for heavy grinding and chewing.	Lesson 13
Monogastrics	Single-stomach animals (e.g., poultry, pigs) that digest concentrates better than coarse fibrous grass.	Lesson 10
Mound	A rounded heap of loose deep topsoil prepared especially for tuber crops like yam and cassava.	Lesson 7
Mulching	Covering topsoil around plants with dry grass or leaves to conserve moisture and suppress weeds.	Lesson 8
Neck (of tooth)	The slightly constricted region at the gumline where a tooth's crown meets its root.	Lesson 12
Non-renewable energy	Finite energy reserves (coal, crude oil, natural gas) that cannot be replaced within human lifetimes once used.	Lesson 17
Pathogen	A microscopic organism (virus, bacterium, protozoan, fungus) capable of causing disease.	Lesson 22
Penstock	A large steep pipe delivering water from a dam reservoir down to spin a hydroelectric turbine.	Lesson 16
Photosynthesis	Process in which green plants absorb CO_2 and water using sunlight to produce glucose and oxygen.	Lessons 3 & 4
Plaque	A sticky invisible film of mouth bacteria, saliva, and food sugars that forms on unbrushed teeth.	Lesson 14
Plasmodium	The single-celled protozoan parasite transmitted by female Anopheles mosquitoes that causes malaria.	Lesson 6
Potential energy	Stored energy due to elevated position (gravitational) or chemical composition.	Lessons 15 & 16
Premolars	Two-cusped teeth behind canines with 1–2 roots used for crushing and grinding food.	Lesson 13
Pulp cavity	Living central chamber inside a tooth containing nerves and blood vessels.	Lesson 12
Pupa (Tumbler)	The comma-shaped, non-feeding aquatic transformation stage between larva and adult mosquito.	Lesson 5

Scientific Term	Definition	Reference
Raised bed	A soil bed built 15–30 cm above ground level to boost water drainage and root aeration in wet conditions.	Lesson 7
Recycle	Reprocessing collected waste materials (plastics, metals, paper, glass) into raw materials for new goods.	Lesson 20
Reduce	Avoiding or minimising waste generation before items ever need disposal (top waste priority).	Lesson 20
Renewable energy	Energy from natural sources (sunlight, wind, flowing water, biogas) that replenish continuously.	Lesson 17
Residue	Solid particles trapped on filter paper during filtration.	Lesson 2
Respiration	Cellular breakdown of food with oxygen in plants, animals, and microbes releasing energy and CO_2.	Lessons 3 & 4
Reuse	Using an intact item again for the same or another purpose without industrial remanufacturing.	Lesson 20
Ridge and furrow	Long parallel raised soil ridges separated by drainage furrows between rows.	Lesson 7
Risk factor	Any environmental, behavioural, biological, or healthcare condition that raises the likelihood of disease.	Lesson 23
Root (of tooth)	The tapered lower part of a tooth anchored inside the jawbone socket.	Lesson 12
Roughage	High-fibre bulky plant feed (pasture grass, hay, crop stover) suitable mainly for ruminants and rabbits.	Lesson 10
Ruminants	Four-chambered stomach herbivores (cattle, sheep, goats) that chew cud and digest fibrous roughage.	Lesson 10
Seed bed	A tilled and shaped soil plot prepared specifically for sowing seeds or planting crop parts.	Lesson 7
Siltation	Accumulation of eroded soil and mud inside a river reservoir, reducing dam water capacity.	Lesson 18
Solution	A uniform (homogeneous) mixture formed when a solute dissolves completely in a solvent.	Lesson 1
Stem cutting	A segment of stem with buds/nodes planted to grow a whole new plant (e.g., cassava, sweet potato).	Lesson 8
Sunken bed	A bed dug 10–20 cm below ground level to capture and conserve moisture during dry seasons.	Lesson 7
Suspension	A cloudy heterogeneous mixture containing visible solid particles that eventually settle on standing.	Lesson 1
Sustainable management	Wisely operating, conserving, and renewing resources so they serve today's needs without harming the future.	Lessons 18 & 20
Tartar (Calculus)	Hardened plaque deposits along the gumline that irritate gums and promote gum disease.	Lesson 14
Temperature	Measure of how hot or cold a body is (average kinetic energy of particles); SI unit is Kelvin (K), commonly °C.	Lesson 19
Tooth decay (Dental caries)	Acid erosion of enamel and dentine caused by plaque bacteria feeding on leftover sugars.	Lesson 14
Turbine	A bladed rotor spun by flowing water, wind, or steam to drive an electrical generator.	Lesson 16
Upcycle	Transforming waste objects creatively into items of equal or higher everyday value.	Lesson 20
Vaccination	Giving a safe vaccine so the body builds protective immunity against specific pathogens (e.g., Hepatitis B).	Lesson 22
Vector	An organism (such as female Anopheles mosquito) that carries a pathogen from one host to another.	Lesson 6
Waste audit	Surveying and sorting the types and quantities of rubbish produced in a school or community.	Lesson 21
Waterlogging	Saturation of soil pores with excess water, driving out oxygen needed by plant roots.	Lesson 9

Student Study Progress Tracker

Track Your Mastery Across All 23 First Term Indicator Lessons

Tick each box after studying the lesson, completing the guided activity, and grading your practice exercise using the Answer Key.

Lsn	Indicator	Lesson Focus	Date	Activity	Score / 8	Mastered?
1	B8.1.1.1.1	Types of Mixtures & Their Properties		☐	/ 8	☐ Yes ☐ Fix
2	B8.1.1.1.2	Separating Mixtures Practically		☐	/ 8	☐ Yes ☐ Fix
3	B8.2.1.1.1	The Process of the Carbon Cycle		☐	/ 8	☐ Yes ☐ Fix
4	B8.2.1.1.2	Environmental Role of the Carbon Cycle		☐	/ 8	☐ Yes ☐ Fix
5	B8.2.2.1.1	Life Cycle & Economic Importance of Anopheles		☐	/ 8	☐ Yes ☐ Fix
6	B8.2.2.1.2	Impact of Anopheles & Malaria Control		☐	/ 8	☐ Yes ☐ Fix
7	B8.2.3.1.1	Types of Seed Beds & Matching Crops		☐	/ 8	☐ Yes ☐ Fix
8	B8.2.3.1.2	Planting Crop Parts on Seed Beds		☐	/ 8	☐ Yes ☐ Fix
9	B8.2.3.2.1	Comparing Growth & Flowering Across Seed Beds		☐	/ 8	☐ Yes ☐ Fix
10	B8.2.4.1.1	Animal Feed Types, Nutrients & Labels		☐	/ 8	☐ Yes ☐ Fix
11	B8.2.4.2.1	Importance of Water & Feed to Animal Growth		☐	/ 8	☐ Yes ☐ Fix
12	B8.3.1.1.1	External & Internal Parts of a Tooth		☐	/ 8	☐ Yes ☐ Fix
13	B8.3.1.1.2	Human Teeth Types & Functions		☐	/ 8	☐ Yes ☐ Fix
14	B8.3.1.1.3	Causes & Prevention of Tooth & Gum Decay		☐	/ 8	☐ Yes ☐ Fix
15	B8.4.1.1.1	Energy Conversion & Everyday Chains		☐	/ 8	☐ Yes ☐ Fix
16	B8.4.1.1.2	Importance of Conversion & Hydroelectric Power		☐	/ 8	☐ Yes ☐ Fix
17	B8.4.1.2.1	Renewable & Non-Renewable Energy + Biogas		☐	/ 8	☐ Yes ☐ Fix
18	B8.4.1.2.2	Sustainable Management of Renewable Energy		☐	/ 8	☐ Yes ☐ Fix
19	B8.4.1.3.1	Heat and Temperature: Differences & Link		☐	/ 8	☐ Yes ☐ Fix
20	B8.5.1.1.1	Sustainable Waste Management Practices		☐	/ 8	☐ Yes ☐ Fix
21	B8.5.1.1.2	Community Waste Management Plan & Evaluation		☐	/ 8	☐ Yes ☐ Fix
22	B8.5.2.1.1	Common Communicable Diseases (Hepatitis etc.)		☐	/ 8	☐ Yes ☐ Fix
23	B8.5.2.1.2	Risk Factors of Communicable Diseases		☐	/ 8	☐ Yes ☐ Fix

Personal Misconceptions & Error Log

Record Mistakes from Practice Exercises & What You Learned to Correct Them

Great science students learn fastest from the mistakes they write down and fix. Use this table during term revision before your end-of-term examination.

Lesson & Q#	Mistake / Wrong Idea I Had	Correct Scientific Explanation	How I Will Avoid It Next Time
Example: L3, Q2	Thought plants only photosynthesise and never respire.	Plants respire 24 hours/day taking in O_2 and releasing CO_2.	Check whether question asks about daytime only or 24-hour respiration.